

Arba Minch University
College of Business and Economics
Department of Economics



Econometrics II (Econ 3051)
MODULE

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INTRODUCTION TO THE MODULE

The course aims to develop the mathematical ability of students up to the level required for a general economics degree course (i.e. one not specializing in mathematical economics) or for a modular degree course in economics.

The module introduces mathematical techniques that will be new to most students through examples of their application to economic concepts. It also tries to get students tackling problems in economics using these techniques as soon as possible so that they can see how useful they are. Students are not required to work through unnecessary proofs, or wrestle with complicated special cases that they are unlikely to encounter again.

In this module, we introduce the basic concepts of differentiation followed by examples of economic applications. Further developments of the topic, such as the second-order conditions for optimization, partial differentiation, and the rules for differentiation of composite functions, are then gradually brought in over in the subsequent chapters, again in the context of economics application.

The module tries to cover those mathematical techniques that will be relevant to students of economics degree program. Most applications are in the field of microeconomics, rather than macroeconomics.

Definition: Mathematical economics is an approach to economic analysis, in which the economist makes use of Mathematical symbols in the statement of the problem and draws upon known mathematical theorems to aid in reasoning.

Mathematical Economics Vs Econometrics

Econometrics:- deals with the study of empirical observations using statistical methods of estimation and hypothesis testing.

Mathematical Economics:- refers to the application of mathematics to the purely theoretical aspects of economic analysis.

Chapter One

Review of Basic Concepts

1.1. Aims and objectives

After completing this chapter students will be able to :

- recall the definition and geometric interpretation of derivative
- use the different rules of differentiation to find the derivative of different types of functions
- apply the concept of derivative in economic applications
- apply the techniques they learn in solving optimization problems in economics
- Recall the antiderivative of a function.
- Explain the different rules of integration.
- Apply the concept of integral calculus in solving economic problems.
- recall the concept of matrix and basic matrix manipulation;

Introduction

This chapter is concerned with revising the basic concepts related with quantitative methods in economics. These concepts are discussed in detail in quantitative method for economists I&II courses. So, the discussions in this chapter will be very brief.

1.1 The concept of calculus.

Definition: calculus, in simple terms, is a mathematical technique which is used to deal with change, motion and growth. It is a result of great advancement in the area of mathematics which is used to solve mathematical problems that can't be solved using ordinary methods. In economics, it is frequently used to solve static and dynamic economic problem. Differential calculus, which indicates the rate of change of the dependent variable with respect to the independent variable, is used in static economic analysis; On the other hand, Integral calculus, difference equation & differential equation are used in dealing with dynamic economic problems. These applications are dealt with in a separate chapter but for this section you will be reminded about only the basic concepts which you have dealt with in your Quantitative methods course.

1.1.1 Differential calculus

1. Derivatives

Given a function $y = f(x)$, the derivative of $f(x)$ at any point 'x' is given by

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x} = f'(x): \text{if this limit exists then } f(x) \text{ is said to be}$$

differentiable function at 'x'.

The term $\frac{f(x + \Delta x) - f(x)}{\Delta x}$, is simply slope of the functions which is an equivalent

expression to $\frac{\Delta y}{\Delta x}$, and it represents average rate of change of a function. But the

derivative approximates the rate of change of the positive function when Δx is infinitesimal i.e. slope of a function at a point or instantaneous rate of change.

Example: find the derivative of $f(x) = x^2$

Solution: $f(x + \Delta x) = (x + \Delta x)^2 = f(x + h) \rightarrow \text{substituting } \Delta x = h$

$$f(x) = x^2$$

$$\frac{df(x)}{dx} = f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h)^2 - f(x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{(x^2 + 2xh + h^2) - x^2}{h}$$

$$= \lim_{h \rightarrow 0} 2x + h = \underline{\underline{2x}}$$

Differentiability and continuity

If the derivative of a function, say $f(x)$, exists at point 'c' then $f(x)$ is said to be differentiable at 'c'.

To say a function $f(x)$ is differentiable at 'c' the following conditions should be fulfilled

1. $f(x)$ should be continuous at 'c'
2. $f(x)$ should have a unique (only one) slope at 'c'

The 1st condition requires the fulfillment of continuity conditions i.e.

1. $\lim_{x \rightarrow c^+} f(x)$ & $\lim_{x \rightarrow c^-} f(x)$ should exist

$$2. \lim_{x \rightarrow c^+} f(x) = \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c} f(x)$$

$$3. \lim_{x \rightarrow c} f(x) = f(c)$$

From this we can understand that continuity doesn't necessarily mean differentiability.

Continuity rules out jumps and gaps at a point while differentiability condition requires smoothness of the curve at the given point in addition to continuity conditions. In other words, differentiability rules out jumps, gaps, sharps & cusp corners, while continuity rules out only the first two

Example show the function $f(x) = |x - 2|$ is not differentiable at $x = 2$.

Steps 1 Testing continuity of the function

$$1. \lim_{x \rightarrow 2^+} |x - 2| = 0$$

$$2. \lim_{x \rightarrow 2^-} |x - 2| = 0$$

$$3. \lim_{x \rightarrow 2} f(x) = f(2) = 0, \text{ so, } f(x) \text{ is continuous at } x = 2$$

Step 2 in this step we will see whether the limit of the difference quotient exists or not.

Since $f(x)$ is an absolute value function, it can be redefined as follows:

$$f(x) = \begin{cases} x - 2, & \text{for } x \geq 2 \\ 2 - x, & \text{for } x < 2 \end{cases}$$

To see the existence of the limit of the difference quotient, we find its left & right side limits .i.e.

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^+} \frac{((x+h)-2) - (x-2)}{h} = 1$$

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^-} \frac{(2-(x+h)) - (2-x)}{h} = -1$$

From the above result, we can see that the left & right side limits are not equal. Therefore the limit of the difference quotient doesn't exist which in turn means that the function $f(x) = |x - 2|$ is not differentiable at $x=2$.

Rules of differentiation

1) Constant function rule

Given $f(x) = k$, where K is a constant,

$$f'(x) = 0$$

2) Power function rule

$$f(x) = x^2$$

$$f'(x) = 2x^{2-1} = 2x$$

$$f(x) = x^n$$

$$f'(x) = nx^{n-1}$$

Rules of differentiation involving two or more functions of the same variables

3) Sum/ difference Rule

Given $f(x)$ and $g(x)$

$$\frac{d}{dx}(f(x) \pm g(x)) = f'(x) \pm g'(x)$$

4) Product Rule :

If $f(x)$ & $g(x)$ are differentiable; then

$$\frac{d[f(x).g(x)]}{dx} = f(x)g'(x) + f'(x)g(x)$$

Corollary:- Given $\phi(x) = f(x).g(x).h(x)$

$$\phi'(x) = f'(x).g(x).h(x) + f(x)g'(x).h(x) + f(x)g(x).h'(x)$$

Note that the above corollary is a direct extension of the product rule of differentiation. The rule follows the same pattern for any number of functions and it can be proved using the same rule.

Example: Given $AR = 25 - Q^2$, find MR?

Dear distance learners, I think you know from preceding economics courses about the relation ship between the marginal, average and total revenue functions. If you have forgotten the relation ships please revise your micro economics module. The solution for the above problem can be found by using the relation ship between the marginal, average and total revenue functions

$AR = \frac{TR}{Q} \Rightarrow AR \cdot Q$, $MR = \frac{dTR}{dQ} \Rightarrow \frac{dAR \cdot Q}{dQ}$ using the product rule of differentiation, we get

$$\begin{aligned} MR &= \frac{dAR}{dQ} Q + AR \frac{dQ}{dQ} \\ &= \frac{d(25 - Q^2)}{dQ} Q + (25 - Q^2) \\ &= -2Q^2 + 25 - Q^2 \\ &= 25 - 3Q^2 \end{aligned}$$

5) Quotient Rule

Given $f(x)$ and $g(x)$, where $g(x) \neq 0$

$$\frac{d\left[\frac{f(x)}{g(x)}\right]}{dx} = \frac{f'(x)g(x) - f(x)g'(x)}{[g(x)]^2}$$

Eg. Given $f(x) = \frac{ax^2 + b}{cx}$, find $f'(x)$?

Since $f(x)$ is a quotient of two functions, let $g(x) = ax^2 + b$ and $h(x) = cx$.

$$\Rightarrow f(x) = \frac{g(x)}{h(x)} \Leftrightarrow f'(x) = \frac{g'h - h'g}{h^2}$$

$$g'(x) = 2ax, h'(x) = c$$

$$\Rightarrow f'(x) = \frac{2ax \cdot cx - c(ax^2 + b)}{(cx)^2}$$

$$= \frac{ax^2 + b}{cx^2}$$

Rules of Differentiation Involving functions of Different Variables

6) Chain Rule (composite function rule)

Given $Z = f(y)$, and $y = g(x)$

$$\frac{dz}{dx} = \frac{dz}{dy} \cdot \frac{dy}{dx}$$

Eg. Given $Z = (x^2 + 3x - 2)^{10}$, find $\frac{dz}{dx}$?

Solution: let $y = x^2 + 3x - 2$ then $Z = (y)^{10}$

$$\begin{aligned}\frac{dz}{dx} &= \frac{dz}{dy} \frac{dy}{dx} = 10y^9 \frac{dy}{dx} \\ &= 10y^9(2x + 3) \\ &= 10(x^2 + 3x - 2)(2x + 3)\end{aligned}$$

7) Inverse function Rule

Given $y = f(x)$

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

Eg. Given $x = \frac{y-1}{y+1}$, find $\frac{dy}{dx}$?

$$\frac{dx}{dy} = \frac{(y+1) - (y-1)}{(y+1)^2} = \frac{2}{(y+1)^2}$$

Using inverse function rule, $\frac{dy}{dx} = \frac{1}{\frac{dx}{dy}}$

$$= \left[\frac{(y+1)^2}{2} \right]$$

Example : find the derivatives of the following functions.

i) $f(x) = 3x^3 + 4x^2 - 3x + 2$ Using rules (1), (2) and 3 above

Solution

$$\begin{aligned}f'(x) &= 3(3)x^{3-1} + 4(2)x^{2-1} - 3(1)x^{1-1} + 0 \\ &= 9x^2 + 8x - 3\end{aligned}$$

ii) $(2x+1)^6$ Here the function is a composite function where the outer function is power function.

$$g(x) = 2x + 1$$

$$f(x) = x^6$$

$$f(g(x)) = (2x+1)^6$$

$$\frac{df(g(x))}{dx} = f'(g(x)) \cdot g'(x) = 6(2x+1)^5 \cdot 2 = 12(2x+1)^5$$

$$\begin{aligned} \text{iii) } g(x) &= \frac{3x+1}{x^2-3x} \Rightarrow g'(x) = \frac{(x^2-3x)(3x+1)' - (3x+1)(x^2-3x)'}{(x^2-3x)^2} \\ &= \frac{3(x^2-3x) - (3x+1)(2x-3)}{(x^2-3x)^2} \\ &= \frac{-3x^2 - 2x + 3}{(x^2-3x)^2} \end{aligned}$$

Derivatives of Exponential and Logarithmic Functions

Before dealing with derivatives of exponential and logarithmic functions, it is important to remind you about some special limits which you have discussed in your Quantitative Methods I.

$$1. \lim_{h \rightarrow 0} \frac{\alpha^h - 1}{h} = \ln \alpha$$

$$2. \lim_{h \rightarrow 0} \frac{e^h - 1}{h} = \ln e = 1: \text{ We will use these special limits in proving the derivatives}$$

of exponential and logarithmic functions.

Derivatives of Exponential functions

$$1. \frac{de^x}{dx} = e^x$$

Proof:- by definition :- $\frac{de^x}{dx} = \lim_{h \rightarrow 0} \frac{e^{x+h} - e^x}{h}$

$$= \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h} \rightarrow \text{factoring out } e^x$$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} e^x \cdot \lim_{h \rightarrow 0} \frac{e^h - 1}{h} \rightarrow \text{Product rule of limits (Quant -I)} \\
 &= e^x \cdot \ln e \\
 &= e^x
 \end{aligned}$$

$$2. \frac{da^x}{dx} = a^x \ln a$$

Proof $\frac{da^x}{dx} = \lim_{h \rightarrow 0} \frac{a^{x+h} - a^x}{h} = \lim_{h \rightarrow 0} a^x \frac{(a^h - 1)}{h} = \lim_{h \rightarrow 0} a^x \lim_{h \rightarrow 0} \frac{(a^h - 1)}{h} = \underline{\underline{a^x \ln a}}$

$$3. \frac{da^{f(x)}}{dx} = f'(x) a^{f(x)} \ln a = a^{f(x)} f'(x) \ln a = \text{Using composite function rule}$$

Derivatives of Logarithmic functions

$$1. \frac{d \ln x}{dx} = (\ln x)' = 1$$

Proof : by definition $(\ln x)' = \lim_{h \rightarrow 0} \frac{\ln(x+h) - \ln x}{h}$

$$\begin{aligned}
 &= \lim_{h \rightarrow 0} \left(\frac{1}{h} \ln \left(\frac{x+h}{x} \right) \right) \rightarrow \text{Using logarithmic properties} \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \ln \left(1 + \frac{h}{x} \right) \\
 &= \lim_{h \rightarrow 0} \frac{1}{h} \cdot \frac{x}{x} \ln \left(1 + \frac{h}{x} \right) - \text{Multiplying by } \frac{x}{x} \rightarrow 1 \\
 &= \lim_{h \rightarrow 0} \frac{1}{x} \ln \left(1 + \frac{h}{x} \right)^{\frac{x}{h}} \\
 &= \frac{1}{x} \ln \cdot \lim_{h \rightarrow 0} \left(1 + \frac{h}{x} \right)^{\frac{x}{h}} \\
 &= \frac{1}{x} \ln e = \frac{1}{x}
 \end{aligned}$$

$$2. \frac{d \log_a^x}{dx} = \frac{1}{x \ln a}$$

Proof :- $\log_a^x = \frac{\ln x}{\ln a}$ so $\left(\log_a^x\right)' = \left(\frac{\ln x}{\ln a}\right)'$

$$\left(\frac{\ln x}{\ln a}\right)' = \frac{1}{\ln a} (\ln x)'. \text{ Because } \frac{1}{\ln a} \text{ is constant}$$

$$\frac{1}{\ln a} \cdot \frac{1}{x} = \frac{1}{x \ln a} = \frac{1}{\ln a} \cdot (\ln x)' = \frac{1}{\ln a} \cdot \frac{1}{x}$$

$$(\ln x)' = \frac{1}{x}$$

Examples

1) Given $y = x^{e^{-x^3}}$, find $\frac{dy}{dx}$?

Solution: taking the logarithm of both sides;

$\ln y = e^{-x^3} \ln x$ Then taking the derivative of both sides, we get

$$\frac{y'}{y} = -3x^2 e^{-x^3} \ln x + \frac{1}{x} (e^{-x^3}) \text{ using product rule}$$

$$y' = y \left(-3x^2 e^{-x^3} \ln x + \frac{1}{x} (e^{-x^3}) \right)$$

$$y' = e^{-x^3} \cdot x^{e^{-x^3}} \left(-3x^2 e^{-x^3} \ln x + \frac{1}{x} (e^{-x^3}) \right)$$

2) Given $y = x^x$ find $\frac{dy}{dx}$? $\rightarrow \underline{\underline{x^x(1 + \ln x)}}$

Solution

$$y = x^x \Rightarrow \ln y = x \ln x$$

$$(\ln y)' = (x \ln x)'$$

$$\frac{y'}{y} = \ln x + 1 \Rightarrow y' = y(\ln x + 1)$$

$$= x^x (\ln x + 1)$$

3) Given $y = e^{x^2 \ln 3x}$, find $\frac{dy}{dx}$?

$$\ln y = x^2 \ln 3x,$$

$$(\ln y)' = (x^2 \ln 3x)' \Rightarrow \frac{y'}{y} = 2x \ln 3x + \frac{x^2}{x}$$

$$y' = y(2x \ln 3x + x) = xy(2 \ln 3x + 1) \text{ because } (3x)' = \frac{3}{3x} = \frac{1}{x}$$

$$= \underline{\underline{xe^{x^2 \ln 3x} (1 + 2 \ln 3x)}}$$

Higher order Derivatives

Let $f(x)$ be a function which is differentiable at a point n in its domain, then $f(x)$ is also a function in the same domain as $f(x)$ is differentiable. The derivative of $f'(x)$, if it exists, is called the 2nd derivative of $f''(x)$ provided that it is continuous and smooth. The derivatives found through this iteration process are called higher order derivatives.

N.B 1. the 2nd, 3rd, 4th etc. derivatives are collectively called higher order derivatives.

2. higher order derivatives are symbolized along the same line as 2nd derivative. i.e. the 2nd, 3rd & 4th derivatives are denoted by

$$f''(x), f'''(x), f^{(4)}(x) \dots \text{etc or}$$

$$\frac{d^2 y}{dx^2}, \frac{d^3 y}{dx^3}, \frac{d^4 y}{dx^4} \text{ Respectively, where } \frac{dy}{dx} \text{ is used as an operator symbol and the super}$$

scripts show the number of times that the function should be differentiated with respect to 'x'.

Formulas for n^{th} Derivatives of some functions	
$f(x)$	$f^{(n)}(x)$
1. x^n	$n(n-1)(n-2) \dots (n(n-1)) = n!$
2. x^m	$m(m-1)(m-2) \dots (m-n)x^{m-n}$
3. e^x	e^x
4. a^x	$(\ln a)a^x$
5. $(ax+b)^n$	$n!$

Self test exercise 1.1

1. check the differentiability of $f(x) = \begin{cases} x+1, & \text{for } x \leq 2 \\ ax-5, & \text{for } x > 2 \end{cases}$ at $x=2$
2. Find the first order derivatives of the following functions
 - a). $y = x^2 \log_3^x$
 - b). $g(x) = e^{(5x+4)}$
 - c). $f(x) = 2x^3 - 15x^2 + 36x + 4$
3. Evaluate the 2nd order derivatives of the function $g(x) = 4\sqrt{x} - x$

1.1.2 Integral calculus

In elementary algebra we come across opposite algebraic operations. The opposite of addition is subtraction and that of division is multiplication in the same way, the opposite of differential calculus is Integration. It is also called anti differentiation, i. e. the process of finding the integral of a function.

If $F(x)$ is a function such that $\frac{dF(x)}{dx} = f(x)$, then $F(x)$ is the anti derivative (Integral) of $f(x)$ symbolically this is represented as $\int f(x)dx = F(x) + c$ where 'c' is a constant. This type of integral is called indefinite integral because it represents a family of functions. If we give definite value for 'c', it represents definite integral (to be discussed latter.). As opposed to definite integrals, indefinite integrals do not possess a definite numerical value.

Rules of Integration

Since integration is the reverse process of differentiation, each rule of integration is associated with some derivative formal.

$$1. \quad \int k dx = kx + c \Rightarrow \frac{d(kx + c)}{dx} = k$$

$$2. \int x^n dx = \frac{x^{n+1}}{n+1} + c, n \neq -1$$

$$3. \int kf(x)dx = k \int f(x)dx$$

$$4. \int (f(x) \pm g(x))dx = \int f(x)dx \pm \int g(x)dx$$

$$5. \int e^x dx = e^x + k$$

$$6. \int \left(\frac{1}{e^x} \right) dx = \int e^{-x} dx = -e^{-x} + k$$

$$7. \int e^{mx} dx = \frac{e^{mx} + c}{m}$$

$$8. \int a^{mx} dx = \left(\frac{1}{m \ln a} \right) a^{mx} + k$$

$$9. \int \frac{1}{x} dx = \ln|x| + k$$

Techniques of integration

Although we have seen some rules of integration, some functions are much more complex for such functions it is difficult to know their integrals using the above rules. There are different techniques of finding integrals of these functions. For this chapter, we will focus on 3 of them. I.e. Integration by substitution, integration by parts and integration partial fraction.

Integration by substitution

One of the methods (techniques) which are used to find integrals of complex functions is substitution method or change of variables. This method consists of transforming the integral and the variable of integration to some standard form say 'u' & 'du' completely eliminating 'x' & 'dx'.

Illustration evaluate $\int x\sqrt{2x+1}dx \rightarrow$ since the integrand is the product of two terms $x\sqrt{2x+1}$. Where x is factor of $2x+1$ we can use substitution.

$$\text{Let } U = 2x + 1$$

$$\frac{du}{dx} = 2 = dx = \frac{du}{2}$$

$\Rightarrow$ substituting the value of 'U' & du in place of $2x+1$ & dx we get.

$\int x\sqrt{u} \frac{du}{2}$, but here we have two variables 'x' & U. so, we should completely eliminate 'x'

$$\Rightarrow U = 2x + 1$$

$$2x = U - 1$$

$$x = \frac{U - 1}{2}$$

$$\frac{\int (U - 1)\sqrt{U} du}{2} - \text{substituting the value } x = \frac{U - 1}{2}$$

$$= \frac{1}{4} \int (U - 1)\sqrt{u} du \text{ Factoring out the constants}$$

$$= \frac{1}{4} \int \left(u^{3/2} - u^{1/2} \right) du$$

$$= \frac{1}{4} \int u^{3/2} du - \int u^{1/2} du$$

$$= \frac{1}{4} \int \left(\frac{2}{5} u^{5/2} - \frac{2}{3} u^{3/2} \right) + k$$

$$= \frac{1}{2} \left(\frac{u^{5/2}}{5} - \frac{u^{3/2}}{3} \right) + k$$

$$= \frac{1}{2} \left(\frac{(2x+1)^{5/2}}{5} \right) - \left(\frac{(2x+1)^{3/2}}{3} \right) + k$$

Note:- $\rightarrow$ mostly, the functions to be substituted by 'u' (important 'u' s) are found in

- i) in Exponents
- ii) under roots

- iii) inside brackets
- iv) in the denominators

Integration by parts

Some times, we may not be able to find integrals of products of functions by using substitution rule. We use a technique if integration called integration by parts. It is derived from the product rule of differentiation.

$\int f(x).g(x)dx = f(x)G(x) - \int f'(x)G(x)dx$ where $G(x)$ is anti- derivative of $g(x)$. Letting $G(x) = v$ & $f(x) = u$ we get $\int u dv = uv - \int v du$.

To see how it is derived from the product rule of differentiation, let us start from the rule

$$\frac{d(f(x)G(x))}{dx} = f'(x)G(x) + g'(x)f(x)$$

$$f(x)G(x) = \int f'(x)G(x)dx + \int g'(x)f(x)dx \rightarrow \text{Taking the integral of both sides.}$$

$$\int f(x).g(x)dx = f(x).G(x) - \int f'(x)G(x)dx$$

$$\int u dv = uv - \int v du$$

Illustration evaluate $\int x e^x dx$

For simplicity in using integration by parts we should select the one which is easy to integrate as 'dv' & the one which is easy to differentiate as 'u' / In the above example, $e^x dx$ is easier to integrate & 'x' is easier to differentiate

Let $u = x$ & $dv = e^x dx$

$$du = dx \quad v = \int dv = \int e^x dx = e^x$$

Substituting, this values in the formula, we get

$$\int x e^x dx = x e^x - \int e^x dx$$

$$= x e^x - e^x + k$$

$$= e^x (x - 1) + k$$

Integration by partial fraction

This technique is used to find integrals of rational functions which can be decomposed (reduced). The point of partial fraction integration is that integration of a rational function can be reduced as follows by determining the roots of the polynomial in the denominator.

$$\int \frac{dx}{x-a} = \ln|x-a| + c$$

Illustration :- use partial fraction method to integrate

$$\int \frac{x dx}{(x-1)(x-2)}$$

First we should decompose it as follows

$$\frac{x}{(x-1)(x-2)} = \frac{A}{x-1} + \frac{B}{x-2}$$

→ now multiplying this equation by (x-1)(x-2) we get

$$x = A(x-2) + B(x-1) \rightarrow \text{comparing the terms on the left and right}$$

$$x = Ax - 2A + Bx - B$$

$$x = (A+B)x - 2A - B$$

The coefficient of 'x' on the right is A+B & on the left 1 so

$$A+B=1$$

$$2A-B=0 \rightarrow$$

The constant on the left is 0

Solving the above equation simultaneously we get A = -1 & B=2

$$\frac{x}{(x-1)(x-2)} = \frac{-1}{x-1} + \frac{2}{x-2}$$

$$\int \frac{x dx}{(x-1)(x-2)} = \int \frac{-1 dx}{(x-1)} + \int \frac{2 dx}{x-2}$$

$$-\ln|x-1| + 2\ln|x-2| + c = \ln \left| \frac{x-2}{x-1} \right|^2 + c$$

DEFINITE INTEGRALS

1 Meaning of Definite Integrals

In the preceeding sections, we have discussed that the integral of a function is a family of functions due to the fact that functions having different constants have the same derivative.

In this section we will revise another form of integral called the **definite integral**.

Let $f(x)$ is a continuous function and if $F(x)$ is the antiderivative of $f(x)$ then

$$\int f(x)dx = F(x) + c$$

If we choose two values of x in the domain, say $x = a$ and, $x = b$ and substitute these values in to the right side of the equation and taking the difference we get

$$[F(b) + c] - [F(a) + c] = F(b) - F(a)$$

This value is a specific numerical value free of the variable x as well as the arbitrary constant c . It is called the **definite integral of $f(x)$ from a to b** .

Notation:

$$\int_a^b f(x)dx \text{ - read as the definite integral of } f \text{ from } a \text{ to } b$$

The numbers a and b are called **limits of integration**.

Letter a is the lower limit of integration.

Letter b is the upper limit of the integration.

$$\text{Therefore, } \int_a^b f(x)dx = F(x) \Big|_a^b = F(b) - F(a)$$

Let us see some examples on how to find the definite integral of a function.

Example 1: Evaluate $\int_0^2 x^2 dx$

Solution: First find the antiderivative of x^2

$$\int x^2 dx = \frac{1}{3} x^3 + c$$

Then evaluate the indefinite integral at the upper and lower limits and take the difference to get

$$\int_0^2 x^2 dx = \frac{1}{3} x^3 \Big|_0^2 = \frac{1}{3} (2)^3 - \frac{1}{3} (0)^3 = \frac{8}{3} - 0 = \frac{8}{3}$$

Definite integrals of marginal cost functions

This concept of the definite integral has several applications in economics. To evaluate TVC from an MC function for a given value of output one simply substitutes the given quantity into the TVC function. This is the same as evaluating the definite integral between zero and the given quantity. For example, assume that you wished to find the value of TVC when $q = 8$ and you are given the function $MC = 7.5 + 0.3q^2$. This value would be

$$TVC = \int_0^8 MC = \int_0^8 7.5 + 0.3Q^2 = 7.5Q + 0.1Q^3 \Big|_0^8 = 111.2$$

Therefore, TVC is equal to the area under the MC schedule between zero and the given level of output.

Another application of definite integrals is to find consumers and producer's surplus.

Recall from your Micro-economics course that, consumer surplus is the difference between what the consumers are willing to pay and what they actually paid.

Geometrically, this is defined as the area below a demand schedule but above the ruling price. This difference is often used as a measure of welfare.

For this part, you are advised to revisit your quantitative method for economists I module

Self Test Exercise 1.2

1. Evaluate the following indefinite integrals. (5 pts each)

a). $\int \frac{3x+11}{x^2-x-6} dx$

b). $\int x\sqrt{2x+1} dx$

2. find the definite integral of the following functions

a. $\int_1^3 (20 + 4x)$

b. $\int_0^8 MC$ where $MC = 7.5 + 0.3Q^2$,

- c. What does the definite integral of MC represent? If the total fixed cost is given by $TFC=300$, find the total cost function.

1.2 Basic matrix manipulation

A matrix is defined as an array of numbers (algebraic symbols) set out in rows & columns. Usually Matrices are denoted by a capital letter in bold type.

The algebra of matrices is a suitable mathematics method which enable s us to handle large systems of simultaneous equations. It can be used to

1. write large systems of equation in a compact way
2. Test the existence of solutions by using determinates
3. find the solutions (if exists)

each of the numbers (elements) in a matrix have their own definite position. If a matrix has m rows & n columns, the order of the matrix become $m \times n$ read 'm' by 'n'

$$A = \begin{matrix} a_{11} & a_{12} & a_{13} & a_{1n} \\ a_{21} & a_{22} & a_{23} & a_{2n} \\ \vdots & & & \\ a_{m1} & a_{m2} & a_{m3} & a_{mn} \end{matrix}$$

$\Rightarrow a_{11}, a_{12} \dots$ ect are called the elements of the matrix

- The number of rows and columns of a matrix together define the dimension of a matrix. If $m=n$ then the matrix is called square matrix

If a matrix contains only one column (row) it is called column(row) matrix

Matrix operation

Since matrix contain a all of numbers (a whole block of numbers) the families algebraic operations which are used for single numbers are not adoptively applicable, and there is a need for new set of operation rules.

1. equality of matrices

$A=B$ only if $a_{ij}=b_{ij}$

Addition & subtraction of matrices

$A + B = [a_{ij}] + [b_{ij}] = [a_{ij} \pm b_{ij}]$ For each element b_{ij} corresponding to a a_{ij}

i.e. addition and subtraction of matrices A & B is defined as addition(subtraction) of each pair of corresponding elements

Note equality, addition & subtraction of matrices is applicable only if the two matrices have equal dimension

Example

$$\begin{bmatrix} 4 & 9 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 0 \\ 0 & 7 \end{bmatrix} = \begin{bmatrix} 4+2 & 9+0 \\ 2+0 & 1+7 \end{bmatrix} = \begin{bmatrix} 6 & 9 \\ 2 & 8 \end{bmatrix}$$

Scalar multiplication

Multiplying a matrix 'A' by a number (scalar) is simply multiplying each element of the matrix by the given scalar i.e. $\lambda A = \lambda[a_{ij}] = [\lambda a_{ij}]$

Example: The numbers of units of a product sold by a retailer for the last 2 weeks are shown in matrix **A** below, where the columns represent weeks and the rows correspond to the two different shop units that sold them.

$$\mathbf{A} = \begin{pmatrix} 12 & 30 \\ 8 & 15 \end{pmatrix}$$

If each item sells for £4, derive a matrix for total sales revenue for this retailer for these two shop units over this two-week period.

Solution

Total revenue is calculated by multiplying each element in matrix of sales quantities **A** by the scalar value 4, the price that each unit is sold at. Thus total revenue can be represented (in £) by the matrix

$$\mathbf{R} = 4\mathbf{A} = \begin{pmatrix} 4 \times 12 & 4 \times 30 \\ 4 \times 8 & 4 \times 15 \end{pmatrix} = \begin{pmatrix} 48 & 120 \\ 32 & 60 \end{pmatrix}$$

Multiplication of matrices

Given two matrices $A_{m \times n}$ and $B_{n \times p}$, the conformability condition for multiplication AB is that the column dimension of A should be equal to the row dimension of B, i.e. AB will be defined only if $n=p$. That is, in order to multiply two matrices, the first matrix must have as many columns as the second matrix has rows. Matrices that fulfill this condition are said to be conformable for multiplication. if the product AB is defined it will have the dimension $m \times p$.

Matrices	A	x	B	=	AB
Orders	m x n		n x p	=	m x p

For example, if A is a 2 x 3 matrix and B is a 3 x 5 matrix, and then the order of the new matrix AB will be 2 x 5. Note how ever, that BA is not defined.

The rule of matrix multiplication: If one matrix is multiplied by another matrix, the basic rule is to multiply elements along the rows of the first matrix by the corresponding elements down the columns of the second matrix.

$$\text{If } \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{pmatrix} \quad 2 \times 3 \quad \times \quad \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \end{pmatrix} \quad 3 \times 2 \quad = \quad \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{pmatrix} \quad 2 \times 2, \text{ then}$$

$$C_{11} = a_{11}(b_{11}) + a_{12}(b_{21}) + a_{13}(b_{31})$$

$$C_{12} = a_{11}(b_{12}) + a_{12}(b_{22}) + a_{13}(b_{32})$$

$$C_{21} = a_{21}(b_{11}) + a_{22}(b_{21}) + a_{23}(b_{31})$$

$$C_{22} = a_{21}(b_{12}) + a_{22}(b_{22}) + a_{23}(b_{32})$$

Self test exercise 1.3

$$1. \text{ Let } M = \begin{pmatrix} 2 & 4 & 6 \\ 7 & 8 & 9 \\ -5 & 2 & 0 \end{pmatrix}, \quad N = \begin{pmatrix} 1 & 2 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}, \text{ and } L = \begin{pmatrix} 1 & 3 \\ 4 & 5 \end{pmatrix}$$

Then find the following if they are conformable? (2 points each)

i) $M + N$ iii) $M \times N$

ii) $M + L$ iv) $N + M$

Chapter -2

2. Derivatives in use

Aims and objectives

After completing this chapter students should be able to:

- Find the maximum or minimum point of a single variable function by differentiation and checking first-order and second-order conditions.
- Use calculus to solve optimization problems in economics
- Derive marginal revenue and marginal cost functions using differentiation and relate them to the slopes of the corresponding total revenue and cost functions.
- Calculate point elasticity for non-linear demand functions and relate it with revenue function.
- Use linear and polynomial approximations to solve complex functions.
- Use Euler's theorem to identify the properties of multivariable functions in economics

Introduction

In the 1st chapter of this module, we have been discussing about derivatives and rules of differentiation. This chapter on the other hand, focuses on the application of derivatives with a special emphasis on their use in economics. Derivatives are widely used in static economic analysis. This chapter focuses on some typical applications of derivatives. However; it may not exhaustively address all of the applications.

2.1 Application of Derivatives

1. **First order Derivatives:-** the sign of first order derivative is important to determine the behavior of functions in a given interval, i.e. whether a function is increasing or decreasing can be determined using the sign of its first derivative.

Definition:- consider a function $f(x)$ which is continuous in an interval $I [a, b]$ where $b > a$, $f(x)$ is said to be

- increasing if $f(b) > f(a)$ or $f(a) < f(b)$

≥ decreasing if $f(a) > f(b)$ or $f(b) < f(a)$

suppose that $f(x)$ is differentiable on the given Interval,

1. If $f'(x) > 0$ for all x in I , then $f(x)$ is increasing on I
2. If $f'(x) < 0$ for all x in I , then $f(x)$ is decreasing on I
3. More generally, the conclusion in (1) and (2) still hold if in each finite interval say MTI there are only finitely many points, at which the assumptions of $f'(x) > 0$ and $f'(x) < 0$ are not satisfied.

Illustrative Example

Given $TR = 20q - q^2$ find the intervals for which TR functions is

- a) increasing
- b) decreasing

$$MR = \frac{dTR}{dQ} = 20 - 2q$$

$$20 - 2q > 0$$

$$20 - 2q < 0$$

$$20 > 2q$$

$$20 < 2q$$

$$10 > q$$

$$q > 10$$

$q < 10$; i.e TR is increasing for $q < 10$, and it is decreasing for $q > 10$.

Application of derivatives in Economics

1. The concept of marginal analysis:- marginal analysis is very important in dealing with most economic functions. Mathematically, marginal functions are the 1st derivatives of total functions.

$$\frac{dTR}{dQ} = MR$$

$$\frac{dTC}{dQ} = MC$$

$$\frac{d\Pi}{dQ} = M\Pi \dots \text{etc.}$$

Derivative can also be used to determine the relationship between marginal & average Economics functions. Consider the cost function.

$$TR = C(Q)$$

$$AC = \frac{TC}{Q} \Rightarrow \frac{dAC}{dQ} = \frac{\frac{dTC}{dQ}Q - TC}{Q^2} \text{ (using the quotient rule).}$$

$$MC = \frac{dTC}{dQ} \quad \frac{1(MC - AC)}{Q} \text{ - from this we can observe that , will be increasing when}$$

$$MC > AC .$$

For TR function

$$TR = PQ \text{ \& } AR = \frac{PQ}{Q} = P$$

$$MR = \frac{d(PQ)}{dQ} = p \cdot \frac{dQ}{dQ} + \frac{Qdp}{dQ} \text{ (Using Quotient rule)}$$

$$MR = p + Q \frac{dp}{dQ}$$

The equation for a) above is a representative relation between AR & MR if the market is perfectly competitive price will be constant &

$$\frac{dp}{dQ} = 0 \text{ \& } MR = p = AR$$

If the market is monopoly, $\frac{dp}{dQ} < 0$, so $MR < AR$.

In general, MR can never be greater than AR. At extreme $MR=AR$ when the market is perfectly competitive.

2.2. Elasticities as Logarithmic Derivatives

i) Elasticity Function

Given any function $y = f(x)$

Let x change from x to x+h, then the proportionate change in x is given by

$$\frac{\Delta x}{x} = \frac{(x+h) - x}{x} = \frac{h}{x} \text{ : similarly, the proportionate change in y is given by}$$

$$\frac{\Delta y}{y} = \frac{f(x+h) - f(x)}{f(x)} .$$

The ratio of the proportionate change in y per unit proportionate change in x

is given by:
$$\frac{\frac{\Delta y}{y}}{\frac{\Delta x}{x}} = \frac{\frac{f(x+h) - f(x)}{f(x)}}{\frac{h}{x}}$$

$$\frac{f(x+h) - f(x)}{f(x)} \cdot \frac{x}{h}$$

$$\epsilon_{yx} = \left(\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \right) \cdot \frac{x}{y}$$

$$\epsilon_{yx} = \boxed{\frac{dy}{dx} \cdot \frac{x}{y}} \quad \text{Point elasticity of y with respect to x.}$$

ii) Elasticity as logarithmic Derivatives

Given a function, $y = f(x)$

Let $u = \ln y$

$$v = \ln x \Rightarrow x = e^v$$

$$\frac{d(\ln y)}{d(\ln x)} = \frac{du}{dv} = \frac{du}{dy} \cdot \frac{dy}{dx} \cdot \frac{dx}{dv}$$

$$= \frac{1}{y} \cdot \frac{dy}{dx} \cdot e^v$$

$$= \frac{1}{y} \cdot \frac{dy}{dx} \cdot x$$

$$\underline{\underline{\epsilon_{yx} = \frac{dy}{dx} \cdot \frac{x}{y}}}$$

iii) Elasticity of Demand and Marginal Revenue (MR)

Assume: $TR = PQ$

$$MR = P + Q \cdot \frac{dP}{dQ}$$

$$MR = P \left(1 + \frac{1}{e_d} \right) \text{ or}$$

$$MR = AR \left(1 - \frac{1}{e_d} \right)$$

From this we conclude that:

i) $MR > 0$, if $1 - \frac{1}{e_d} > 0$ or $e_d > 1 \rightarrow$ elastic demand

In this case TR increases as Q increases (or as p decreases)

ii) $MR < 0$, if $1 - \frac{1}{e_d} < 0$ or $e_d < 1 \rightarrow$ inelastic demand

In this case TR decrease as Q increases (or as P decreases)

iii) $MR = 0$, if $1 - \frac{1}{e_d} = 0$ or $e_d = 1 \rightarrow$ unitary elastic demand

In this case TR does not change as Q(or P) changes.

Example: verify $\eta_d = \frac{AR}{AR - MR}$ for the demand function

$$Q = \left(\frac{a - p}{b} \right), b \neq 0, \text{ Where } Q = \text{output}$$

P = price

Relationship between Average cost and Marginal cost functions

Given $C(Q)$, $AC(Q)$ and $AC(Q)$, $MC(Q)$, for $Q > 0$; $TC(Q) = AC(Q) \cdot Q$

$$\frac{d[AC(Q)]}{dQ} = \frac{d\left(\frac{C(Q)}{Q}\right)}{dQ} = \frac{1}{Q} [MC(Q) - AC(Q)]$$

- from this it follows that:

i) At the minimum point AC curve

The slope is zero:

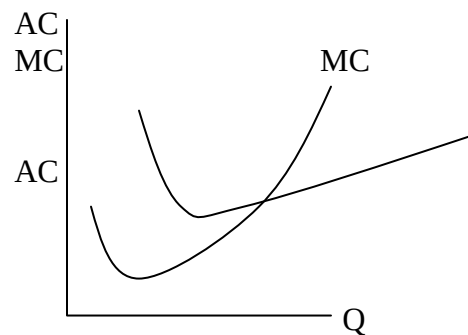
$$\Rightarrow AC(Q) = MC(Q)$$

ii) When AC is falling, $AC'(Q) < 0$

$$\Rightarrow MC(Q) < AC(Q)$$

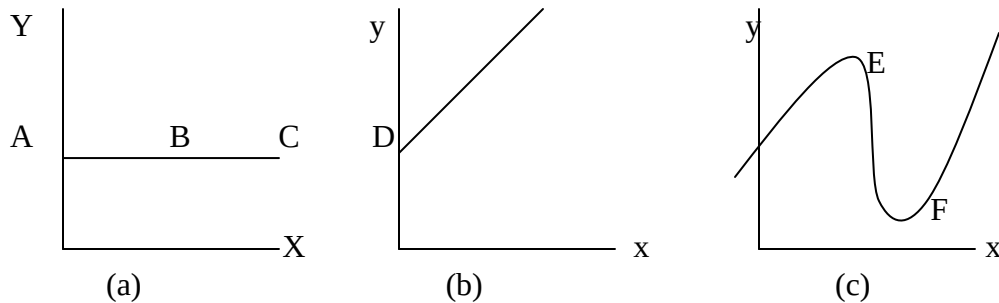
iii) When AC is rising, $AC'(Q) > 0$

$$\Rightarrow MC(Q) > AC(Q)$$



Relative Vs Absolute Extremum

- consider the following three cases:



In figure (a), no significant choice regarding the value of x is made for the maximum or minimum of y . Hence the points represent neither a maximum nor minimum.

In figure (b), given the set of non-negative real numbers as the domain, the function is strictly increasing. However we may consider point D representing an absolute (global) minimum in the range of the functions.

In figure c, points E and F are examples of a relative (or local) maximum and minimum respectively.

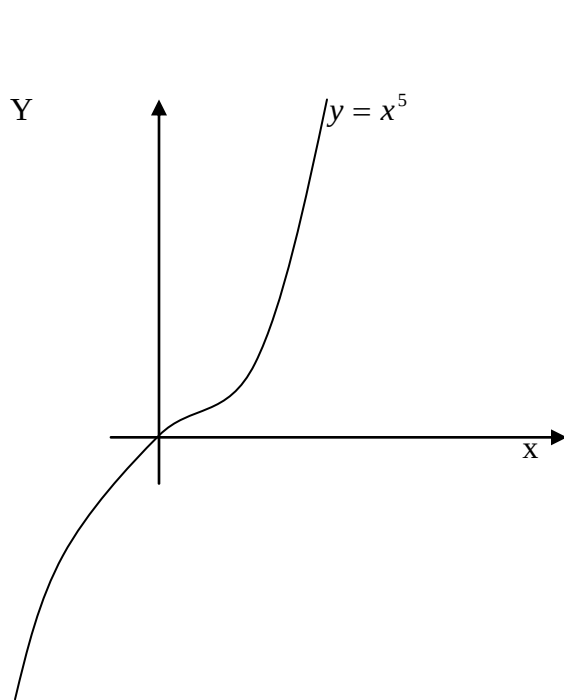
<u>Summary:</u>	<u>Condition</u>	<u>Relative max</u>	<u>Relative min</u>
	1st order necessary condition	$f'(x_0) = 0$	$f'(x_0) = 0$
	2nd order sufficient condition	$f''(x_0) < 0$	$f''(x_0) > 0$

Note that : For $f^n(x_0) \neq 0$, where n is higher derivative of order n ;

- i) if $f^n(x_0) \neq 0$ and n is **even**, the point is an extremum point
- ii) If $f^n(x_0) \neq 0$ and n is **odd**, the point is an inflection point

Example:

1) Given $y = x^5 = f(x)$



$$f'(X) = 5x^4$$

⋮

$f^{(5)}(X) = 120 \rightarrow$ The function is at a point of inflection, at $x = 0$

2.3. Higher order Derivatives

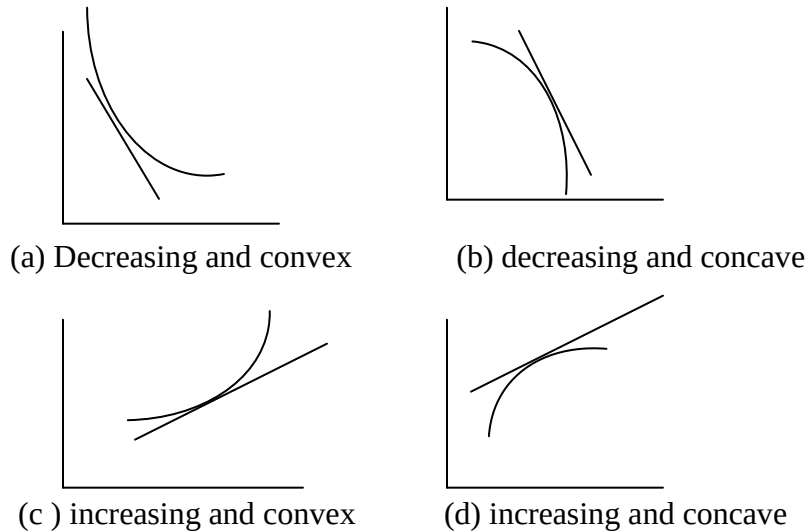
2.3.1. Concavity and Convexity

Definition: let $f(x)$ be a function which is defined on an interval I , we say that $f(x)$ is convex (concave upward) on I if $f(c) < L(c)$, where $L(x) = f(a) + \frac{f(b) - f(a)}{b - a}(x - a)$ for all $a, b, c \in I$ with $a < c < b$. Here, $L(x)$ is the line through $(a, f(a))$ and $(b, f(b))$. we say that it is concave down on I if $f(c) > L(c)$ for all $a, b, c \in I$ with $a < c < b$.

Geometrically: the line tangent to convex (concave upward) functions is below the graph of the functions (tangent from below).

The tangent line of concave down ward functions is tangent from i.e. the line is above the graph.

Fig.



For a function $f(x)$ defined on an interval I ,

- 1) If $f(x)$ is differentiable and $f(x)$ is increasing on I , then $f(x)$ is concave upward on I and if $f'(x)$ is decreasing on I , then $f(x)$ is concave down on I .
- 2) If $f(x)$ is twice differentiable on I , then
 - $f(x)$ is concave upward on I if $f''(x) > 0$ for all $x \in I$
 - $f(x)$ is concave down on I if $f''(x) < 0$ for all $x \in I$

Illustrative Example

Given the cost function $TC = 0.5Q^3 - 0.7Q^2 - 30Q + 256$, then

- a) find the values MC
- b) find the interval for which AC & MC are concave upward & concave down ward.

Solution:-

$$MC = \frac{dTC}{dQ} = 1.5Q^2 - 1.4Q - 30$$

$$AC = \frac{TC}{Q} = \frac{0.5Q^3 - 0.7Q^2 - 30Q + 256}{Q}$$

$$= 0.5Q^2 - 0.7Q - 30 + \frac{256}{Q}$$

$\Rightarrow$ AC will be concave upward if $\frac{d^2 AC}{dQ^2} > 0$

$$\frac{dAC}{da} = Q - 0.7 - \frac{256}{Q}$$

$$\frac{d^2 AC}{dQ^2} = 1 - \frac{512}{Q^3} > 0$$

$$1 > \frac{512}{Q^3}$$

$$Q^3 > 512$$

$$Q > 8 \rightarrow \text{Concave upward (convex)}$$

$$Q < 8 \rightarrow \text{concave downward}$$

Example :1 Given, $f(x) = ax^2 + bx + c$, for $a > 0$

Given, check the concavity or convexity of the function?

→ since $f''(x) = 2a > 0$ it is a strictly convex function

Examples: check for concavity and /or convexity

a) $f(x) = ax^2 + bx + c$ Answer $f''(x) = 2a$:- concave when $2a < 0$

$$a < 0$$

- Convex when $2a > 0$

$$a > 0$$

b. Given the production function $y = AK^\alpha$, $A > 0, 0 < \alpha < 1$

$$\frac{d^2 y}{dk^2} = \alpha(\alpha - 1)AK^{\alpha-2} < 0 \rightarrow \text{concave}$$

c. Suppose the functions U & g are both increasing and concave so that

$U' \geq 0, U'' \leq 0, g' \geq 0$, and $g'' \leq 0$. prove that the composite function

$f(x) = g(U(x))$ is also increasing and concave?

Solution : $f(x) = g(u(x))$

$$f'(x) = \frac{d[g(u(x))]}{du(x)} \cdot \frac{du(x)}{dx}$$

$$= \underset{\substack{\geq 0 \\ \geq 0(\text{increasing})}}{g'(u(x))} \cdot \underset{\geq 0}{u'(x)}$$

→ The increasing transformation of an increasing function is increasing.

$$\begin{aligned} f''(x) &= [g'(u(x)) \cdot u'(x)]' \\ &= [g'(u(x))]' \cdot u'(x) + g'(u(x)) \cdot u''(x) \\ &= g''(u(x)) \cdot \underset{\leq 0}{[u'(x)]^2} + \underset{\leq 0}{u''(x)} \cdot \underset{\geq 0}{g'(u(x))} \\ &\quad (-) \quad \quad \quad (-) = \leq 0 \text{ concave.} \end{aligned}$$

→ An increasing concave transformation of a concave function is concave

Self test Exercise 2.1

1) Given $Q = \alpha p^\beta$, where α and β are constants: find price elasticity of demand?

2) Given $y = f(x)$, show that the derivative

$$\frac{d(\log_a^y)}{d(\log_a^x)}, a \neq c = \frac{dy}{dx} \cdot \frac{x}{y}?$$

3) Given $\frac{1}{9}x^3 - \frac{1}{6}x^2 - \frac{2}{3}x + 1$, find the critical points and determine whether they are maximum or minimum.

4) Given the total cost function:

$$C(Q) = aQ^2 + bQ + c, (Q > 0). \text{ find the minimum point of the average cost function?}$$

5) Find the intervals where the following cubic cost function is convex and where it is concave and find the point of inflection?

$$C(Q) = aQ^3 + bQ^2 + cQ + d$$

6) for the function $y = -(x-8)^4$

a) Find the critical value(s)?

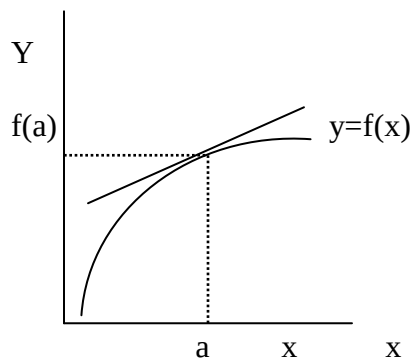
b) Test to see if at the critical values the function is at a relative max, min or possible inflection point?

2.4 Linear Approximations

To avoid working with a complicated function we sometimes try to find a simpler function that approximates the original one.

Since linear functions are easy to work with, we first try to find a linear approximation for any given function.

Suppose the function $y=f(x)$ is differentiable at $x= a$, what is the equation of the tangent to the graph of the function at the point $(a, f(a))$?



- Let us approximate the function $y= f(x)$ by the tangent line passing through the point $(a, f(a))$.

- Given the equation of the tangent line as :

$$y = mx + b, \text{ the slope of the function}$$

At a is given by $f'(a) = m$

$$y = f'(a).x + b$$

$$\text{Thus, } f(a) = f'(a).a + b$$

Thus, we have : $\Rightarrow b = f(a) - f'(a).a$

At point a,

$$\Rightarrow y = f'(a).x + f(a)$$

$$y = \underline{\underline{f(a) + f'(a)(x - a)}}$$

Example

1) Find the linear approximation of $f(x) = \sqrt[3]{x}$ about $x = 1$

$$y = f(a) + f'(a)(x - a)$$

$$\Rightarrow f(1) = 1$$

$$f'(1) = \frac{1}{3}(1)^{-2/3} = \frac{1}{3}$$

$$\Rightarrow \underline{\underline{\sqrt[3]{x} = 1 + \frac{1}{3}(x - 1) \text{ for } x \text{ close to } 1}}$$

2) Find the linear approximation of $f(\varepsilon) = (1 + \frac{3}{2}\varepsilon + \frac{1}{2}\varepsilon^2)^{1/2}$ about $\varepsilon = 0$?

Solution $f(0) = 1$

$$f'(\varepsilon) = \left(1 + \frac{3}{2}\varepsilon + \frac{1}{2}\varepsilon^2\right)^{-1/2} \left(\frac{3}{2} + \varepsilon\right) \frac{1}{2}$$

$$\text{And } f'(0) = \frac{1}{2} \left(\frac{3}{2}\right) = \frac{3}{4}$$

$$\Rightarrow f(\varepsilon) = (1 + \frac{3}{2}\varepsilon + \frac{1}{2}\varepsilon^2)^{1/2} \approx f(0) + f'(0)(\varepsilon - 0)$$

$$\approx \underline{\underline{1 + \frac{3}{4}\varepsilon}}$$

3) Find an approximate value for: $(1.001)^{50}$?

$$\text{Let } f(x) = x^{50}, x \rightarrow 1$$

$$x^{50} \approx f(1) + f'(1)(x - 1)$$

$$\approx 1 + 50(1.001 - 1)$$

$$\approx \underline{\underline{1.05}}$$

2.5 Polynomial Approximations

Approximations by linear functions may be insufficiently accurate. So, it is normal to try quadratic approximations or approximations by polynomial of higher order.

1) Quadratic Approximations:- How can a function $f(x)$ be approximated near $x=a$, by a quadratic approximation?

$$\text{Given } p(x) = f(x) = A + B(x-a) + C(x-a)^2$$

$$p'(x) = B + 2C(x-a)$$

$$p''(x) = 2C$$

When $x=a$

$$p(a) = A = f(a)$$

$$p'(a) = B = f'(a)$$

$$p''(a) = 2C \Rightarrow C = \frac{1}{2} p''(a) = \frac{1}{2} f''(a)$$

Hence $p(x) = A + B(x-a) + C(x-a)^2$ is approximated about $x = a$ as:

$$f(x) \approx f(a) + f'(a)(x-a) + \frac{1}{2} f''(a)(x-a)^2$$

Example. Find the quadratic approximation for:

$$f(x) = \sqrt[3]{x}, \text{ about } x=1$$

Solution: $f(x) = x^{1/3}$, and $f(1) = 1$

$$f'(x) = \frac{1}{3} x^{-2/3}, \quad f'(1) = \frac{1}{3}$$

$$f''(x) = -\frac{2}{9} x^{-5/3}, \quad f''(1) = -\frac{2}{9}$$

$$\text{Thus, } f(x) \approx 1 + \frac{1}{3}(x-1) - \frac{1}{9}(x-1)^2$$

2) Higher order approximations

Suppose we want to approximate a function $f(x)$ around $x=a$ by an n^{th} degree polynomial of the form:

$$p(x) = A_0 + A_1(x-a) + A_2(x-a)^2 + A_3(x-a)^3 + \dots + A_n(x-a)^n$$

$$p'(x) = A_1 + 2A_2(x-a) + 3A_3(x-a)^2 + \dots + nA_n(x-a)^{n-1}$$

$$p''(x) = 2A_2 + 6A_3(x-a) + \dots + (n-1)(n)A_n(x-a)^{n-2}$$

$$\vdots$$

$$p^{(n)}(x) = n(n-1)(n-2)\dots 2.1.A_n$$

When $x=a$

$$p(a) = A_0 = f(a)$$

$$p'(a) = A_1 = 1!A_1 \Rightarrow A_1 = \frac{p'(a)}{1!} = \frac{f'(a)}{1!}$$

$$p''(a) = 2A_2 = 2!A_2 \Rightarrow A_2 = \frac{p''(a)}{2!} = \frac{f''(a)}{2!}$$

$$p'''(a) = 6A_3 = 3!A_3 \Rightarrow A_3 = \frac{p'''(a)}{3!} = \frac{f'''(a)}{3!}$$

$$\vdots$$

$$p^{(n)}(a) = n(n-1)(n-2)\dots 2.1.A_n = n!A_n \Rightarrow A_n = \frac{p^{(n)}(a)}{n!} = \frac{f^{(n)}(a)}{n!}$$

$$\Rightarrow f(x) \approx f(a) + \frac{f'(a)(x-a)}{1!} + \frac{f''(a)(x-a)^2}{2!} + \dots + \frac{f^{(n)}(a)(x-a)^n}{n!}$$

e.g Find the third order approximation of $f(x) = \sqrt{1+x}$ about $x=0$?

Solution: $f(0) = 1$

$$f'(x) = \frac{1}{2}(1+x)^{-1/2} \Rightarrow f'(0) = \frac{1}{2}$$

$$f''(x) = -\frac{1}{4}(1+x)^{-3/2} \Rightarrow f''(0) = -\frac{1}{4}$$

$$f'''(x) = \frac{3}{8}(1+x)^{-5/2} \Rightarrow f'''(0) = \frac{3}{8}$$

Therefore, $f(x) \approx \frac{f'(0)(x-0)}{1!} + \frac{f''(0)(x-0)^2}{2!} + \frac{f'''(0)(x-0)^3}{3!}$

$$\underline{\underline{f(x) \approx 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3}}$$

Self test Exercise 2.2:

1. Find the quadratic approximation for $f(x) = (5x + 3)^{-2}$, for x close to 0.
2. Find quadratic approximation of $F(K) = AK^\alpha$ about $K = 1$?
3. Prove that $(1 + x)^m \approx \underline{1 + mx}$ for x close to zero, and use this approximation to the following numbers?

a) $3\sqrt{1.1} = \left(1 + \frac{1}{10}\right)^{\frac{1}{3}}$ c) $(1.02)^{25}$

b) $\sqrt[5]{33} = 2\left(1 + \frac{1}{32}\right)^{\frac{1}{5}}$ d) $\sqrt{37} = \sqrt{36 + 1}$

2.6 Expansion of Functions using Maclaurin and Taylor series

a) Maclaurin series of a polynomial function_ (Expansion around the point $x=0$)

- Consider expanding an n^{th} degree polynomial function:

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots + a_nx^n$$

- Successive differentiation of $f(x)$ yields:

$$f'(x) = a_1 + 2a_2x + 3a_3x^2 + \dots + na_nx^{n-1}$$

$$f''(x) = 2a_2 + 6a_3x + \dots + n(n-1)a_nx^{n-2}$$

$\vdots$

$$f^{(n)}(x) = n(n-1)(n-2)\dots(2)(1)a_n$$

Evaluate at $x=0$

$$f(0) = a_0 \Rightarrow a_0 = \frac{f(0)}{0!}$$

$$f'(0) = a_1 \Rightarrow a_1 = \frac{f'(0)}{1!}$$

$$f''(0) = 2a_2 \Rightarrow a_2 = \frac{f''(0)}{2!}$$

$$f^{(n)}(0) = n(n-1)(n-2)\dots(2)(1)a_n = n!a_n \Rightarrow a_n = \frac{f^{(n)}(0)}{n!}$$

Now, substituting these values into the original function we have.

$$f(x) = \frac{f(0)}{0!} + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \dots + \frac{f^{(n)}(0)x^n}{n!}$$

Example -1: Find the Maclaurin series for the function: $f(x) = 8 + 36x + 12x^2 + x^3$

Solution: $f(0) = 8$

$$f'(x) = 36 + 24x + 3x^2$$

$$f''(x) = 24 + 6x$$

$$f'''(x) = 6$$

$$\begin{cases} f'(0) = 36 \\ f''(0) = 24 \\ f'''(0) = 6 \end{cases}$$

Thus, the Maclaurin series is:

$$f(x) = f(0) + \frac{f'(0)x}{1!} + \frac{f''(0)x^2}{2!} + \frac{f'''(0)x^3}{3!}$$

$\Rightarrow f(x) = 8 + 36x + 12x^2 + x^3$ } This verifies that the Maclaurin series correctly represents the given function.

b) Taylor series of a polynomial function (Expansion around any point $x=x_0$)

For the purpose of expansion around a specific point x_0 , we may first interpret any given value of x as a deviation from x_0 .

Let $X = X_0 + \delta$, where δ represents this deviation so that $\delta = x - x_0$.

Consider the function: $f(x) = 2 + 4x + 3x^2$

$$f(x) = 2 + 4(x_0 + \delta) + 3(x_0 + \delta)^2$$

$$f'(x) = 4 + 6(x_0 + \delta)$$

$$f''(x) = 6$$

- Since x_0 in the present context is fixed, the only variable in the expression $x = x_0 + \delta$ is δ . Therefore, we can rewrite $f(x)$ as a function of δ as:

$$g(\delta) = 2 + 4(X_0 + \delta) + 3(X_0 + \delta)^2 \equiv f(x)$$

$$g'(\delta) = 4 + 6(X_0 + \delta) \quad [\equiv f'(x)]$$

$$g''(\delta) = 6 \quad [\equiv f''(x)]$$

- Expanding $g(\delta)$ around zero ($\delta=0$) yields the following Maclaurin series

$$g(\delta) = g(0) + \frac{g'(0)\delta}{1!} + \frac{g''(0)\delta^2}{2!}$$

→ The fact that $\delta = 0$ implies $X = X_0$, hence

$$g(0) = 2 + 4x_0 + 3x_0^2 = f(x_0)$$

$$g'(0) = 4 + 6x_0 = f'(x_0)$$

$$g''(0) = 6 = f''(x_0)$$

→ substituting these in to the function $g(\delta)$, we have

$$f(x) = [g(\delta)] = \underline{\underline{f(x_0) + \frac{f'(x_0)(x-x_0)}{1!} + \frac{f''(x_0)(x-x_0)^2}{2!}}}$$

Example:- Find a Taylor series expansion for the function $f(x) = 5 + 3x + 2x^2 + 4x^3$, around $x_0 = 3$.

Expansion of an arbitrary function: From Taylor's Theorem so far, we have seen how n^{th} degree polynomial functions can be expressed in another equivalent n^{th} degree polynomial form. But it is also possible to express any arbitrary function $f(X)$ (one that is not necessarily polynomial) in a polynomial form.

- According to Taylor's Theorem, given an arbitrary function $\phi(x)$, which is continuous with a finite power n , its expansion around the point X_0 yields

$$\phi(x) = \frac{\phi(X_0)}{0!} + \frac{\phi'(X_0)(X-X_0)}{1!} + \frac{\phi''(X_0)(X-X_0)^2}{2!} + \dots + \frac{\phi^{(n)}(X_0)(X-X_0)^n}{n!} + R_n$$

or $\phi(X) = P_n + R_n$, where P_n = the polynomial approach X_2

R_n = the remainder.

Examples

- 1) Expand the non – polynomial function:

$$\phi(x) = \frac{1}{1+x}, \text{ Around the point } x_0 = 1, \text{ with } n = 4?$$

We need the first four derivatives;

$$\phi'(x) = -1(1+x)^{-2} \text{ So that } \phi'(1) = -1/4$$

$$\phi''(x) = 2(1+x)^{-3} \quad \phi''(1) = 1/4$$

$$\phi'''(x) = -6(1+x)^{-4} \quad \phi'''(1) = -3/8$$

$$\phi^{(4)}(x) = 24(1+x)^{-5} \quad \phi^{(4)}(1) = 3/4$$

$$\text{And } \phi(1) = 1/2$$

$$\phi(x) \approx \frac{1}{2} - \frac{1}{4}(x-1) + \frac{1}{8}(x-1)^2 - \frac{1}{16}(x-1)^3 + \frac{1}{32}(x-1)^4 + R_4$$

$$\phi(x) \approx \frac{31}{32} - \frac{13}{16}x + \frac{1}{2}x^2 - \frac{3}{16}x^3 + \frac{1}{32}x^4 + R_4$$

- 2) Expand the quadratic function $\phi(x) = 5 + 2x + x^2$, around $X_0 = 1$, with $n = 1$?

$$\phi(x) \approx \underline{\underline{8 + 4(x-1) + R_1}} \quad \phi(1) = 5 + 2 + 1 = 8$$

$$\phi'(1) = 2 + 2(1) = 4$$

Lagrange form of the Remainder

- According to Lagrange form of the remainder, we can express R_n as:

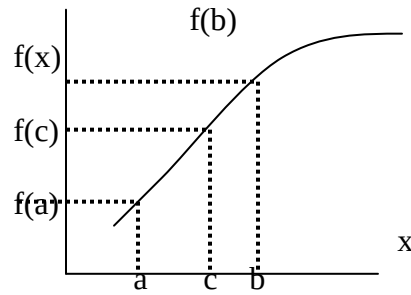
$$R_n + \frac{\phi^{(n+1)}(c)}{(n+1)!} (x - x_0)^{n+1}$$

Where C is the value at which the remainder is evaluated and lies between X & X_0 .

(For the detail, Read A. Chiang)

2.7 Intermediate Value Theorem (mean)

Let $f(x)$ is a function that is continuous for all x in the closed interval $[a, b]$, and assume that $f(a) \neq f(b)$



The theorem implies that, if $f(x)$ is continuous, the graph of $f(x)$ must intersect the line $y = f(c)$ at least at one point. i.e., point $(c, f(c))$, where $a < c < b$.

An important implication of the theorem is the following. Let $f(x)$ be a function continuous in $[a, b]$ and assume that $f(a)$ & $f(b)$ have different signs. Then there is at least one point $c \in (a, b)$ such that $f(c) = 0$.

Example : prove the equation $X^6 + 3x^2 - 2x - 1 = 0$, has at least one solution between 0 and 1?

Solution: - Evaluate the function at the two extreme values

$$\Rightarrow f(0) = -1$$

$$f(1) = 1$$

So, According to the Intermediate Value Theorem, there is at least one solution $c \in (0, 1)$ such that $f(c) = 0$

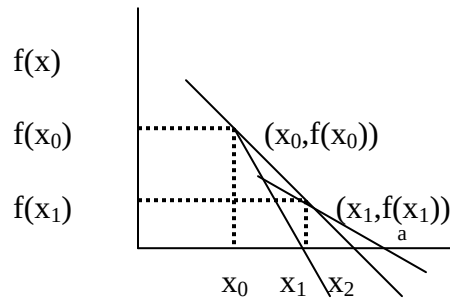
We can further narrow down the interval in which the solution is found. For example, if we take the mid-point of the interval $(0, 1) = \frac{1}{2}$

$$\Rightarrow f\left(\frac{1}{2}\right) < 0, \text{ Therefore, there is at least one solution in } \left(\frac{1}{2}, 1\right)$$

$\Rightarrow$ Take, the mid-point of $\left(\frac{1}{2}, 1\right)$ i.e, $x = \frac{3}{4}$. again $f\left(\frac{3}{4}\right) < 0$, implying that there is a solution between $\frac{3}{4}$ & 1.

Newton's Method

The Intermediate Value Theorem shows that a given equation $f(x) = 0$ has a solution in a given interval. However, it doesn't provide additional information about the location of the solution, Newton's method leads to a better approximate solution.



If $f(x)=0$ at point a , estimate a by starting an initial estimate x_0 of a .

By constructing a tangent line at $(x_0, f(x_0))$, we obtain x_1 . Hence, x_1 is a better approximation of ' a ' than x_0 . Repeat the procedure by constructing another tangent line at the point $(x_1, f(x_1))$ and then find x_2 .

We can easily find the formula (equation) for X_i 's derived by Newton's Method

Example 2: What is the equation of the line is given by:

$$m = \frac{y - y_0}{x - x_0}$$

$$\Rightarrow y - y_0 = m(x - x_0)$$

$$\Rightarrow y - f(x_0) = f'(x_0)(x - x_0)$$

At $x=x_1, y=0$

$$\Rightarrow 0 - f(x_0) = f'(x_0)(x_1 - x_0)$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}, \text{ etc In general, the}$$

Point generated by Newton's method are given by:

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

Example 2: - find an approximate value of 'a' for the equation:
 $f(x)x^6 + 3x^2 - 2x - 1 = 0$, in the interval $[0,1]$, using Newton's method once

Solution: Here $n = 0$ and $x_0 = 1$ or 0 (use either extreme) $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$, where

$$f'(x) = 6x^5 + 6x - 2$$

$$f'(x) = 10$$

$$f(x) = f(1) = 1$$

$$\Rightarrow x_1 = 1 - \frac{1}{10} = \underline{\underline{0.9}}$$

*(check for $f(x_1) = f(0.9) = \underline{\underline{0.1614}}$ which is close to zero.)

Apply Newton's Method twice?

Solution: $x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$

$$\Rightarrow x_2 = 0.9 - \frac{f(0.9)}{f'(0.9)} = \underline{\underline{0.028}}$$

2.8 Multivariate Calculus

Partial Derivatives

Given a function $y = f(x_1, x_2, \dots, x_n)$, if the variable x_1 undergoes a change Δx_1 while all other independent variables remain fixed,

The difference quotient in this case can be expressed as:

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_1}$$

Thus, $\frac{\partial y}{\partial x_1} = f_1 = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x_1}$ } First order partial derivative

Second order direct partial

$$Z = f(x, y)$$

$$f_{xx} = \frac{\partial^2 Z}{\partial x^2} (f_x) = f_{xx}(x, y)$$

$$f_{yy} = \frac{\partial^2 Z}{\partial y^2} (f_y) = f_{yy}(x, y)$$

Second order mixed partials (cross – partials)

$$\frac{\partial f_x}{\partial y} = \frac{\partial \left(\frac{\partial Z}{\partial x} \right)}{\partial y} = \frac{\partial^2 Z}{\partial y \partial x} = f_{xy}$$

$$\frac{\partial f_y}{\partial x} = \frac{\partial \left(\frac{\partial Z}{\partial y} \right)}{\partial x} = \frac{\partial^2 Z}{\partial x \partial y} = f_{yx}$$

Young's Theorem

It states that the mixed (or cross –partials) for a given function will always be equal if both cross partials exist. i.e.

Given $Y=f(x_1, x_2)$

$$\underline{f_{12}=f_{21}}$$

Ex-1 consider $u(x_1, x_2) = x_1 \ln x_2$

Solution :-

$$u_1 = \ln x_2$$

$$u_2 = \frac{x_1}{x_2}$$

$$u_{12} = \frac{1}{x_2}$$

$$u_{21} = \frac{1}{x_2}$$

$$\Rightarrow \underline{u_{12} = u_{21}}$$

Ex-2: for $y = f(x_1, x_2) = x_1 e^{x_1 + x_2^2}$, show that $f_{12} = f_{21}$?

Solution: $f_1 = (x_1 + 1)e^{x_1 x_2^2}$

$$f_2 = 2x_1 x_2 e^{x_1 + x_2^2}$$

$$\underline{f_{12} = (x_1 + 1)(2x_2) e^{x_1 + x_2^2}}$$

$$\underline{f_{21} = (x_1 + 1)(2x_2) e^{x_1 + x_2^2}}$$

}

Generalization of Young's Theorem:

Given $y = (x_1, x_2, \dots, x_n)$

$$f_{123\dots n} = f_{213\dots n} = f_{312\dots n} \dots\dots = f_{n-312}$$

2.4.3. The Multivariate Chain Rule

Many economic models involve composite functions. Differentiation of composite functions requires the application of chain rule.

-suppose $Z = F(x, y)$ and $x = f(t), y = g(t)$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

Example:

1) $y = \ln(x_1 + x_2)$, where $x_1 = 2t$, and $x_2 = t^2$

$$\begin{aligned} \frac{dy}{dt} &= \frac{\partial y}{\partial x_1} \cdot \frac{\partial x_1}{\partial t} + \frac{\partial y}{\partial x_2} \cdot \frac{\partial x_2}{\partial t} \\ &= \frac{2}{x_1 + x_2} + \frac{2t}{x_1 + x_2} \\ &= \frac{2(t+1)}{2t+t^2} \end{aligned}$$

Or by direct substitution: $y = \ln(2t + t^2)$

$$\Rightarrow \frac{dy}{dt} = \frac{2+2t}{2t+t^2}$$

Suppose $z = F(x, y) = x^2 e^y$

$$x = \ln t, y = t^2, \text{ find } \frac{dz}{dt}?$$

The partial derivative case.

Given $z = F(x, y)$, where $x = f(s, t), y = g(s, t)$

$$\frac{\partial z}{\partial s} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial s} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial s}$$

$$\text{Similarly: } \frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

Example:

Suppose $z = F(x, y) = x^2 + 2y^2$

Where $x = t - s^2$

$$y = ts$$

$$\frac{\partial z}{\partial t} = \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial t} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial t}$$

$$= 2x \cdot 1 + 4y \cdot s$$

$$= 2(t - s^2) + 4(ts)s$$

$$= \underline{\underline{2t - 2s^2 + 4ts^2}}$$

2.9 Homogenous functions & Euler's Theorem

Definition: A function $y = f(x_1, x_2, \dots, x_n)$ is said to be homogenous of degree r if and only if $f(tx_1, tx_2, \dots, tx_n) = t^r f(x_1, x_2, \dots, x_n) = t^r y$, where r measures the degree of homogeneity. In economic application the constant r is usually taken to be positive, as most economic variables do not admit negative values.

Examples: Determine whether the following functions are homogenous. If so, of what degree?

1) $f(x, y, z) = \frac{x}{y} + \frac{2z}{3x}$

Solution: $f(tx, ty, tz) = \frac{tx}{ty} + \frac{2tz}{3tx}$

$$\begin{aligned} \frac{t}{t} \left(\frac{x}{y} + \frac{2z}{3x} \right) &= t^0 \left(\frac{x}{y} + \frac{2z}{3x} \right) \\ &= \underline{\underline{t^0 f(x, y, z)}} \end{aligned}$$

→ Here f is homogenous of degree 0 (H^0).

2) $y = f(L, K) = L^\alpha K^{1-\alpha}$

Solution: $f(tL, tK) = (tL)^\alpha (tK)^{1-\alpha}$

$$= t^\alpha L^\alpha \cdot t^{1-\alpha} K^{1-\alpha}$$

$$= tL^\alpha K^{1-\alpha} = \underline{\underline{t^1 f(L, K) = H^1}}$$

3) $f(x, y) = 3x^2y - y^3$

Solution: $f(tx, ty) = 3(tx^2)(ty) - (ty)^3$
 $= 3t^2x^2(ty) - t^3y^3$
 $t^3(2xy^2y - y^3) = \underline{\underline{t^3 f(x, y)}} = H^3$

4) $f(x, y) = x^3 - xy + y^3$

Euler's Theorems

If $f(x_1, x_2, x_3, \dots, x_n)$ is homogenous of degree r , then:

$$\frac{\partial f}{\partial x_1} \cdot x_1 + \frac{\partial f}{\partial x_2} \cdot x_2 + \dots + \frac{\partial f}{\partial x_n} \cdot x_n = rf(x_1, x_2, \dots, x_n)$$

Proof: by the definition of homogeneity:

$$f(tx_1, tx_2, \dots, tx_n) = t^r f(x_1, x_2, \dots, x_n)$$

-Differentiating both sides of the identity with respect to t :

$$\begin{aligned} \frac{\partial f}{\partial (tx_1)} \cdot \frac{\partial (tx_1)}{\partial t} + \dots + \frac{\partial f}{\partial (tx_n)} \cdot \frac{\partial (tx_n)}{\partial t} &= rt^{r-1} f(x_1, x_2, \dots, x_n) \\ &= \frac{\partial f}{\partial (tx_1)} \cdot x_1 + \dots + \frac{\partial f}{\partial (tx_n)} \cdot x_n = rt^{r-1} f(x_1, x_2, \dots, x_n) \end{aligned}$$

Putting $t = 1$, results the Euler's Theorem:

$$\frac{\partial f}{\partial x_1} \cdot x_1 + \frac{\partial f}{\partial x_2} \cdot x_2 + \dots + \frac{\partial f}{\partial x_n} \cdot x_n \equiv rf(x_1, x_2, \dots, x_n)$$

Example Find the degree of homogeneity of : $f(x, y) = x^4 + x^2y^2$ using Euler's Theorem?

- According to Euler's theorem:

$$f_x X + f_y Y = rf(x, y)$$

$$f_x = 4x^3 + 2x^2y^2, \text{ and } f_y = 2yx^2$$

$$\Rightarrow f_x X + f_y Y = (4x^3 + 2xy^2)x + (2yx^2)y$$

$$= 4x^4 + 2x^2y^2 + 2y^2x^2$$

$$= 4x^4 + 4x^2y^2 = \underline{\underline{4f(x, y)}}, \text{ where } r = 4$$

Extension of Euler's Theorem: If $f(x_1, x_2 \dots x_n)$ is homogenous of degree r , then the first partials $f_1, f_2 \dots f_n$ are homogenous of degree $r-1$.

Proof: By the definition of homogeneity:

$$f(tx_1, tx_2 \dots tx_n) = t^r f(x_1, x_2 \dots x_n)$$

- Differentiating both sides of the identity with respect to X_i :

$$\frac{\partial f}{\partial (x_i)} \cdot \frac{\partial (tx_i)}{\partial x_i} = t^r \frac{\partial f}{\partial x_i}$$

$$\Rightarrow \frac{\partial f}{\partial (tx_i)} \cdot t = t^r \frac{\partial f}{\partial x_i}$$

- Dividing both sides by t :

$$\boxed{\frac{\partial f}{\partial (tx_i)} = t^{r-1} \frac{\partial f}{\partial x_i}}$$

Linear Homogeneity

An important special case of homogeneity is that of homogeneity of degree one, which is often referred to as linear homogeneity. Linearly homogenous functions are widely applied in the theory of production and imply the economic assumption of constant returns to scale. Because linear homogeneity implies raising all inputs t -fold, it will always raise output exactly t -fold.

Properties of Linearly homogenous production functions

Given the Linearly homogenous production function $Q = f(L, K) = L^\alpha K^{1-\alpha}$:

1) The average products of each factor can be expressed as functions of the two input ratios.

$$AP_L = \frac{Q}{L} = \frac{L^\alpha K^{1-\alpha}}{L} = \frac{K^{1-\alpha}}{L^{1-\alpha}} = \left(\frac{K}{L} \right)^{1-\alpha}$$

$$AP_L = \frac{Q}{K} = \frac{L^\alpha K^{1-\alpha}}{k} = \frac{K^{-\alpha}}{L^{-\alpha}} = \left(\frac{K}{L}\right)^{-\alpha}$$

- 2) Similarly, the marginal products can be expressed as functions of the ratio of the two inputs.

$$\text{i.e. } MP_L = \frac{\partial Q}{\partial L} = \alpha L^{\alpha-1} K^{1-\alpha} = \frac{\alpha K^{1-\alpha}}{L^{1-\alpha}} = \alpha \left(\frac{K}{L}\right)^{1-\alpha}$$

$$\begin{aligned} MP_K \frac{\partial Q}{\partial k} &= (1-\alpha) L^\alpha k^{-\alpha} \\ &= (1-\alpha) \left(\frac{k}{L}\right)^{-\alpha} \end{aligned}$$

Therefore, the marginal products also depend only on ratios of the two inputs.

Conclusion: If labour & capital are changed by the same proportion, the marginal and average products of the inputs will not be affected.

Example: consider a firm with a linearly homogenous production function:

$$Q = f(L, K)$$

By Euler's theorem

$f_L L + f_K K = Q \rightarrow$ Economically, this property means that under conditions of constant returns to scale, if each input is paid the amount of its marginal product, the total product will be exactly exhausted by the distributive shares from all the input factors, or equivalently, the pure economic profit will be zero.

From

$$Q = f_L L + f_K K = MP_L L + MP_K K, \text{ Dividing both sides by } L, \text{ we have}$$

$$\frac{Q}{L} = MP_L + MP_K \cdot \frac{K}{L}$$

$$AP_L = MP_L + MP_K \cdot \frac{K}{L}$$

$$\Rightarrow MP_K \cdot \frac{K}{L} = AP_L - MP_L$$

Assuming K/L is positive:

- i) when APL is rising ($MPL > APL$) $\Rightarrow MP_K$ is negative (excess capital)
- ii) When APL is falling ($MPL < APL$) $\Rightarrow MP_K$ is positive (capital is utilized)

Exercise 2.3

1. Consider two goods X_1 and X_2 with prices P_1 & P_2 . Suppose that a consumer with money income M has the following demand function for X_1 :

$$X_1 = \frac{MP_2}{P_1^2}$$

- a. Show the effect of balanced or neutral inflation on the demand of those goods?
 - b. Show whether Euler's Theorem holds for this demand function?
2. Find the Maclaurin series for the function: $f(x) = 6 + 8x - 7x^2 + 3x^3 + 2x^4$

Chapter -3

Unconstrained Optimization

Aims and objectives:

After successfully completing this chapter, students are expected to:

- ❖ Know how to solve Functions of one variable where there is no constraint
- ❖ solve Functions of several variables

Introduction

When finding the relative extrema of objective functions of more than one choice variable which are independent of one another), the decision made regarding one variable does not affect the choice of the other variable. Such functions which involve no constraints are called unconstrained functions & the optimization process is called unconstrained or free optimization.

From the point of view of economics, extreme (equilibrium) points may be of two types. These are goal equilibrium and non-goal equilibrium. The former is equilibrium point which the economic agent is deliberately striving to achieve, and entails maximization of any form of benefit and minimization of costs (of any form). The general term for this type of economic problem is optimization. In this chapter, we will focus on optimization problems using the classical method of optimization (the concept of derivatives)

3.1. Functions of one variable

Minimum and maximum values of a function occur either at stationary points or at end points. So, to find the extreme values of a continuous function, the 1st requirement is to find stationary points. This is called 1st order condition and tested using 1st derivatives. i.e.

$$\frac{dy}{dx} = 0$$

However; the 1st order condition is not sufficient to say a function achieves maximum or minimum, because it tells us only that the function has stationary point (zero slopes). The sufficient condition is the rate of change of the slope (the 1st derivative).

Given a two variable function $y = f(x)$, the extreme values can be fixed using the following general rule.

F.O.C: The derivative version

$$\frac{dy}{dx} = f'(x) = 0$$

The differential version

$$dy = f'(x)dx = 0, dx \neq 0$$

S.O.C: $\frac{d^2y}{dx^2} = f''(CV) > 0 \rightarrow \min$
 $< 0 \rightarrow \max$

$$d^2y = f''(CV)dx^2 > \rightarrow \min$$

$$> 0 \rightarrow \text{Max}$$

$$= 0 \text{ test inconclusive}$$

Example

3.2. Functions of several independent Variables

Given $z = f(x, y)$:

Using the differential version of optimization, we have:

1. F.O.C:

$$dz = f_x dx + f_y dy = 0, \text{ for } dx \neq 0 \text{ and } dy \neq 0$$

$$\Rightarrow dz = 0, \text{ iff } f_x \text{ and } f_y \text{ are zero}$$

Therefore, the F.O.C. will be, $f_x = f_y = 0$

2. S.O.C; once the 1st order condition is satisfied, the next step is to check the 2nd order (sufficient) condition. Given

$$dz = f_x dx + f_y dy,$$

$$\begin{aligned} d^2z &= d(dz) = \frac{\partial(dz)}{\partial x} dx + \frac{\partial(dz)}{\partial y} dy \\ &= \frac{\partial(f_x dx + f_y dy)}{\partial x} dx + \frac{\partial(f_x dx + f_y dy)}{\partial y} dy \end{aligned}$$

$$= (f_{xx}d_x + f_{yx}d_y)d_x + (f_{xy}d_x + f_{yy}d_y)d_y$$

But by young's theorem $f_{xy} = f_{yx}$. Substituting this value, we get the following equation.

$$d^2z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$$

For any dx and dy, both different from zero, the following rule is used to identify the maximum and minimum points:

$$\Rightarrow d^2z < 0 \rightarrow \text{Maximum}$$

$$\Rightarrow d^2z > 0 \rightarrow \text{Minimum}$$

But for any values of dx and dy, not both zero:

$$\rightarrow d^2z < 0, \text{iff. } f_{xx} < 0, f_{yy} < 0, \text{and. } f_{xx}f_{yy} > (f_{xy})^2$$

$$\rightarrow d^2z > 0, \text{iff. } f_{xx} > 0, f_{yy} > 0, \text{and. } f_{xx}f_{yy} > (f_{xy})^2$$

So, the conditions for optimum values of a two variable function can be summarized as follows.

Conditions	Maximum	Minimum
F.O.C	$f_x = f_y = 0$	$f_x = f_y = 0$
S.O.C	$f_{xx} < 0, f_{yy} < 0$ $f_{xx}f_{yy} > (f_{xy})^2$	$f_{xx} > 0, f_{yy} > 0$ $f_{xx}f_{yy} > (f_{xy})^2$

Violation (failure) of the S.O.C

If the second order conditions for maximum and minimum are not fulfilled, the points will be either of the following.

i) If $f_{xx}f_{yy} < (f_{xy})^2$ and f_{xx} and f_{yy} have different signs, then the point is saddle point

ii) If $f_{xx}f_{yy} < (f_{xy})^2$ and f_{xx} and f_{yy} have the same signs, then the point is inflection point

iii) If $f_{xx} \cdot f_{yy} = (f_{xy})^2$, Test inconclusive

Note: - always, we test a point for second order condition only if it passes the first order condition (first order partial derivative test)

Self test Exercise

For each of the following functions, find the critical values, and determine whether the function is at a relative max, min, an inflection or saddle point?

1) $z = f(x, y) = 3x^2 + 2y^2 - 5x - 4y + 12$

2) $z = f(x, y) = x + 2y - e^x - e^{2y}$

3) $z = e^{2x} - 2x + 2y^2 + 3$

Determinant test for the S.O.C

The expression for d^2z in our derivation of second order total differential exemplifies what is known as quadratic forms.

Definition a form is a polynomial expression in which each component term has a uniform degree. i.e. the sum of exponents in each term is uniform.

Example:- $f(x, y, z) = ax - by + z = \text{a linear form}$

$$f(x, y) = ax^2 + 2bxy + Cy^2 = \text{a quadratic form, etc.}$$

Second order Total Differential as a Quadratic form

Consider $d^2z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$

Let $d^2z = q$ $f_{xx} = a$

$dx = u$ $f_{yy} = b$

$dy = V$ $f_{xy} = f_{yx} = h$

If we further consider dx & dy as variable and the partial derivatives as coefficients, then: $q = au^2 + 2huv + bv^2$

Algebraic Method and Second Order Condition

Having represented the second order partial differentials as quadratic forms, let us see how to determine maximum and minimum points using a relatively easy method.

In the case of optimization with one variable, it is easy to test both 1st order and 2nd order conditions. But in the case of multi-variable functions, it may be very complex to test the second order conditions. The complexity increases as the number of variables increase since we will have many direct and cross partials. In such cases, we resort to the concept of matrix algebra to simplify our computation.

Do you remember sign definiteness of matrices? good. If you forget it, please revise your Quantitative Method for Economists' II module. The concept of linear algebra is used in the following way.

The second- order sufficient condition for extremum stipulates d^2z to be definitely positive (for a minimum) and definitely negative (for a maximum)

The major question, then is that: what restrictions must be placed upon a, b and h, while u and v are allowed to take any values (so long as they are not both zero), in order to ensure a definite sign for q?

Positive and Negative Definiteness

A quadratic form q is said to be:

- Positive definite, if $q > 0 \rightarrow$ S.O.sufficient condition for Min.
- Positive semi-definite , if $q \geq 0 \rightarrow$ S.O necessary condition for Min
- Negative definite , if $q < 0 \rightarrow$ S.O.sufficient condition for Max
- Negative semi-definite, if $q \leq 0$. This is a necessary condition for Max

Note:- If q changes signs when the variables assume different values, on the other hand, q is said to be indefinite (an indication of a saddle point).

Determinant test for sign Definiteness

$$\begin{aligned}
 q &= au^2 + 2huv + bv^2 \\
 &= au^2 + huv + huv + bv^2 \\
 &= (au + hv)u + (hu + bv)v \\
 &= (au + hv.hu + bv) \begin{pmatrix} u \\ v \end{pmatrix} \\
 &= \begin{bmatrix} u & v \end{bmatrix} \begin{bmatrix} a & h \\ h & b \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}
 \end{aligned}$$

Here the determinant of $\begin{bmatrix} a & h \\ h & b \end{bmatrix}$ is referred to as **Discriminant of the quadratic form q** and is denoted by /D/.

In the particular case of the quadratic form $d^2z = f_{xx}dx^2 + 2f_{xy}dxdy + f_{yy}dy^2$, the discriminant is a determinant with the second-order partial derivatives as its elements.

Such a determinant is called **a Hessian determinant** (or simply a Hessian) In the two variable case, the Hessian is

$$|H| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

From this Hessian determinant (or simply a Hessian), $|f_{xx}| = f_{xx}$, represents the first principal minor of $|H|$ and is denoted by $|H_1|$.

The second principal minor of $|H|$ is

$$|H_2| = H = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2$$

The sign definiteness of d^2z is determined by the sign of the two principle minors. Thus,

i) $d^2z > 0$ (positive definite) iff:

$$|H_1| = f_{xx} > 0, \text{ and } |H_2| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2 > 0$$

ii) $d^2z < 0$ (negative definite) iff:

$$|H_1| = f_{xx} < 0, \text{ and } |H_2| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} = f_{xx} f_{yy} - (f_{xy})^2 > 0.$$

iii) d^2z is indefinite iff:

$$|H_2| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix} < 0,$$

Example : optimize the following functions

$$z = f(x, y) = 6x^2 - 9x - 3xy - 7y + 5y^2$$

Solution: **F.O.C** $f_x = 12x - 9 - 3y = 0$

$$f_y = -3x - 7 + 10y = 0$$

Solving simultaneously yields the critical values $x = 1$ & $y = 1$

S.O.C $f_{xx} = 12$, $f_{yy} = 10$ and $f_{xy} = -3$

Determinant Test

$$|H_1| = 12 > 0, \text{ and } |H_2| = \begin{vmatrix} 12 & -3 \\ -3 & 10 \end{vmatrix} = 111 > 0$$

Thus, $d^2z > 0$ (positive definite). i.e. the equation is minimized at the critical values.

The case of three variable quadratic form

Given $z = f(x_1, x_2, x_3)$

$$\text{Given } |H| = \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}$$

→ the sign definiteness of d^2z depends on the signs of the first, second and third principal minors, thus;

i) $d^2z > 0$ (positive definite) iff:

$$|H_1| > 0, |H_2| > 0, \text{ and } |H| > 0$$

ii) $d^2z < 0$ (negative definite) iff:

$$|H_1| < 0, |H_2| > 0, \text{ and } |H_3| < 0 \quad \text{the three principal minors alternate in sign.}$$

Example:

A firm produces three products Q_1, Q_2 , and Q_3 . The profit function is given by:

$$\Pi(Q_1, Q_2, Q_3) = 180Q_1 + 200Q_2 + 150Q_3 - 3Q_1Q_2 - 2Q_2Q_3 - 2Q_1Q_3 - 4Q_1^2 - 5Q_2^2 - 4Q_3^2,$$

then optimize the profit function?

Solution:

$$\text{F.O.C: } \Pi_1 = 0, \quad \Pi_1 = 180 - 3Q_2 - 2Q_3 - 8Q_1 = 0$$

$$\Pi_2 = 0, \quad \Pi_2 = 200 - 3Q_1 - 2Q_3 - 10Q_2 = 0$$

$$\Pi_3 = 0, \quad \Pi_3 = 150 - 2Q_2 - 2Q_1 - 8Q_3 = 0$$

Critical values: $Q_1^* = 14.53, Q_2^* = 13.27, \text{ and } Q_3^* = 11.79$

$$\text{S.O.C (Second Order Condition): } |H| = \begin{vmatrix} -8 & -3 & -2 \\ -3 & -10 & -2 \\ -2 & -2 & -8 \end{vmatrix}$$

$$|H_1| = |-8| = -8 < 0$$

$$|H_2| = \begin{vmatrix} -8 & -3 \\ -3 & -10 \end{vmatrix} = 71 > 0$$

Therefore, the profit function is maximized at the critical values.

$$|H_3| = |H| = -520 < 0$$

the case of n- variable Quadratic forms

the algebraic method of testing second order condition can be directly extended for the case of n-variable quadratic forms as follows.

Given $z = f(x_1, x_2 \dots x_n)$

F.O.C. $f_1 = f_2 = \dots = f_n = 0$

$$\textbf{S.O.C. } |H| = \begin{vmatrix} f_{11} & f_{12} \cdots & f_{1n} \\ f_{21} & f_{22} \cdots & f_{2n} \\ f_{n1} & f_{n2} \cdots & f_{nn} \end{vmatrix}$$

i) **Positive Definite (minimum)**

$$|H_1| > 0, |H_2| > 0, \dots, |H_n| > 0$$

ii) **Negative definite (Maximum)**

$|H_1| < 0, |H_2| > 0, |H_3| < 0, \dots, (-1)^n |H_n| > 0 \rightarrow$ all odd numbered principal minors are negative, and all even numbered ones are positive.

Exercises

- 1) A firm produces two products x & y. the cost of producing the products is :
 $c(x, y) = 0.04x^2 + 0.01xy + 0.01y^2 + 4x + 2y + 500$. Suppose that the firm sells all its output at a per unit price of birr 15 for X, and birr 9 for y. optimize the profit function of the firm.
- 2) The production function for product Q is given by: $Q = f(L, K) = 6\sqrt{K}^3\sqrt{L}$. Let the price per unit of output be birr 0.5. The cost per unit of capital is birr 0.1 and the price per unit of labor be birr 1. Optimize the profit function of the firm?
- 3) Consider a profit maximizing firm that sales its output Q at a constant unit price, P and purchases two inputs X_1 & X_2 at constant unit factor prices w_1 & w_2 respectively. The firm faces competitive input & output market and the production function is given by: $Q(X_1, X_2)$, then optimize the profit function of

the firm and provide economic interpretation of the first & second order conditions.

- 4) Consider a monopolist firm that produces two products Q_1 & Q_2 . Suppose that the demands facing the firm are as follows:

$$P_1 = 55 - Q_1 - Q_2$$

$$P_2 = 70 - Q_1 - 2Q_2$$

The total cost function is given by: $c(Q_1, Q_2) = Q_1^2 + Q_1Q_2 + Q_2^2$, then optimize the profit function of the firm and determine the optimal levels of output, price and profit.

- 5) Suppose a monopolist firm sales a certain product Q in three separate markets. The demands facing the firm are as follows.

$$P_1 = 63 - 4Q_1$$

$$P_2 = 105 - 5Q_2$$

$$P_3 = 75 - 6Q_3$$

The total cost function is: $C(Q) = 20 + 15Q$

$$\text{Where } Q = Q_1 + Q_2 + Q_3$$

Then optimize the profit function in of the firm, i.e. determine the prices charged, the quantities sold as well as the optimal profit.

- 6) Suppose a monopolist is practicing price discrimination in the sale of product Q by charging different prices in two separate markets. The demand functions in the two markets are given by:

$$P_1 = 100 - Q_1$$

$$P_2 = 80 - Q_2$$

Suppose the cost function is: $C = 6Q$

$$\text{Where } Q = Q_1 + Q_2$$

- i)
 - a) How much would be sold in the two markets to maximize profit?
 - b) What are the prices charged?
 - c) How much is the maximum profit?
- ii) If price discrimination is made illegal:
 - a) How much would be sold in the two markets?

- b) What prices would be charged?
- c) How much profit would be lost?

Chapter 4

Constrained Optimization

Aims and objectives:

Up on successful completion of this chapter, students will be able to solve

- . One Variable constrained optimization
- . Two variable problems with equality constraint
- . Inequality Constraints & The theorem of Kuhn and Tucker

Introduction

In economic optimization problems, the variables involved are often required to satisfy certain constraints. In the case of unconstrained optimization problems, no restrictions have been made regarding the value of the choice variables. But in reality, optimization of a certain economic function should be in line with certain resource requirement or availability. This emanates from the problem of scarcity of resources. For example; maximization of production should be subject to the availability of inputs. Minimization of costs should also satisfy a certain level of output. The other constraint in economics is the non negativity restriction. Although sometimes negative values may be admissible, most functions in economics are meaningful only in the first quadrant. So, these constraints should be considered in the optimization problems.

Constrained optimization deals with optimization of the objective function (the function to be optimized) subject to constraints (restrictions). In the case of linear objective and constraint functions, the problems are solved using linear programming model. But when we face a non linear function, we use the concept of derivatives for optimization. This chapter focuses on optimization of non linear constrained functions.

4.1. One Variable constrained optimization

i) With equality constraint:- optimization of one variable function subject to an equality constraint takes the form,

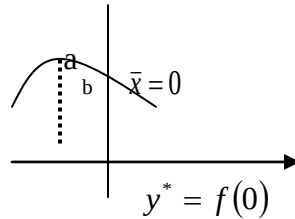
$$\text{Max (min): } y = f(x), \text{ subject to: } x = \bar{x}$$

The solution for this type of problem is $\Rightarrow y^* = f(\bar{x})$

iii) Non- negativity constraint:

a) Max(min): $y = f(x)$, *subjecto* : $x \geq 0$.

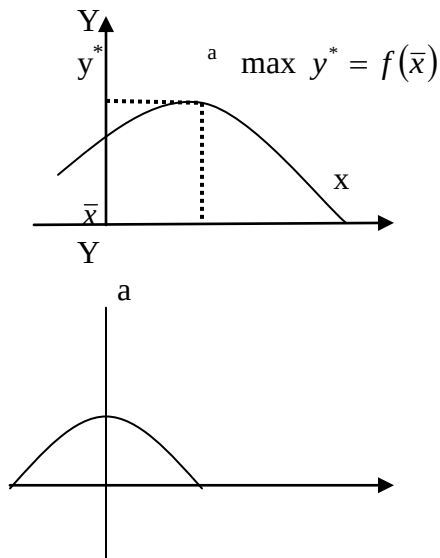
Consider the following graph.



In the above graph, y attains its unconstrained maximum at point $a(x < 0)$. But when a non negativity restriction is imposed on the choice variable, the maximum point will be at $x=0$.

$(\bar{x} = 0)$, At $\bar{x} = 0$, $f'(x) < 0$

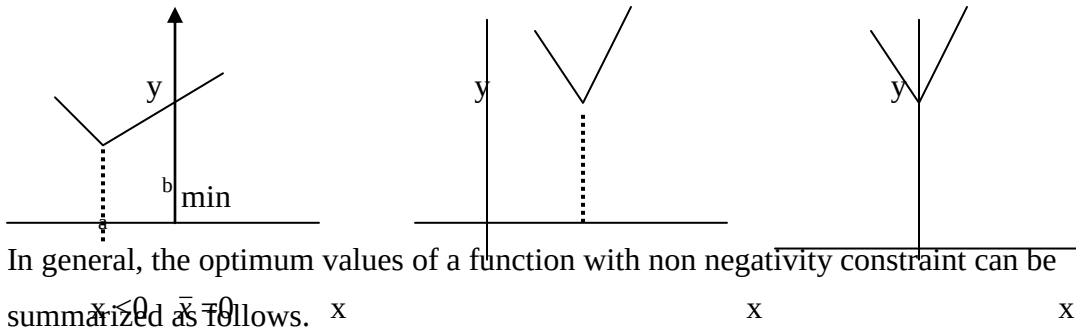
b)



Here both the constrained & unconstrained max is at point 'a' $\Rightarrow f'(x) = 0$.

-Both constrained & unconstrained max coincide, and $f'(x) = 0$

Minimum:- minimization of functions with non-negativity constraint can be viewed graphically as follows



$$\begin{array}{ll}
 \text{i)} & \text{Max: } f'(x) \leq 0 \left\{ \begin{array}{l} \text{if } f'(x) = 0, \bar{x} \geq 0 \\ \text{if } f'(x) < 0, \bar{x} = 0 \end{array} \right. \\
 y^* = f(0) & \\
 f'(x) > 0 & \\
 \text{ii)} & \text{Min: } f'(x) \geq 0 \left\{ \begin{array}{l} \text{if } f'(x) = 0, \bar{x} \geq 0 \\ \text{if } f'(x) > 0, \bar{x} = 0 \end{array} \right. \\
 & \bar{x} = 0 \\
 & y^* = f(0) \\
 & f'(x) = 0
 \end{array}$$

Example:1

$$\text{Max } y = -3x^2 - 7x + 2, \text{ subject to, } x \geq 0$$

In unconstrained operation:

F.O.C. $f'(x) = -6x - 7 = 0$

$$x = -\frac{7}{6}$$

But imposing the non – negative constraint we have:

$$x^* = 0, \text{ and } f(0) = 2$$

$$f'(0) = -7 < 0$$

4.2. Two variable problems with equality constraint

In the case of two choice variables, optimization problem with equality constraint takes the form $\text{Max/Min: } y = f(x_1, x_2), \text{ Subject to: } g(x_1, x_2) = c$. This type of

optimization problem is commonly used in economics. Because, for the purpose of simplification, two variable cases are assumed in finding optimum values. For example in maximization of utility using indifference approach, the consumer is assumed to consume two bundles of goods.

Example: Max $u(x_1, x_2)$, subject to $p_1x_1 + p_2x_2 = M$

In this section, we will see two methods of solving two variable optimization problems with equality constraints.

i) Direct substitution Method:

Direct substitution method is used for a two variable optimization problem with only one constraint. It is relatively simple method. In this method, one variable is eliminated using substitution before calculating the 1st order condition. Consider the consumer problem in the above example.

$$P_2x_2 = M - P_1x_1$$

$$x_2 = \frac{m}{P_2} - \frac{p_1}{p_2} x_1$$

Now x_2 is expressed as a function of x_1 . Substituting this value, we can eliminate x_2 from the equation.

$$\Rightarrow \text{Max}, u = u(x_1, x_2(x_1))$$

$$\frac{du}{dx_1} = \frac{\partial u}{\partial x_1} + \frac{\partial u}{\partial x_2} \cdot \frac{\partial x_2}{\partial x_1} = 0$$

$$= Mu_1 + Mu_2 \left(-\frac{p_1}{p_2} \right) = 0$$

$$\Rightarrow Mu_1 = Mu_2 \cdot \frac{p_1}{p_2}$$

$$\Rightarrow \frac{Mu_1}{Mu_2} = \frac{p_1}{p_2}$$

Example $u = x_1x_2$, subject to, $x_1 + 4x_2 = 120$

$$4x_2 = 120 - x_1$$

$$x_2 = 30 - \frac{1}{4}x_1$$

$$\Rightarrow u = x_1 \left(30 - \frac{1}{4}x_1 \right) = 30x_1 - \frac{1}{4}x_1^2$$

F.O.C. (First Order Condition) $\frac{du}{dx_1} = 30 - \frac{1}{2}x_1 = 0$

$$\Rightarrow \underline{x_1 = 60, \text{ and, } x_2 = 15}$$

iii) Lagrange Multiplier Method

When the constraint is a complicated function or when there are several constraints, we resort to the method of Lagrange.

$$\text{Max: } f(x_1, x_2) \text{ Subject to } g(x_1, x_2) = c$$

$$L = f(x_1, x_2) + \lambda(c - g(x_1, x_2)) \text{ L denotes the Lagrange.}$$

An interpretation of the Lagrange Multiplier

The Lagrange multiplier, λ , measures the effect on the objective function of a one unit change in the constant of the constraint function (stock of a resource). To show this; given the Lagrangian function;

$$L = f(x_1, x_2) + \lambda(c - g(x_1, x_2))$$

F.O.C.

$$L_1 = f_1(x_1, x_2) - \lambda g_1(x_1, x_2) = 0$$

$$L_2 = f_2(x_1, x_2) - \lambda g_2(x_1, x_2) = 0$$

$$L_\lambda = c - g(x_1, x_2) = 0$$

We can express the optimal choice of variables λ^* , x_1^* and x_2^* as implicit functions of the parameter c:

$$x_1^* = x_1^*(c)$$

$$x_2^* = x_2^*(c)$$

$$\lambda^* = \lambda^*(c)$$

Now since the optimal value of L depends on λ^* , x_1^* and x_2^* we may consider L^* to be a function of c alone. That is,

$$L^* = f(x_1^*(c), x_2^*(c)) + \lambda^*(c)[c - g(x_1^*(c), x_2^*(c))]$$

Differentiating L^* with respect to c, we have:

$$\begin{aligned} \frac{dL^*}{dc} &= f_{x_1} \frac{dx_1^*}{dc} + f_{x_2} \frac{dx_2^*}{dc} + [c - g(x_1^*, x_2^*)] \frac{d\lambda^*}{dc} + \lambda^* \left[1 - g_{x_1} \frac{dx_1^*}{dc} - g_{x_2} \frac{dx_2^*}{dc} \right] \\ &= \cancel{(f_{x_1} - g_{x_1} \lambda^*) \frac{dx_1^*}{dc}} + \cancel{(f_{x_2} - \lambda^* g_{x_2}) \frac{dx_2^*}{dc}} + \cancel{[c - g(x_1^*, x_2^*)] \frac{d\lambda^*}{dc}} + \lambda^* \end{aligned}$$

The expressions with a sign of cancellation are all equal to zero.

$$\underline{\underline{\frac{dL^*}{dc} = \lambda^*}}$$

Note :- If $\lambda > 0$, it means that for every one unit increase (decrease) in the constant of the constraining function, the objective function will decrease (increase by a value approximately equal to λ .

If $\lambda < 0$, a unit increase (decrease) in the constant of the constraint will lead to an increase (decrease) in the value of the objective function by approximately equal to λ .

S.O.C. Conditions for a constrained optimization problem

Max: $f(x, y)$ subject to, $g(x, y)$

$$L = f(x, y) + \lambda g(x, y)$$

F.O.C.

$$\begin{array}{l} L_{\lambda} = g(x, y) = 0 \quad \left| \begin{array}{l} \bar{x} \\ \bar{y} \\ \bar{\lambda} \end{array} \right. \\ L_x = f_x + \lambda g_x = 0 \\ L_y = f_y + \lambda g_y = 0 \end{array}$$

S.O.C: $L_{\lambda\lambda} = 0, L_{\lambda x} = g_x, L_{\lambda y} = g_y$

$$L_{xx} = f_{xx} + \lambda g_{xx}, L_{xy} = f_{xy} + \lambda g_{xy}$$

$$L_{yy} = f_{yy} + \lambda g_{yy}$$

In matrix form:

$$\begin{pmatrix} L_{\lambda\lambda} & L_{\lambda x} & L_{\lambda y} \\ L_{x\lambda} & L_{xx} & L_{xy} \\ L_{y\lambda} & L_{yx} & L_{yy} \end{pmatrix} = \begin{pmatrix} 0 & g_x & g_y \\ g_x & L_{xx} & L_{xy} \\ g_y & L_{yx} & L_{yy} \end{pmatrix}$$

This matrix is called Bordered Hessian and its determinant is denoted by $|\overline{H}|$.

The bordered Hessian $|\overline{H}|$ is simply the plain Hessian $|H| = \begin{vmatrix} L_{xx} & L_{xy} \\ L_{yx} & L_{yy} \end{vmatrix}$ bordered by

the first order derivatives of the constraint with zero on the principal diagonal.

Determinant Criterion for the sign definiteness of d^2z .

$$d^2z \text{ is } \begin{cases} \text{Positivedefinite} \\ \text{Negativedefinite} \end{cases} \text{ subject to } dg = 0 \text{ iff } \begin{vmatrix} 0 & g_x & g_y \\ g_x & L_{xx} & L_{xy} \\ g_y & L_{yx} & L_{yy} \end{vmatrix} < 0 \rightarrow \min$$

$$> 0 \rightarrow \text{Max}$$

n- variable case

Given $f(x_1, x_2, \dots, x_n)$

Subject to $g(x_1, x_2, \dots, x_n) = c$

$$|\bar{H}| = \begin{vmatrix} 0 & g_1 & g_2 & \dots & g_n \\ g_1 & L_{11} & L_{12} & & L_{1n} \\ \vdots & & & & \\ g_n & L_{n1} & L_{n2} & & L_{nn} \end{vmatrix}$$

Its bordered leading principal minors can be defined as:

$$|\bar{H}_2| = \begin{vmatrix} 0 & g_1 & g_2 \\ g_1 & L_{11} & L_{12} \\ g_2 & L_{21} & L_{22} \end{vmatrix}, |\bar{H}_3| = \begin{vmatrix} 0 & g_1 & g_2 & g_3 \\ g_1 & L_{11} & L_{12} & L_{13} \\ g_2 & L_{21} & L_{22} & L_{23} \\ g_3 & L_{31} & L_{32} & L_{33} \end{vmatrix} \quad \text{etc.}$$

Conditions

Maximization

Minimization

F.O.C.

$$L_\lambda = L_1 = L_2 \dots L_n = 0$$

$$L_\lambda = L_1 = L_2 \dots L_n = 0$$

S.O.C.

$$|\bar{H}_2| > 0, |\bar{H}_3| < 0, \dots$$

$$|\bar{H}_2|, |\bar{H}_3|, \dots, |\bar{H}_n| < 0$$

Example: $\text{Max}: z(x, y) = 2xy \text{ subject to } 3x + 4y = 90$

$$L = 2xy + \lambda(90 - 3x - 4y)$$

The subsequent equations can be solved by either of the following ways.

a) direct substitution

b) inverse method

c) Cramer's rule

d) Gauss-Jordan Elimination Method

$$\begin{array}{l} L_\lambda = 90 - 3x - 4y = 0 \\ L_x = 2y - 3\lambda = 0 \\ L_y = 2x - 4\lambda = 0 \end{array} \quad \begin{array}{l} x^* = 15 \\ y^* = 11.25 \\ \lambda^* = 7.5 \end{array}$$

F.O.C.:-

$$\text{S.O.C.: } \begin{pmatrix} 0 & g_1 & g_2 \\ g_1 & L_{xx} & L_{xy} \\ g_2 & L_{yx} & L_{yy} \end{pmatrix} \Rightarrow |\bar{H}| = \begin{vmatrix} 0 & 3 & 4 \\ 3 & 0 & 2 \\ 4 & 2 & 0 \end{vmatrix}$$

$$|\bar{H}| = |\bar{H}_2| = 48 > 0 \rightarrow \text{Negative definite (Max)}$$

More than one Equality constraint

Max/Min: $f(x_1, x_2, x_3)$, subject to, $g^1(x_1, x_2, x_3) = c^1$ and

$$g^2(x_1, x_2, x_3) = c^2$$

$$L = f(x_1, x_2, x_3) + \lambda_1 (c^1 - g^1(x_1, x_2, x_3)) + \lambda_2 (C^2 - g^2(x_1, x_2, x_3))$$

$$\text{F.O.C.: } L_1 = f_1 - \lambda_1 g_1^1 - \lambda_2 g_1^2 = 0$$

$$L_2 = f_2 - \lambda_1 g_2^1 - \lambda_2 g_2^2 = 0$$

$$L_3 = f_3 - \lambda_1 g_3^1 - \lambda_2 g_3^2 = 0$$

$$L_{\lambda_1} = C^1 - g^1(x_1, x_2, x_3) = 0$$

$$L_{\lambda_2} = C^2 - g^2(x_1, x_2, x_3) = 0$$

$$\text{S.O.C. } \bar{H} = \begin{vmatrix} 0 & 0 & g_1^1 & g_2^1 & g_3^1 \\ 0 & 0 & g_1^2 & g_2^2 & g_3^2 \\ g_1^1 & g_1^2 & L_{11} & L_{12} & L_{13} \\ g_2^1 & g_2^2 & L_{21} & L_{22} & L_{23} \\ g_3^1 & g_3^2 & L_{31} & L_{32} & L_{33} \end{vmatrix}$$

$|\bar{H}_2|$ = is one that contains L_{22} as the last element of its principal diagonal

$|\bar{H}_3|$ = is one that contains L_{33} as the last element of its principal diagonal.

A Jacobian Determinant

A Jacobian determinant permits testing for functional dependency for linear & non – linear functions. A Jacobian determinant is composed of all the first order partial derivatives of a system of equations arranged in ordered sequence.

Example given $y_1 = f_1(x_1, x_2, x_3)$

$$y_2 = f_2(x_1, x_2, x_3)$$

$$y_3 = f_3(x_1, x_2, x_3)$$

$$|J| = \begin{vmatrix} \frac{\partial y_1}{\partial x_1} & \frac{\partial y_1}{\partial x_2} & \frac{\partial y_1}{\partial x_3} \\ \frac{\partial y_2}{\partial x_1} & \frac{\partial y_2}{\partial x_2} & \frac{\partial y_2}{\partial x_3} \\ \frac{\partial y_3}{\partial x_1} & \frac{\partial y_3}{\partial x_2} & \frac{\partial y_3}{\partial x_3} \end{vmatrix}$$

- If $|J| = 0$, then there is functional dependency among equations and hence no unique solution exists.

E.g $y_1 = 4x_1 - x_2 \Rightarrow y_1^1 = 4$ & $y_2^1 = -1$ y_2^1 means $\frac{\partial y_1}{\partial x_2}$ and y_1^1 means $\frac{\partial y_1}{\partial x_1}$

$$y_2 = 16x_1^2 + 8x_1x_2 + x_2^2 \Rightarrow \frac{\partial y_2}{\partial x_1} = 32x_1 + 8x_2$$

$$\frac{\partial y_2}{\partial x_2} = 8x_1 + 2x_2$$

$$|J| = \begin{vmatrix} 4 & -1 \\ 32x_1 + 8x_2 & 8x_1 + 2x_2 \end{vmatrix} = 64x_1 + 16x_2 \neq 0, \text{ thus, the system has a solution because the}$$

value of the Jacobian determinant is non-singular, i.e., different from zero.

As stated in the preceding paragraphs, partial derivatives can provide a means of testing whether there exists functional (linear or nonlinear) dependence among a set of n functions in n variables. This is related to the notion of Jacobian determinant.

Exercise

1. Optimize: $y = 3x_1^2 - 5x_1 - x_1x_2 + 6x_2^2 - 4x_2 + 2x_2x_3 + 4x_3^2 + 2x_3 - 3x_1x_3$, using.
 - a) The Cramer's rule for the first order condition.
 - b) The Hessian for the S.O.C.
2. A monopolistic firm produced two related goods. The demand and total cost functions are:

$$Q_1 = 100 - 2p_1 + 2p_2$$

$$Q_2 = 75 + 0.5p_1 - p_2$$

$$TC = Q_1^2 + 2Q_1Q_2 + Q_2^2$$

Optimize the firms profit function by:

- a) Finding the inverse demand function.
 - b) Using cramer's rule for the f.o.c.
 - c) Using the Hessian for the s.o.c.
3. Maximize a firm's total cost:
- $$c = 45x^2 + 90xy + 90y^2, \text{ when the firm has to meet a production quota:}$$
- $$2x + 3y = 60, \text{ by:}$$
- a) finding the critical values
 - b) using the Bordered Hessian to test the S.O.C.
4. A monopolist firm produces two substitute goods and faces the following demand functions.
- $$X = 80 - P_x + 2p_y$$
- $$y = 110 + 2p_x - 6p_y$$
- Its total cost function is : $TC = 0.5x^2 + xy + y^2$ what is the profit maximizing level of output if the firm has to meet a production quota of $3x + 4y = 350$?
5. Maximize utility: $u(x, y) = x^{0.25}y^{0.4}$, subject to, $2x + 8y = 104$ (use any method).

4.3. Inequality Constraints & The theorem of Kuhn and Tucker

Inequality constraints are those requiring certain variables to be non-negative. These often have to be imposed in order that the solution should make economic sense. In addition, bounds on the availability of resources are often expressed as inequalities rather than equalities. Consider the following examples.

Ex-1: Max: $\Pi = 5x_1 + 3x_2$, Π stands for profit.

$$\text{subject to } \left. \begin{array}{l} 6x_1 + 2x_2 \leq 36 \\ 5x_1 + 5x_2 \leq 40 \\ 2x_1 + 4x_2 \leq 28 \\ x_1, x_2 \geq 0 \end{array} \right\} \text{Here we add slack variables to make the constraint binding}$$

(active)

$$\text{Example-2: Min : } C = 2x_1 + 4x_2 : \text{subject to } \left. \begin{array}{l} 2x_1 + x_2 \geq 14 \\ x_1 + x_2 \geq 12 \\ x_1 + 3x_2 \geq 18 \\ x_1, x_2 \geq 0 \end{array} \right\} \text{Here we subtract surplus}$$

values to make the constant binding. C denotes cost.

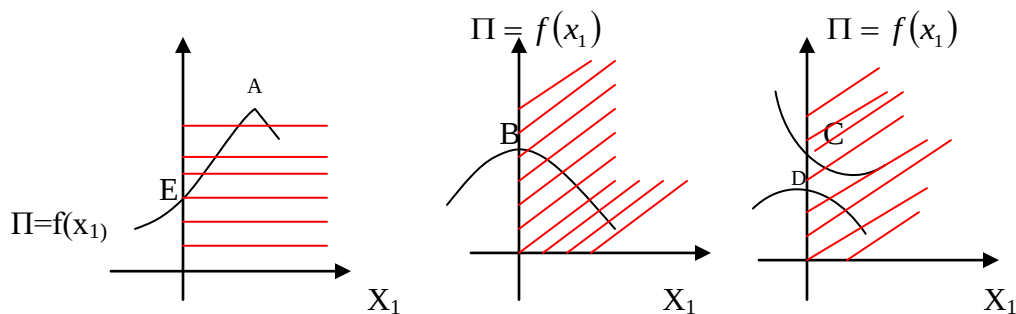
Note :- If at optimality the slack/surplus of a constraint is zero, then the constraint is said to be active or binding. Otherwise, it is inactive.

Effects of non – negativity Restrictions

i. single variable case

$$\text{Max : } \Pi = f(x_1), \text{ subject to, } x_1 \geq 0$$

In view of the restriction $x_1 \geq 0$, three possible situations may arise.



$$\left. \begin{array}{l} f'(x_1) = 0, \text{ and } x_1 > 0 (\text{point .A}) \\ f'(x_1) = 0, \text{ and } x_1 = 0 (\text{point .B}) \\ f'(x_1) < 0, \text{ and } x_1 = 0 (\text{point .c \& D}) \end{array} \right\}$$

These three conditions can be consolidated into a single statement:

$$f'(x_1) \leq 0, x_1 \geq 0, \text{ and } x_1 f'(x_1) = 0 \text{ (for max) and}$$

$$f'(x_1) \geq 0, x_1 \geq 0, \text{ and } x_1 f'(x_1) = 0 \text{ for minimum}$$

- the equation $x_1 f'(x_1) = 0$ is referred to as the complimentary slackness between x_1 & $f'(x_1)$.

ii. Two variable, one constraint case

$$\max \Pi = f(x_1, x_2), \text{ subject to, } g(x_1, x_2) \leq r$$

$$x_1, x_2 \geq 0$$

And slack variable to change the constraint into strict equality, thus .

$$\max \Pi = f(x_1, x_2), \text{ subject to, } g(x_1, x_2) + s = r$$

$$x_1, x_2, s \geq 0$$

Step1: Form the Lagrangian function

$$L = f(x_1, x_2) + \lambda[r - g(x_1, x_2) - s]$$

Step2: Find the first order partials with respect to each of the choice variables and set them equal to zero.

$$L_{x_1} \leq 0, x_1 \geq 0, \text{ and } x_1 \frac{\partial L}{\partial x_1} = 0 \dots \dots (1)$$

$$L_{x_2} \leq 0, x_2 \geq 0, \text{ and } x_2 \frac{\partial L}{\partial x_2} = 0 \dots \dots (2)$$

$$L_s \leq 0, s \geq 0, \text{ and } s \frac{\partial L}{\partial s} = 0 \dots \dots (3)$$

$$L_\lambda = r - g(x_1, x_2) - s = 0 \dots \dots (4)$$

Now, we can consolidate equation (3) and (4) and in the process, eliminate the dummy variables. i.e. $\frac{\partial L}{\partial s} = -\lambda = 0$, thus equation (3) tells us that $-\lambda \leq 0, s \geq 0$, and $-\lambda s = 0$, or equivalently

$$\lambda \geq 0, s \geq 0, \text{ and } \lambda s = 0 \dots \dots (5)$$

→ From equation (4) we have

$$s = r - g(x_1, x_2)$$

→ substituting (4) in (5) we have

$$\lambda \geq 0, r - g(x_1, x_2) \geq 0, \text{ and } \lambda[r - g(x_1, x_2)] = 0$$

- thus, the Kuhn-Tucker (K-T) conditions for maximization problem with inequality constraint are:

$$\frac{\partial L_1}{\partial x_1} = f_1 - \lambda g_1 \leq 0, x_1 \geq 0, \text{ and } x_1 \frac{\partial L}{\partial x_1} = 0$$

$$\frac{\partial L_2}{\partial x_2} = f_2 - \lambda g_2 \leq 0, x_2 \geq 0, \text{ and } x_2 \frac{\partial L}{\partial x_2} = 0$$

$$\frac{\partial L}{\partial \lambda} = r - g(x_1, x_2) \geq 0, \lambda \geq 0, \text{ and } \lambda [r - g(x_1, x_2)] = 0$$

- Kuhn-Tucker(K-T) conditions for minimization problem with inequality constraint:

$$L_{x_1} = f_1 - \lambda g_1 \geq 0, x_1 \geq 0, \text{ and } x_1 L_{x_1} = 0$$

$$L_{x_2} = f_2 - \lambda g_2 \geq 0, x_2 \geq 0, \text{ and } x_2 L_{x_2} = 0$$

$$L_{\lambda} = r - g(x_1, x_2) \leq 0, \lambda \geq 0, \text{ and } \lambda L_{\lambda} = 0$$

[Students' activity:- For n- variable, m-constraint case, please read A.C . Chiang]

Example- 1

$$\max y = 10x_1 - x_1^2 + 180x_2 - x_2^2, \text{ subject to, } x_1 + x_2 \leq 80$$

$$x_1, x_2 \geq 0$$

Step -1:- Form the Lagrangian function

$$L = 10x_1 - x_1^2 + 180x_2 - x_2^2 + \lambda(80 - x_1 - x_2)$$

Step -2: solve for the first order condition (F.O.C.) and set the partials with respect to each of the choice variables equal to zero.

$$L_{x_1} = 10 - 2x_1 - \lambda \leq 0 \dots \dots \dots (1)$$

$$L_{x_2} = 180 - 2x_2 - \lambda \leq 0 \dots \dots \dots (2)$$

$$L_{\lambda} = 80 - x_1 - x_2 = 0 \dots \dots \dots (3)$$

$$\rightarrow \text{From (1) \& (2) we have, } x_2 = x_1 + 85 \dots \dots \dots (4)$$

$\rightarrow$ Substituting (4) in (3), we have $x_1 = -\frac{5}{2}$ (not feasible) because the x's should be non-negative.

$\rightarrow$ Hence setting $x_1^* = 0$, we have $x_2^* = 80$

$$\text{And } \lambda^* = 20$$

Step -3: Check whether the constraints are satisfied

i) non – negativity constraint $x_1^* = 0$

$$x_2^* = 80$$

$$\lambda^* = 20$$

ii) inequality constraint : $x_1 + x_2 \leq 80$

$$0 + 80 \leq 80$$

Step -4: check the complementary slackness condition $-x_1 \frac{\partial L}{\partial x_1} = 0 \rightarrow$ since x_1 is zero,

$$x_1 \frac{\partial L}{\partial x_1} = 0.$$

$$-x_2 \frac{\partial L}{\partial x_1} = 0, x_2^* = 80 \Rightarrow \frac{\partial L}{\partial x_2} = 0$$

$$180 - 2x_2 - \lambda = 0$$

$$180 - 2(80) - 20 = 0$$

$$0 = 0$$

$$-\lambda \frac{\partial L}{\partial \lambda} = 0, \lambda^* = 20 \Rightarrow \frac{\partial L}{\partial \lambda} = 0$$

$$\Rightarrow 80 - x_1 - x_2 = 0$$

$$80 - 0 - 80 = 0$$

$$0 = 0$$

* Since all the K- T conditions are satisfied, the function is maximized at

$$x_1^* = 0 \text{ \& } x_2^* = 80.$$

Example -2

$$\text{Minimize } C = (x_1 - 4)^2 + (x_2 - 4)^2$$

$$2x_1 + 3x_2 \geq 6$$

$$\text{Subject to: } -3x_1 - 2x_2 \geq -12$$

$$x_1, x_2 \geq 0$$

Solution:

Step1: Formulate the Lagrange

$$L = (x_1 - 4)^2 + (x_2 - 4)^2 + \lambda_1(6 - 2x_1 - 3x_2) + \lambda_2(-12 + 3x_1 + 2x_2)$$

F.O.C.(First Order Condition)

$$L_1 = 2(x_1 - 4) - 2\lambda_1 + 3\lambda_2 \geq 0 \dots\dots\dots(1)$$

$$L_2 = 2(x_2 - 4) - 3\lambda_1 + 2\lambda_2 \geq 0 \dots\dots\dots(2)$$

$$L_{\lambda_1} = 6 - 2x_1 - 3x_2 \leq 0 \dots\dots\dots(3)$$

$$L_{\lambda_2} = -12 + 3x_1 + 2x_2 \leq 0 \dots\dots\dots(4)$$

To solve such multivariable non –linear programming problem, let us use iteration or trial & error), method

Iteration-1 $\lambda_1 = 0, \text{ and } \lambda_2 > 0$

Thus we have $\frac{\partial L}{\partial \lambda_1} = 0$ and $\frac{\partial L}{\partial \lambda_2} = 0$ i.e. $2x_1 + 3x_2 = 6$ and $3x_1 + 2x_2 = 12$

Solving simultaneously yields $x_1 = \frac{24}{5}$ and $x_2 = \frac{-6}{5}$ (this is not feasible).

From equation (1) & (2), we have:

$$2x_1 - 2\lambda_1 + 3\lambda_2 = 8, \text{ and } \dots\dots\dots(5)$$

$$2x_2 - 3\lambda_1 + 2\lambda_2 = 8 \dots\dots\dots(6)$$

$$\text{And } \lambda_2 > 0 \Rightarrow \frac{\partial L}{\partial \lambda_2} = 3x_1 + 2x_2 = 12 \dots\dots\dots(7)$$

$\Rightarrow$ solving equations (5) , (6), and (7) assuming $\lambda_1 = 0$, yields

$$x_1^* = \frac{28}{13}$$

$$\lambda_2^* = \frac{36}{13}$$

$$\lambda_1 = 0$$

$$\lambda_2 = \frac{16}{13}$$

$\rightarrow$ check whether the constraints are satisfied

- i) Non – negative constraint $\rightarrow$ satisfied
- ii) Inequality constraint:

$$2\left(\frac{18}{13}\right) + 3\left(\frac{36}{13}\right) \geq 6$$

$$-3\left(\frac{28}{13}\right) - 2\left(\frac{36}{13}\right) \geq -12$$

→ Check the complementary slackness conditions. This is satisfied.

* Therefore, since the solution values for the four variables satisfy all the K-T conditions, they are acceptable as the final solutions.

Chapter -5

5. Comparative static analysis

After successfully completing this chapter, students should be able to:

- Appreciate the importance of comparative statics in economics
- Understand differentiation and *its application to comparative statics*
- Analyze Comparative Statics of General function model
- Identify the limitations of comparative static analysis

Introduction

Dear Distant Learner in comparative statics, we simply compare the initial (or pre-change) equilibrium state with the final (or post change) equilibrium state by disregarding the process of adjustment of the variables over times. Thus comparative statics is essentially concerned with finding the rate of change of the equilibrium value of a choice(endogenous) variable with respect to a change in a particular parameter (exogenous) variable.

Student activity- what happens to the quantity demanded of a good when the price of related goods or income changes?

In general, the logical simulation of the testing of theories in economics is called the theory of comparative statics.

5.1. Differentiation and its application to comparative static Analysis

Let $R(x)$ = total revenue function

$C(x)$ = total cost function

$t(x)$ = per unit tax (beyond the control of the firm)

$$\text{Max: } \Pi(x) = R(x) - C(x) - tx$$

Find the first order condition. We all know that the necessary condition for profit maximization stipulates that marginal revenue should be equal to marginal cost. (MR=MC)

$$\Pi'(x) = R'(x) - C'(x) - t = 0 \dots \dots (1)$$

S.O.C $\Pi''(x) < 0 \rightarrow$

$$\Rightarrow R''(x) - C''(x) < 0 \dots \dots (2)$$

Now, the profit maximizing level of output for the firm depends upon the tax rate (an exogenous variable).

i.e. $X = X^*(t) \dots \dots \dots (3)$

substituting (3) in (1):

$$R'(x^*(t)) - C'(x^*(t)) - t = 0$$

$$\Rightarrow R''(x^*) \frac{dx^*}{dt} - C''(x^*) \frac{dx^*}{dt} - 1 = 0$$

$$\Rightarrow \frac{dx^*}{dt} [R''(x^*) - C''(x^*)] = 1$$

$$\Rightarrow \frac{dx^*}{dt} = \frac{1}{R''(x^*) - C''(x^*)} < 0$$

Output X will decline as the tax rate the firm faces increases. Thus, a prediction about changes in the choice variable, i.e. marginal adjustment of output, when the parameter facing the decision maker changes is easily derived, this is the goal of comparative statics.

Example - 2 $\Pi(x) = p \cdot x - c(x)$

F.O.C: $\Pi'(x) = p - c'(x) \dots \dots (1)$

S.O.C $\Pi''(x) = -c''(x) < 0 \Rightarrow c''(x) > 0 \dots \dots (2)$

$X = X^*(p) \dots \dots \dots (3)$

Substituting (3) into (1).

$$p - c'(x^*(p)) = 0$$

$$\Rightarrow \frac{dp}{dp} - c''(x^*) \frac{dx^*}{dp} = 0 \Rightarrow \frac{dx^*}{dp} = \frac{1}{c''(x)} > 0 \rightarrow$$

This means that if the output price to a competitive firm is raised, output level is increased. i.e. the supply function is upward sloping.

Market model

Consider one commodity market model.

$$D: Q = a - bp, (a, b > 0)$$

$$S: Q = -c + dp, (c, d > 0)$$

At equilibrium $Q_d = Q_s$

With solutions:

$$P^* = \frac{a + c}{b + d}$$

$$Q^* = \frac{ad - bc}{b + d}$$

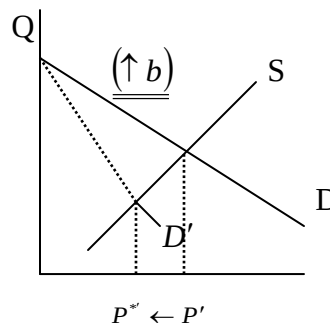
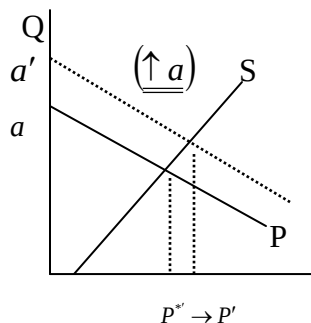
What would happen to the equilibrium P&Q, if one or more of the parameters changes?

$$\frac{\partial p^*}{\partial a} = \frac{1}{b + d} > 0$$

$$\frac{\partial p^*}{\partial b} = \frac{-(a + c)}{(b + d)^2} < 0$$

$$\frac{\partial p^*}{\partial c} = \frac{1}{b + d} > 0$$

$$\frac{\partial p^*}{\partial d} = \frac{-(a + c)}{(b + d)^2} < 0$$



Notional Income model

Consider the following simple national income model:

$$y = c + I_0 + G_0$$

$$C = \alpha + \beta(Y - T), (\alpha > 0, 0 < \beta < 1)$$

$$T = \gamma + \delta y, (r > 0, 0 < \delta < 1)$$

Where, γ = non – income tax revenue

δy = income tax revenue

δ = income tax rate.

-Endogenous variable = Y, C, & T

- Exogenous variable = I_0 & G_0

-Exogenous parameters = α, β, γ and δ .

The model can be solved for y^* as:

$$y^* = \frac{\alpha - \beta\gamma + I_0 + G_0}{1 - \beta + \beta\delta}, \text{ this is the reduced form of the above national income model.}$$

From this equation income value, we can have six comparative static derivatives as we have six exogenous variables.

$$1. \frac{\partial y^*}{\partial G_0} = \frac{1}{1 - \beta + \beta\delta} > 0$$

$$2. \frac{\partial y^*}{\partial \gamma} = \frac{-\beta}{1 - \beta + \beta\delta} < 0$$

$$3. \frac{\partial y^*}{\partial \delta} = \frac{-\beta y^*}{1 - \beta + \beta\delta} < 0$$

5.2. Comparative Statics of General function model

In the comparative static problems considered before, equilibrium values of endogenous variables of the model could be explicitly expressed in terms of the exogenous variables; accordingly, the technique of simple partial differentiation was all we need to obtain the desired comparative static information.

However, when a model contains functions expressed in general form, explicit solutions are not available. In such cases, a new technique must be employed that makes use of such concepts as implicit function rule to find the comparative static derivatives directly from the given general function model.

Example :- Consider a market model:

$$Qd = D(p, y_0), \frac{\partial D}{\partial p} < 0, \frac{\partial D}{\partial y_0} > 0$$

$$Qs = s(p), \frac{\partial s}{\partial p} > 0$$

At equilibrium $D(P, y_0) = S(P) \Rightarrow D(P, y_0) - S(P) = 0$

Where, p = endogenous

y_0 = exogenous.

We know that every equation price is a function of income. i.e. $p = p^*(y_0)$. Therefore, the equilibrium condition can be taken to be an identity in the equilibrium solution.

$$\Rightarrow D(p^*, y_0) - s(p^*) \equiv 0$$

$$\Rightarrow F(p^*, y_0) \equiv 0$$

→ The comparative static analysis of this model will therefore be concerned with how a change in y_0 will affect the equilibrium position of the model.

i.e.i) what is the effect of a change in y_0 on p^* ?

$$\frac{dp^*}{dy_0} = \frac{-\frac{\partial F}{\partial y_0}}{\frac{\partial F}{\partial p^*}} = \frac{-\frac{\partial D}{\partial y_0}}{\frac{\partial D}{\partial p^*}(-) - \frac{\partial s}{\partial p^*}(+)} > 0$$

Thus, increase $y \rightarrow P^*$ or the other way.

iii) what is the effect of a change in y_0 on Q^* ?

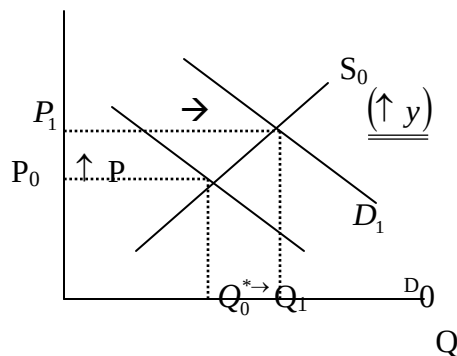
At equilibrium, $Q^* = Qd = Qs$

We can write,

$$Q^* = S(p^*) \text{ and } p = p^*(y_0)$$

$$\Rightarrow \frac{dQ^*}{dy_0} = \frac{ds}{dp^*} \cdot \frac{dp^*}{dy_0} > 0$$

Thus, the comparative static results convey the proposition that an up-ward shift of the demand curve (due to a rise in income) will result in a higher equilibrium price as well as a higher equilibrium quantity.



5.3. Limitations of comparative static analysis

Comparative static is helpful in finding out how a disequilibrating change in a parameter will affect the equilibrium state of a model. However, by its very nature, comparative statics has the following limitations.

- 1) Ignores the process of adjustment from the old equilibrium to the new one.
- 2) Neglects the time element (length of time) involved in the adjustment process from one to another equilibrium
- 3) Assumes that a new equilibrium can be defined and attained after a disequilibrating change in a parameter, i.e. disregards the possibility that the new equilibrium may not be attained ever because of the inherent instability of the model. All these limitations are addressed by dynamic analysis which will be dealt in the next chapter.

CHAPTER 6

6. Dynamic equilibrium analysis

Aims and objectives:-

After completing this chapter students should be able to:

- ▲ Demonstrate how a time lag can affect the pattern of adjustment to equilibrium in some basic economic models.
- ▲ Distinguish between continuous time and discrete time models
- ▲ Set up and solve linear first-order difference and differential equations.
- ▲ Apply the differential equation to find solutions to Domar's economic growth model, Solow growth model and Market equilibrium models.
- ▲ Apply the difference equation solution method to the cobweb iterative model
- ▲ Identify the stability conditions in the above model

Introduction

In the previous chapters much of the economic analysis used has been comparative statics. This entails the comparison of different (static) equilibrium situations, with no regard on how variables adjust to their new equilibrium values. The study of the mechanism by which variables adjust between equilibrium values is called 'dynamics'.

The way markets adjust to equilibrium over time shows a great variation. If the task of adjusting to equilibrium from a given point is left for market, the adjustment process will be completely different from the way static equilibrium is achieved. In some markets, the adjustment process is very fast (instantaneous). In other markets, the adjustment process is slow, and even we may come across functions which follow a path which diverge from their dynamic equilibrium point. But, in static equilibrium, there is no model that can explain these dynamic adjustments. This chapter will introduce you to such models that explain dynamic equilibrium.

Dynamic equations may be viewed in two ways. Based on the dynamic behavior of variables, time may be considered as continuous or discrete quantity. In the language of Mathematics, the former is called differential equation, and the latter difference equation.

Dear Distance Learners, we think that you have some idea about simple dynamic equations in your Quantitative Methods course. For this course we will discuss simple dynamic equilibrium analysis in the cases of both continuous and discrete time. But it will be a bit deeper than the former.

6.1 Definition of differential equation: -

An equation $F(x; y)$ which assumes a functional relationship between independent variable x and dependent variable y and includes x , y and derivatives of y with respect to x is called an ordinary differential equation. It is an implicit functional relation ship between functions and their derivatives.

The order of a differential equation is determined by the order of the highest derivative in the equation. The highest power to which the highest order derivative is raised is called the degree of the differential equation.

Example 1. $y \frac{d^2 y}{dx^2} + 2xy = 0$ Is a second order differential equation of degree one.

Because the highest order derivative in the equation is $\frac{d^2 y}{dx^2}$ ¹ (2nd order derivative) and the power to which this derivative is raised is 1.

Example 2 $\left[\frac{d^3 y}{dx^3} \right]^4 + y \left[\frac{d^2 y}{dx^2} \right]^6 + 4x = 0$ is a 3rd order differential equation of degree 4

Any differential equations course will concern itself with answering one or more of the following questions.

1. Given a differential equation, will a solution exist?

Not all differential equations will have solutions so it's useful to know ahead of time if there is a solution or not. We proceed to solving a differential equation only if it has a solution. This question is usually called the **existence question**

2. If a differential equation does have a solution how many solutions are there?

It is possible for a differential equation to have more than one solution. We would like to know how many solutions there will be for a given differential equation.

¹ $\frac{d^2 y}{dx^2}$ Is an operator symbol indicating 2nd derivative and should not be confused with the power to which the function is raised.

It would be nice if, during the derivation of our differential equation, we could make sure that our assumptions would give us a differential equation that upon solving will yield a single unique solution.

This question is usually called the **uniqueness question** in a differential equations course.

3. If a differential equation does have a solution can we find it?

This may seem like an odd question to ask and yet the answer is not always yes. Just because we know that a solution to a differential equations exists does not mean that we will be able to find it.

6.1.2 First order linear differential equation

The first special case of first order differential equations that we will look at in this module is the linear first order differential equation. In this case, the highest order derivative appearing in the equation is first order derivative. More over it is linear for that the highest power to which the derivative is raised is 1.

A first-order linear differential equation is an equation of the form

$$\frac{dy}{dt} + p(t)y = Q(t)$$

Where P and Q are continuous functions of t . This first-order linear differential equation is said to be in **standard form**.

6.2 Methods of solving differential equation

1. Direct integration: - for a differential equation of the form $\frac{dy}{dx} = f(x)$ solution

can be found using our knowledge of integral calculus i.e.

$$\frac{dy}{dx} = f(x)$$

$$dy = f(x)dx$$

$$\int dy = \int f(x)dx, \text{ taking the integral of both sides}$$

$$y = F(x) + c$$

2. Separable Differential Equations

A separable differential equation is any differential equation that we can write in the following form.

$$N(y)\frac{dy}{dx} = M(x)$$

Note that in order for a differential equation to be separable all the y 's in the differential equation must be multiplied by the derivative and all the x 's in the differential equation must be on the other side of the equal sign.

The solution for the above type of equation is found first by rewriting it as

$$N(y)dy = M(x)dx$$

Then integrating both sides we get $\int N(y)dy = \int M(x)dx$

Example 1

Consider the differential equation

$$M'(t) = -2(M(t))^2 t$$

After a simple rearrangement, we may write this equation as

$$\frac{-dM}{M^2} = 2tdt \quad \text{Then, taking the integral of both sides, we get the following}$$

result

$$-\int \frac{dM}{M^2} = \int 2tdt \rightarrow \frac{1}{M} = t^2 + c$$

Finally, our solution set becomes $\underline{\underline{M(t) = \frac{1}{t^2 + c}}}$

Example 2

Consider the differential equation

$$M'(t) = \frac{t}{(x(t))^4} + 1$$

Separating the variables yields

$$(M^4 + 1)dM = tdt$$

The value of $M^2 + 1$ is not 0 for any value of M , so all the solutions of the equation are obtained by integrating both sides of this equation, which leads to

$$\frac{(M(t))^5}{5} + M(t) = \frac{t^2}{2} + c$$

2. Exact Differential Equation

Given a function of two variables $F(y, t)$, its total differential is

$dF(y, t) = \frac{dF}{dy}dy + \frac{dF}{dt}dt$. When the left hand side of the equation is set equal to 0, the

equation becomes $\frac{dF}{dy}dy + \frac{dF}{dt}dt = 0$ (a)

This is called **exact differential equation**.

Generally, an equation of the form $Mdy + Ndt = 0$ is said to be exact if and only if there

exists a function $F(y, t)$, such that, $M = \frac{\partial F}{\partial y}$ and $N = \frac{\partial F}{\partial t}$ (b)

Using Young's theorem, this may also be written as $\frac{\partial M}{\partial t} = \frac{\partial N}{\partial y}$ ²

Solution to exact differential equation:-

$dF(y, t) = 0 \rightarrow F(y, t) = c$. But the main task is to find the primitive function $F(y, t)$. to

find the primitive function, we should resort to integration. Since $M = \frac{\partial F}{\partial y}$, $F(y, t)$ must

contain the **partial integral**³ of M with respect to y . this leads us to

$$F(y, t) = \int Mdy + \lambda(t).$$

² By Young's theorem, $\frac{\partial^2 F}{\partial y \partial t} = \frac{\partial^2 F}{\partial t \partial y} \Rightarrow \frac{\partial}{\partial t} \left(\frac{\partial F}{\partial y} \right) = \frac{\partial}{\partial y} \left(\frac{\partial F}{\partial t} \right) \Rightarrow \frac{\partial M}{\partial t} = \frac{\partial N}{\partial y}$

³ Partial integration is the opposite of partial differentiation, and in the integration process, all terms that don't contain the variable of integration should be considered as constants.

Where $\lambda(t)$ represents those additive terms containing only t and lost during partial differentiation with respect to y . the first term on the right side may be easy to find. But the difficulty is to find $\lambda(t)$. We use the second partial differential $\frac{dN}{dt}$ to find $\lambda(t)$.

3. Using Integrating factor

To solve a first-order linear differential equation, we can use an integrating factor $u(x)$ which converts the left side into the derivative of the product $u(x)y$. that is, we need a factor $u(x)$ such that

$$\begin{aligned} u(x)\frac{dy}{dx} + u(x)p(x)y &= \frac{d[u(x)y]}{dx} \\ u(x)y' + u(x)p(x)y &= u(x)y' + yu'(x) \\ p(x) &= \frac{u'(x)}{u(x)} \\ \ln|u(x)| &= \int p(x)dx + c \\ u(x) &= ce^{\int p(x)dx} \end{aligned}$$

Because you don't need the most general integrating factor, let $c=1$ Multiplying the original equation $\frac{dy}{dx} + p(x)y = Q(x)$ by $u(x) = e^{\int p(x)dx}$ produces

$$\begin{aligned} y'e^{\int p(x)dx} + yp(x)e^{\int p(x)dx} &= Q(x)e^{\int p(x)dx} \\ \frac{d}{dx}\left[ye^{\int p(x)dx}\right] &= Q(x)e^{\int p(x)dx} \end{aligned} \quad \text{So the general solution becomes}$$

$$ye^{\int p(x)dx} = \int Q(x)e^{\int p(x)dx} dx + c$$

THEOREM 6.1 Solution of a First-Order Linear Differential Equation

An integrating factor for the first-order linear differential equation $\frac{dy}{dx} + p(x)y = Q(x)$ is

$$u(x) = e^{\int p(x)dx} \text{ and the general solution is } ye^{\int p(x)dx} = \int Q(x)e^{\int p(x)dx} dx + c^4$$

The use of integrating factor changes the function in to an exact differential equation.

4. Bernoulli equation

A well-known *nonlinear* equation that reduces to a linear one with an appropriate substitution is the **Bernoulli equation**, named after James Bernoulli (1654-1705)

The equation takes the form of $\frac{dy}{dt} + P(t)y = Q(t)y^n$.

When the power in the right hand side (n) is zero the equation become linear and can be solved using the ordinary methods. If n is one the equation becomes separable differential equation discussed above. Since we have discussed these issues above, our focus here will be when $n \neq 0$ and $n \neq 1$.

The solution for the above equation may be found as follows

$$y^{-n} \frac{dy}{dt} + P(t)y^{1-n} = Q(t) \text{ Multiplying both sides by } y^{-n}$$

Let $z = y^{1-n}$ then $\frac{dz}{dt} = (1-n)y^{-n} \frac{dy}{dt}$. By substitution we get,

$$\frac{1}{(1-n)} \frac{dz}{dt} + P(t)z = Q(t)$$

$$\frac{dz}{dt} + (1-n)P(t)z = (1-n)Q(t). \text{ Now it is very easy to find the solution}$$

since the equation is linearized. Using integrating factor, the solution becomes

$$ze^{\int (1-n)P(t)dt} = \int (1-n)Q(t)e^{\int (1-n)P(t)dt} dt + c$$

$$y^{1-n}e^{\int (1-n)P(t)dt} = \int (1-n)Q(t)e^{\int (1-n)P(t)dt} dt + c$$

⁴ You had better memorize the integrating factor $u(x) = e^{\int p(x)dx}$ used to transform the left hand side into $ye^{\int p(x)dx}$ rather than the process.

Example 1:- find the general solution for $\frac{dy}{dt} + ty = te^{-t^2} y^{-3}$

The equation is Bernoulli equation with $n=-3$.

Let $z = y^{1-n} = y^{1-(-3)} = y^4$ and $\frac{dz}{dt} = 4y^3 \frac{dy}{dt}$ (1)

Multiplying both sides of the original equation by $4y^3$ produces

$$4y^3 \frac{dy}{dt} + 4ty^4 = 4te^{-t^2} \quad \text{Substituting from (1) above}$$

$$\frac{dz}{dt} + 4tz = 4te^{-t^2} \quad \text{..... (2)}$$

The last equation is linear in z and the value of $P(t)$ is $4t$.

$\int p(t)dt = \int 4tdt = 2t^2$. So the integrating factor is e^{2t^2}

Multiplying both sides of equation 2 by the integrating factor e^{2t^2} produces an exact differential equation of the form

$$\frac{dz}{dt} e^{2t^2} + 4e^{2t^2} tz = 4te^{t^2}. \quad \text{The left hand side } \frac{dz}{dt} e^{2t^2} + 4e^{2t^2} tz \text{ can be written in total}$$

differential form as $\frac{d}{dt}(ze^{2t^2})$

$$\frac{d}{dt}(ze^{2t^2}) = 4te^{t^2}$$

$$\int \frac{d}{dt}(ze^{2t^2}) dt = \int 4te^{t^2} dt \quad \rightarrow \text{taking the integral of both sides.}$$

$$ze^{2t^2} = \int 4te^{t^2} dt$$

$$ze^{2t^2} = 2e^{t^2} + c$$

$$z = 2e^{-t^2} + ce^{-2t^2}$$

Finally we should transform the solution we get in terms of z , back to a solution in terms of y using $z = y^4$

⁵ Note:- $\frac{d}{dt}(ze^{2t^2}) = e^{2t^2} \frac{dz}{dt} + 4te^{2t^2} z$

$y^4 = 2e^{-t^2} + ce^{-2t^2}$ is a general solution.

6.3 First order linear differential equation with constant coefficient and constant term

To make the basic idea progressively clear, let us start with the simplest case i.e. when $P(t)$ and $Q(t)$ are both constants. The equation of this type is called a first-order **linear differential equation with constant coefficient and constant term**, and takes the form

$$\frac{dy}{dt} + vy = z, \text{ where } Z \text{ and } V \text{ are constants. Still we may have another}$$

classification based on the value that the constants take.

The case of $z=0$: - in this case the equation takes the following form

$$\frac{dy}{dt} + vy = 0. \text{ This equation is called Homogeneous Linear differential}$$

equation. The solution can be found easily as follows.

$$\frac{dy}{dt} = -vy \quad \rightarrow \quad \frac{dy}{y} = -vdt$$

$$\int \frac{dy}{y} = \int -vdt$$

$$= \ln(y) = -vt + c$$

$$= y = e^{-vt+c} = e^c \times e^{-vt}. \text{ Let } e^c = A$$

$$= \underline{\underline{y = Ae^{-vt}}}$$

The above solution is a general solution for the homogeneous equation because the value of A is not known. To get definite solution we should return back to the initial condition (the value of y when $t=0$)

$$y(0) = Ae^{-v \times 0} = A. \text{ Substituting this value, we get the definite solution which is}$$

$$\underline{\underline{y(t) = y(0)e^{-vt}}}$$

The case of $z \neq 0$: - in this the equation is called non homogeneous equation, and it is of

the form $\frac{dy}{dt} + vy = z$. This equation has two components i.e., the particular integral

and the complementary equation. So, to find the solution of the complete equation, we should find the solution of each of the components.

The solution of the complementary equation is the same to the solution of the homogeneous equation we have seen above.

To find the solution of the particular integral, we mostly use iteration. We start from the simplest form that y can take. The simplest form for our case is when $y=k$ where k is a constant function. In this case the non homogeneous equation become

$$vy = z$$
$$y_p = \frac{z}{v} \text{ Since } y \text{ is a constant function, } \frac{dy}{dt} = 0$$

So, $y_p = \frac{z}{v}$, is the solution for the particular integral.

The general solution for the complete system will be

$y(t) = y_p + y_c$, where y_c is the solution for the complementary equation derived from the homogeneous equation.

$y(t) = Ae^{-vt} + \frac{z}{v}$ to get definite solution we see the initial value of y at $t=0$

$$y(0) = Ae^{-v \times 0} + \frac{z}{v} = A + \frac{z}{v}$$

$$A = y(0) - \frac{z}{v}$$

Substituting the value for A we get the definite solution $y(t) = \left(y(0) - \frac{z}{v} \right) e^{-vt} + \frac{z}{v}$

The case of $v=0$: in this case, the equation becomes $\frac{dy}{dt} = z$ and the solution is easily

derived using direct integration as

$$\int \frac{dy}{dt} dt = \int z dt \Rightarrow y = zt + c \quad \text{is the general solution and the}$$

definite solution may be found by substituting c with a definite value using the initial condition, $y(0) = c$

$y_t = zt + y(0)$ is the definite solution.

6.4 Practical Application of First Order Linear Difference Equation in Economics

Introduction:- Many economic models with a temporal dimension involve relationships between the values of variables at a given point in time and the changes in these values over time. A model of economic growth, for example, typically contains a relationship between the change in the capital stock and the value of output. When time is modeled as a continuous variable, these economic relationships are modeled using **differential equations**.

Dear distance learners, in this section we will try to discuss some typical applications of differential equation in modeling economic relationships. The focus of this module will be entirely on the mathematical part of the models. The theoretical parts may be found in your Microeconomics, Macroeconomics, and Development Economics Modules. So, if you have any difficulty regarding the theoretical framework, you would probably consult the aforementioned modules.

1. Domar's Growth Model

Premises: - 1. A change in $I(t)$ has a dual effect i.e., both on the demand and supply side

2. $I(t)$ has a multiplier effect on the aggregate demand and the multiplier is

$$\frac{1}{1-c} \text{ or } \frac{1}{s} \quad {}^6 \text{Where } c \text{ is marginal propensity to consume and } s \text{ marginal}$$

propensity to save.

Suppose investment is the only parametric expenditure in the economy.

$$\frac{dy}{dt} = \frac{dI}{dt} \cdot \frac{dy}{dI} \Rightarrow \text{chainrule}$$

$$\frac{dy}{dt} = \frac{dI}{dt} \cdot \frac{1}{s} \Rightarrow b/c \frac{dy}{dI} = \frac{1}{s} \dots\dots\dots (1)$$

The effect of investment on the supply side, according to this model, is through the increase in capacity capital ratio. Let δ be the capacity of the economy (measured as potential output) & $k(t)$ be capital stock $\left(= \int_0^t I(t) \right)$ then;

⁶ The multiplier may be derived from the IS-LM model of a closed economy by taking investment as an exogenous variable of time.

$$\frac{\delta}{k} = \rho \text{ Where } \rho \text{ (rho) is the capacity capital ratio.}$$

$$\delta = \rho k$$

If we assume ρ to be constant $\frac{d\delta}{dt} = \rho \frac{dk}{dt}$ but $\frac{dk}{dt} = I(t)$

$$\frac{d\delta}{dt} = \rho I(t) \dots\dots\dots (2)$$

At equilibrium, the aggregate demand should be exactly equal to the potential output

i.e., $y = \delta$. Taking the derivative of both sides, $\frac{dy}{dt} = \frac{d\delta}{dt} = \rho I(t)$

Substituting $\frac{dy}{dt} = \frac{dI}{dt} \cdot \frac{1}{s}$ from equation (1) we get

$$\frac{1}{s} \frac{dI}{dt} = \rho I(t) \Rightarrow \frac{dI}{dt} - \rho s I(t) = 0 \dots\dots\dots (3)$$

The last equation above is a homogeneous first order linear differential equation with the constant coefficient being ρs . So the solution becomes

$$I(t) = Ae^{-\rho s t} \text{ general solution}$$

$$I(t) = I(0)e^{-\rho s t} \text{ definite solution}$$

2. Dynamic price adjustment model

From comparative statistics if
$$\begin{aligned} Q_d &= a - bP \\ Q_s &= -c + dP \end{aligned}$$

At market clearing price level

$$Q_d = Q_s$$

$$a - bP = -c + dP$$

$$\bar{P} = \frac{a + c}{b + d}$$

But the value doesn't mean inter-temporal equilibrium. If the task of adjustment is left for market with out any interference, the adjustment entirely depends on the relative strength of supply and demand. We should note, however; that it is not only p that adjusts; but Q_s and Q_d too. Let's assume that the rate of change of price is a linear function of the excess demand/supply ($Q_d - Q_s$); then

$$\frac{dp}{dt} = \gamma(Q_d - Q_s) \text{ Where } \gamma \text{ is adjustment coefficient?}$$

$$\frac{dp}{dt} = \gamma(a - bp - (-c + dp))$$

$$\frac{dp}{dt} + \gamma(b + d)p = \gamma(a + c)$$

Where a, b, c and d are constants. The above equation is a linear non homogeneous difference equation where $v = \gamma(b + d)$ and $z = \gamma(a + c)$. So, the general solution becomes

$$p(t) = Ae^{-vt} + \frac{z}{v}$$

$$P(t) = Ae^{-\gamma(b+d)t} + \frac{a+c}{b+d} \text{ general solution}$$

$$p(t) = \left(p(0) - \frac{a+c}{b+d} \right) e^{-\gamma(b+d)t} + \frac{a+c}{b+d} \text{ definite solution}$$

But $\frac{a+c}{b+d} = \bar{p}$, substituting this value in the above equation, we get the following.

$$p(t) = \left(p(0) - \bar{p} \right) e^{-\gamma(b+d)t} + \bar{p} \text{ Where } b \text{ and } d \text{ are slopes of demand and supply curves.}$$

The last equation tells us that the inter-temporal equilibrium depends on the slopes of supply and demand curves. i.e., $b+d > 0$ to get a dynamically stable path (time path which converges to equilibrium). *Actually, for normal demand and supply curves this requirement is always fulfilled because, since sign is considered in the equation, both b and d are positive.* If the conditions are fulfilled, the dynamic time path will be convergent.

If $p(0) > \bar{p}$, the time path of the function asymptotically approaches $\bar{p}$ from above

If $p(0) < \bar{p}$, the time path of the function asymptotically approaches $\bar{p}$ from below

If $p(0) = \bar{p}$ the time path will be a horizontal line at $\bar{p}$

3. Solow growth model

Basic Assumptions

1. Continuous time.
2. Single good produced with a constant technology.
3. No government or international trade.

4. All factors of production are fully employed.

5. Labor force grows at constant rate $n = \frac{\dot{L}}{L}$

6. Initial values for capital, K_0 and labor, L_0 given.

The neoclassical cobb-duglas production function is given by

$$Y(t) = AK(t)^a L(t)^{1-a} \dots\dots\dots(1).$$

The production function assumes a constant return to scale. More over, the marginal productivities of inputs are positive. I.e.,

$$\frac{\partial Q}{\partial K} = aAK^{a-1}L^{1-a} > 0$$

$$\frac{\partial Q}{\partial L} = (1-a)AK^aL^{-a} > 0$$

And decreasing (diminishing) i.e,

$$\frac{\partial^2 Q}{\partial K^2} = (a-1)aAK^{a-2}L^{1-a} < 0$$

$$\frac{\partial^2 Q}{\partial L^2} = -a(1-a)AK^aL^{-a-1} < 0$$

If we define all the terms of the equation in per worker terms;

$$y = \frac{Y}{L} = \frac{AK^aL^{1-a}}{L} = A\left(\frac{K}{L}\right)^a \left(\frac{L}{L}\right)^{1-a} \dots\dots\dots(2)$$

$$y = Ak^a$$

The capital accumulation in the economy at a given period depends on the net investment in the period. So capital accumulation grows at a rate investment net of depreciation. If δ is the rate at which capital depreciates, δK will be the total investment for replacement of depreciated capital. So the net investment is given by,

$$I_n = sY - \delta K$$

$$\dot{K} = sY - \delta K \dots\dots\dots (3)$$

Where s is a constant saving rate (by assumption)

Dividing both sides of equation 3 by K gives

$$\frac{\dot{K}^*}{K} = \frac{sY}{K} - \delta$$

$$\frac{\dot{K}^*}{K} = \frac{s \frac{Y}{L}}{\frac{K}{L}} - \delta \Rightarrow \frac{\dot{K}^*}{K} = s \frac{y}{k} - \delta \dots\dots\dots (4) \text{ (Taking the per worker values of the variables)}$$

But $\frac{\dot{k}}{k} = \frac{\dot{K}}{K} - \frac{\dot{L}}{L} = \frac{\dot{K}}{K} - n \Rightarrow \frac{\dot{K}}{K} = \frac{\dot{k}}{k} + n \dots\dots\dots (5)$

Substituting these values in equation 4;

$$\begin{aligned} \frac{\dot{k}}{k} + n &= s \frac{y}{k} - \delta, \text{ but } y = Ak^a \\ \dot{k} &= sAk^a - (\delta + n)k \\ k + (\delta + n)k &= sAk^a \dots\dots\dots (6) \end{aligned}$$

The last equation is a differential equation and it is in the form of Bernoulli equation where $n=a$. the dynamic equilibrium (steady state) can be found by using solution methods of Bernoulli equation as follows;

$$\begin{aligned} k \dot{k}^{-a} + (\delta + n)k^{1-a} &= sA \text{ Multiplying both sides by } k^{-a} \\ \frac{1}{1-a} \frac{dz}{dt} + (\delta + n)z &= sA \text{ Substituting } z = k^{1-a} \text{ and } \frac{dz}{dt} = \frac{dz}{dk} \cdot \dot{k} = (1-a)k^{-a} \dot{k} \\ \frac{dz}{dt} + (1-a)(\delta + n)z &= (1-a)sA \dots\dots\dots (7) \end{aligned}$$

The last equation above is the linearized form of the Bernoulli equation where a, s, δ and A are constants. So, the solution is

$$\begin{aligned} z(t) &= Ae^{-(1-n)(\delta+n)t} + \frac{(1-a)sA}{(1-a)(\delta+n)} = Ae^{-(1-n)(\delta+n)t} + \frac{sA}{(\delta+n)} \Rightarrow \text{generalsolution} \\ &= \left(z(0) - \frac{sA}{(\delta+n)} \right) e^{-(1-n)(\delta+n)t} + \frac{sA}{(\delta+n)} \text{definitesdution} \end{aligned}$$

Since the solution is in terms of z, we should transform it back to k.

$$k^{1-a}(t) = \left(k^{1-a}(0) - \frac{sA}{(\delta+n)} \right) e^{-(1-n)(\delta+n)t} + \frac{sA}{(\delta+n)} \text{definitesdution}$$

As t becomes larger and larger, $\left(k^{1-a}(0) - \frac{sA}{(\delta+n)} \right) e^{-(1-n)(\delta+n)t}$ approaches to zero.

So, $k^{1-a}(t) = \frac{sA}{(\delta+n)} \Rightarrow k(t) = \left(\frac{sA}{(\delta+n)} \right)^{\frac{1}{1-a}}$, the time path of capital stock is determined

by the values of α , s , δ and A . this leads to the conclusion that the time path of k approaches to a constant (steady state value) and that steady state value is

$k(t) = \left(\frac{sA}{(\delta+n)} \right)^{\frac{1}{1-a}}$ and the steady state value of percapita output is

$$y(t) = A \left(\frac{sA}{(\delta+n)} \right)^{\frac{a}{1-a}}, \text{ because } y = Ak^a$$

$$y(t) = A^{\frac{1}{1-a}} \left(\frac{s}{(\delta+n)} \right)^{\frac{a}{1-a}}$$

6.5 Difference Equation

In the previous sections of our discussion, we take time as a continuous variable. In such cases the change of time can be made infinitesimal. That is why we call it differential equation. When time is considered as a discrete variable such that it takes only integer value, the concept of derivatives can not be applied. In such cases the pattern of change of variables and their functional values are described as *differences* rather than *differentials*. An equation describing the discrete change of variable with respect to time is called difference equation. This discrete time version of economic dynamics is also called period analysis.

In this section we will discuss about the simplest form of difference equation i.e. first order linear difference equation.

Definition: - a difference equation is an equation involving differences. It can be viewed at least from 3 angles; as a sequence of numbers, as a discrete dynamical system and as an iterated function.

1. Difference equation is a sequence of numbers generated recursively using a rule to relate each number to the previous number in a sequence. Example: the sequence $s = \{1, 1, 2, 3, 5, 8, 13, 21\}$ is generated using the rule $y_{k+2} = y_{k+1} + y_k$ where $k=1, 2, 3, \dots$ and the initial value $y(0)=1$
2. a difference equation is an iterated map in such a way that

$$y(k+1) = f(y(k))$$

If we see the sequence in 1 above as an iterated function it can be represented as follows. $y(0), f(y(0)), f(f(y(0))), \dots$. Where $f(y(0))$ is the 1st iterated function of $y(0)$.

A linear difference equation can be defined as an equation that relates a variable y_t to its previous values linearly. The simplest one is a first-order scalar linear difference equation taking the form,

$$y_t = ay_{t-1} + B, \forall t,$$

This is a first-order difference equation, since only first lag of its own values y_{t-1} appears in the equation. Here, we are interested in finding a solution of y in terms of its current, past, or future values.

6.5.1 Solving Difference Equation

A solution of a difference equation is an expression or formula that makes the difference equation true for all values of the integer variable t .

The method of finding solution to difference equation basically follows the same logic to that of solving difference equation. In finding solution to differential equation, we are interested in the time path of a given function. Although we use the concept of derivatives and differentials in the solution procedure, the final answer is free from derivatives and differentials. In the same way, our final answer in solving difference equation should be free from any Δy term although we use the concept of differences in finding solution. In solving differential equation we have intensively used the concept of integration. However; this concept can't be applied for the case at hand since time is considered as a discrete quantity.

ITERATION METHOD

The nature of difference equation allows the solution to be calculated recursively. It is easier to see the solution of difference equation through algebraic method.

Example; given $y(t+1) = ay(t) + \beta$ with initial value $y(0)$, it is possible to determine the solution using algebraic method as follows.

at $t = 0, y(0) = y_0$ initial value

$$t = 1 \Rightarrow y(1) = ay_0 + \beta$$

$$t = 2 \Rightarrow y(2) = ay(1) + \beta = a(ay_0 + \beta) + \beta \Rightarrow a^2 y_0 + (a+1)\beta$$

.....

.....

$$t = n \Rightarrow y(n) = a^n y_0 + (1 + a + \dots + a^{n-1})\beta$$

However; the series in the bracket of the right hand equation is a closed geometric series

taking the values $\sum_{i=0}^{n-1} a^i = \begin{cases} n, \text{ for } a = 1 \\ \frac{1-a^n}{1-a}, \text{ if } a \neq 1 \end{cases}$

Therefore using this recursive approach, the solution to a difference equation $y(t+1) = ay(t) + b$ with initial value $y(0)$ is

$$y(t) = \begin{cases} y_0 + \beta t, \text{ for } a = 1 \\ -y_0 + \beta, \text{ for } a = -1 \text{ \& } t \rightarrow \text{odd} \\ y_0, \text{ for } a = -1 \text{ \& } t \rightarrow \text{even} \\ a^t y_0 + \left(\frac{1-a^t}{1-a} \right) \beta, \text{ if } a \neq 1 \end{cases}$$

General Method

The general method, as the name suggests, provides a systematic general framework for solving First-order difference equations. Suppose you are given a general first order difference equation of the form $y_t = ay_{t-1} + \beta$

The solution to this difference equation consists of two parts: a part named the *particular solution* denoted y_p which is any solution to the first order difference equation given above and a part named the homogenous solution denoted y_c which solves the difference equation $y_t = ay_{t-1}$. The general solution is the sum of y_p and y_c .

Mathematical Economics

Let's begin by thinking about the homogenous solution, i.e. the solution to $y_t = \alpha y_{t-1}$ since at each stage y is α times the previous value, the solution to this would be of the form

$$y_c = \alpha^t A \dots \dots \dots (1)$$

Where A is a constant. The ' A ' term is important because we have to consider all possible solutions to $y_t = \alpha y_{t-1}$

Now let's think of the particular solution, which can be ANY solution to the equation $y_t = \alpha y_{t-1} + \beta$. Since any solution is allowed, we can, for example, find a solution that sets all the y 's equal. If we set $y = \alpha y + \beta$ we get the solution $y = \frac{\beta}{1-\alpha}$.

Combining $y_t = y_p + y_c$ we get the general solution of the form $y_t = \alpha^t A + \frac{\beta}{1-\alpha}$.

If the initial condition is known (y_0 is given), we can definitize the above solution by determining the exact value of ' A '. Setting $t=0$, we get $y_0 = A + \frac{\beta}{1-\alpha} \Rightarrow A = y_0 - \frac{\beta}{1-\alpha}$

So, the definite solution of the difference equation will be of the form

$$\begin{aligned} y_t &= \alpha^t \left[y_0 - \frac{\beta}{1-\alpha} \right] + \frac{\beta}{1-\alpha} \\ y_t &= \alpha^t y_0 + \frac{\beta}{1-\alpha} - \alpha^t \frac{\beta}{1-\alpha} \\ y_t &= \alpha^t y_0 + \left[\frac{1-\alpha^t}{1-\alpha} \right] \beta \end{aligned}$$

Note, the final answer is similar with the one we get using iteration method. However; the general method is used only when $|\alpha| \neq 1$. In the case of $|\alpha| = 1$, the iteration method is used.

Example: solve the difference equation $y_t = 0.2y_{t-1} + 4$ where $y_0 = 10$ using the general method.

Solution; let us start with the solution of the homogeneous system $y_t = 0.2y_{t-1}$. Using the formula in equation (1) above, the solution is $y_t = 0.2^t A$, where A is a constant.

The particular solution is the solution where all y 's are equal; i.e.,

$$y = 0.2y + 4$$

$$y = 5$$

Combining the two solutions, we get $y_t = y_p + y_c$

$$y_t = 0.2^t A + 5$$

Since we know the value of y_0 , we can definitize the solution by substituting a specific value to A . $y_0 = A + 5 \Rightarrow A = y_0 - 5$

$$y_t = 0.2^t (y_0 - 5) + 5$$

$$\underline{y_t = 0.2^t (5) + 5}$$

6.6. DYNAMICS OF FIRST ORDER LINEAR DIFFERENCE EQUATIONS

Steady State Value

Having solved for the time path, we can examine the dynamic behavior of a variable (i.e. its behavior over time). We will examine the stability of the difference equation as well as its steady state value.

The steady state value is a value at which the endogenous variable exhibits no dynamic adjustment. In other words, given the difference equation $y_t = \alpha y_{t-1} + \beta$, a steady state value y^* exhibits the property that $y_t^* = y_{t+1}^*$

A difference equation can have zero, one or more (in the case of non-linear difference equations) steady state values.

In the above difference equation, the steady state value is therefore

$$y^* = \alpha y^* + \beta \Rightarrow y^* = \frac{\beta}{1 - \alpha}$$

Stability

A difference equation is said to be stable if $\lim_{t \rightarrow \infty} y_t$ is a finite quantity. If $\lim_{t \rightarrow \infty} y_t$

doesn't exist, then the difference equation is said to be unstable.

A stable difference equation will converge to a steady state value. For example, let us examine the difference equation in the above example.

$$y_t = 0.2^t (5) + 5$$

$\lim_{t \rightarrow \infty} y_t = \lim_{t \rightarrow \infty} (0.2^t(5) + 5) = 5$. Therefore the difference equation is stable.

For general form $y_t = ay_{t-1} + \beta$ where $a \neq 0$, and the solution $y_t = \alpha^t y_0 + \left[\frac{1 - \alpha^t}{1 - \alpha} \right] \beta$, it

will be stable only if $|a| < 1$, in which case the steady state value will be

$$\lim_{t \rightarrow \infty} y_t = \lim_{t \rightarrow \infty} \left(\alpha^t y_0 + \left[\frac{1 - \alpha^t}{1 - \alpha} \right] \beta \right) = \frac{\beta}{1 - \alpha}$$

The equation will be unstable if $\alpha > 1$ & $\alpha < -1$ because, in such cases

$$\lim_{t \rightarrow \infty} \left(\alpha^t y_0 + \left[\frac{1 - \alpha^t}{1 - \alpha} \right] \beta \right) \text{ Will be undefined approaching either } \infty \text{ or } -\infty$$

If $\alpha = 1$, the solution is $y_t = y_0 + \beta t$ and it will be divergent unless $\beta = 0$

If $\alpha = -1$, the solution is $y_t = y_0, t \rightarrow \text{even}$
 $y_t = -y_0 + \beta, t \rightarrow \text{odd}$

In this case, the equation will oscillate between the two values as t becomes larger and

larger in value. The only case in which this system converges is for $y_0 = \frac{\beta}{2}$. So, it will

not be stable. In general, we can identify 7 cases for the dynamic behaviors of the sequence based on the values of a . this are summarized as follows.

Value of a	$\lim_{t \rightarrow \infty} \alpha^t$	Dynamic behavior of the function
$\alpha > 1$	∞	Diverges from the equilibrium (unstable)
$\alpha = 1$	1	Stable
$0 < \alpha < 1$	0	Stable
$\alpha = 0$	0	Stable
$-1 < \alpha < 0$	0	Alternate b/n positive and negative values but approach to equilibrium) (stable)
$\alpha = -1$	-	Alternate between 1 and -1 for even and odd values of t
$\alpha < -1$	-	Alternate b/n positive and negative values but diverge

		from equilibrium) (unstable)
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6.7 Practical Application of Difference Equation in Economics

The cobweb model

In some markets, particularly agricultural markets, supply cannot immediately expand to meet increased demand. Crops have to be planted and grown and livestock takes time to raise. Some manufactured products can also take a while to produce when orders suddenly increase. The cobweb model takes into account this delayed response on the supply side of a market by assuming that quantity supplied now (Q_{st}) depends on the ruling price in the previous time period (P_{t-1}), i.e. $Q_{st} = f(P_{t-1})$ where the subscripts denote the time period. Consumer demand for the same product (Q_{dt}), however, is assumed to depend on the current price, i.e. $Q_{dt} = f(P_t)$

This is a reasonable picture of many agricultural markets. The quantity offered for sale this year depends on what was planted at the start of the growing season, which in turn depends on last year's price. Consumers look at current prices, though, when deciding what to buy.

The cobweb model also assumes that:

- ▲ the market is perfectly competitive
- ▲ Supply and demand are both linear schedules.

The assumptions of the cobweb model mean that the demand and supply functions can be specified in the format

$$\begin{aligned} Q_{dt} &= a - bp_t \text{ \&} \\ Q_{st} &= -c + dp_{t-1} \end{aligned}$$

Where a , b , c and d are parameters specific to individual markets. Assuming that market clears at each period, we can transform the above two equations as 1st order linear difference equation.

$$Qdt = Qst$$

$$a - bp_t = -c + dp_{t-1}$$

$$dp_{t-1} + bp_t = a + c$$

$$p_t + \frac{d}{b} p_{t-1} = \frac{a+c}{b}$$

The general solution for this equation is $p_t = A\left(-\frac{d}{b}\right)^t + \frac{a+c}{b+d}$.

$$\text{At } t=0 \quad p_0 = A + \frac{a+c}{b+d} \Rightarrow A = p_0 - \frac{a+c}{b+d}$$

Using this initial condition, we can definitize the above solution as

$$p_t = \left(p_0 - \frac{a+c}{b+d}\right)\left(-\frac{d}{b}\right)^t + \frac{a+c}{b+d}. \quad \text{But, } \frac{a+c}{b+d} = \bar{p}.$$

$$p_t = (p_0 - \bar{p})\left(-\frac{d}{b}\right)^t + \bar{p}$$

In the last equation, the $(p_0 - \bar{p})$ term shows whether the time path is above or below the

equilibrium value. On the other hand, the negative sign in the term $\left(-\frac{d}{b}\right)^t$ dictates that

the time path will be oscillatory. Since b & $d > 0$, $\left(-\frac{d}{b}\right)^t$ will have positive and negative

values for even and odd values of t . as far as the values of b & d are concerned, there will be 3 distinct oscillations in the cobweb model. This are

1. When $d > b$, the oscillation will be explosive, and the dynamic value of the price level will centrifugally go farther and farther from the intertemporal equilibrium
2. When $d = b$, the oscillation will be uniform
3. When $d < b$, the oscillation will be damping, and the time path of the price level will centripetally converge.

Answer key

Exercise 1.1

1. At $x = 2$, the function is continuous b/c
- $$\begin{aligned} -x + 5 &= -2 + 5 = 3 \\ x + 1 &= 2 + 1 = 3 \end{aligned}$$

To check the differentiability, let us check the existence of limit of the difference quotient.

$$\begin{aligned} \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0^-} \frac{((x+h)+1) - (x+1)}{h} = 1 \\ \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} &= \lim_{h \rightarrow 0^+} \frac{(-(x+h)+5) - (-x+5)}{h} = -1 \end{aligned}$$

Since the limit of the difference quotient doesn't exist, the function is not differentiable at $x=2$

2. A. $y = x^2 \log_3 x$

$$\frac{dy}{dx} = 2x \log_3 x + x^2 \frac{1}{x \ln 3} = 2x \log_3 x + \frac{x}{\ln 3}$$

b. $g(x) = e^{(5x+4)}$

$$\frac{dg}{dx} = 5e^{5x+4}$$

c. $f(x) = 2x^3 - 15x^2 + 36x + 4$

$$\frac{df}{dx} = 6x^2 - 30x + 36$$

3. $g(x) = 4\sqrt{x} - x \Rightarrow \frac{dg}{dx} = \frac{4}{2\sqrt{x}} - 1 \Rightarrow \frac{d^2g}{dx^2} = -\frac{1}{x}$

Exercise 1.2

a.

$$\int \frac{3x+11}{x^2-x-6} dx \Rightarrow \frac{3x+11}{x^2-x-6} = \frac{a}{x-6} + \frac{b}{x+5}$$

$$\Rightarrow ax + 5a + bx - 6b = 3x + 1$$

$$a + b = 3 \text{ \& } 5a - 6b = 1$$

$$a = \frac{19}{11} \text{ \& } b = \frac{14}{11}$$

$$\int \frac{3x+11}{x^2-x-6} dx = \int \frac{19}{11(x-6)} dx + \int \frac{14}{11(x+5)} dx$$

$$\frac{19}{11} (\ln(x-6)) + \frac{14}{11} \ln(x+5)$$

b. $\int x\sqrt{2x+1} dx$ let $u = 2x+1, x = \frac{u-1}{2}$

$$\int x\sqrt{2x+1} dx = \int \frac{u-1}{2} \sqrt{u} \frac{du}{2} = \frac{1}{2} \left(\frac{u^{\frac{5}{2}}}{\frac{5}{2}} - \frac{u^{\frac{3}{2}}}{\frac{3}{2}} \right)$$

$$2. \int_1^3 (20+4x) = 20x + 2x^2 \Big|_1^3 = (20(3) + 2(3)^2) - (20 + 2) = 56$$

Exercise 2.1

1. $Q = \alpha p^\beta$, where α and β are constants

$$\varepsilon_d = \frac{dQ}{dP} \cdot \frac{P}{Q} = \alpha \beta P^{\beta-1} \frac{P}{\alpha P^\beta} = \beta$$

2. $\frac{d(\log_a^y)}{d(\log_a^x)}, a \neq c = \frac{dy}{dx} \cdot \frac{x}{y}?$

3. Given $\frac{1}{9}x^3 - \frac{1}{6}x^2 - \frac{2}{3}x + 1$, find the critical points and determine whether they are maximum or minimum.

$$\frac{dy}{dx} = \frac{1}{3}(x^2 - x - 2) = 0, \Rightarrow x = -1 \text{ \& } 2$$

$$\frac{d^2y}{dx^2} = \frac{1}{3}(2x - 1), \text{ at } x = -1, \frac{d^2y}{dx^2} = -1 \Rightarrow \text{maximum}$$

$$x = 2, \frac{d^2y}{dx^2} = 1 \Rightarrow \text{minimum}$$

4. $C(Q) = aQ^2 + bQ + c, (Q > 0).$

First order condition: $C'(x) = 2aQ + b = 0 \Rightarrow Q = -\frac{b}{2a}$

Second order condition: $C''(x) = 2a$. **The cost function** will have minimum point only if the second derivative is positive. So $C(x)$ will have minimum iff $a > 0$.

5. $C(Q) = aQ^3 + bQ^2 + cQ + d$

$$C'(x) = 3aQ^2 + 2bQ + c$$

$$C''(x) = 6aQ + 2b = 0 \Rightarrow Q = -\frac{b}{3a}, \text{ inflection}$$

$$C''(x) > 0, Q > -\frac{b}{3a} \text{ concave up}$$

$$C''(x) < 0, Q < -\frac{b}{3a}, \text{ concave down}$$

Exercise 2.2

1. Quadratic approximation of $f(x) = (5x + 3)^{-2}$, about $x=0$

$$f(0) = 3^{-2} \Rightarrow \frac{1}{9}$$

$$f'(x) = -10(5x + 3) \Rightarrow f'(0) = -30$$

$$f''(x) = -10$$

$$f(x) \approx \frac{1}{9} - 30x - 10x^2$$

2. Quadratic approximation of $F(K) = AK^\alpha$ about $K = 1$

$$F(1) = A$$

$$F'(k) = aAk^{a-1} \Rightarrow F(1) = aA$$

$$F''(k) = a(a-1)Ak^{a-2} \Rightarrow F''(1) = a(a-1)A$$

$$F(k) \approx A + aA(k-1) + \frac{a(a-1)A}{2}(k-1)^2$$

3. Prove that $(1+x)^m \approx \underline{\underline{1+mx}}$ for x close to zero

We can easily come up with the proof using linear approximation of the left hand function about x=0

$$f(0) = (1+0)^m = 1,$$

$$f'(x) = m(1+x)^{m-1} \Rightarrow f'(0) = m$$

$$f(x) \approx 1 + mx; \text{ proved}$$

Exercise 2.3

1. $X_1 = \frac{MP_2}{P_1^2},$

- a. if there is a neutral inflation it means that there is a proportional increase in the price of the two commodities. The effect of such an increase in the prices on the demand can be determined as follows. Let's assume that there is a proportional increase in the price of the two commodities by a rate h . the new price will be $P_n = (1+h)P$. Let $r = 1+h \Rightarrow P_n = rP$

$$\frac{MrP_2}{rP_1^2} = \frac{MP_2}{rP_1^2} = r^{-1}x_1 \Rightarrow \frac{x_1}{r}. \text{ So the new quantity demanded is the}$$

quantity demanded before inflation net of inflation.

- b. By Euler theorem if a function is homogeneous of degree r , the first order partials should be homogeneous of degree $r-1$.

$$\frac{\partial x}{\partial P_1} = -\frac{2MP_2P_1}{P_1^4} \Rightarrow \frac{-2M(rP_2)(rP_1)}{(rP_1)^4} = r^{-2} \left(-\frac{2MP_2P_1}{P_1^4} \right)$$

$$\frac{\partial x}{\partial P_2} = \frac{M}{P_1^2} \Rightarrow \frac{M}{(rP_1)^2} = r^{-2} \left(\frac{M}{P_1^2} \right). \text{ So it satisfies Euler's theorem}$$

2. $f(x) = 6 + 8x - 7x^2 + 3x^3 + 2x^4$

Solution to the self test Exercises

Chapter – Three

1. Given

Cost: $C(x, y) = 0.04x^2 + 0.01xy + 0.01y^2$

Price of x: $P_x = 15$

Price of y $P_y = 9$

$\pi = TR - TC$ where π = profit

TR = Total Revenue

TC = Total Cost

$TR = TR_x + TR_y$

$\pi = 15x + 9y - (0.04x^2 + 0.01xy + 0.01y^2 + 4x + 2y + 500)$

F.O Partial

$\frac{\partial \pi}{\partial x} = 0: 15 - 0.08x - 0.01y - 4 = 0 \dots \dots \dots (1)$

$\frac{\partial \pi}{\partial y} = 0: 9 - 0.01x - 0.02y - 2 = 0 \dots \dots \dots (2)$

From (1), we have $11 = 0.08x + 0.01y$ or $0.08x + 0.01y = 11 \dots \dots \dots (3)$

From (2), we have $0.01x + 0.02y = 7 \dots \dots \dots (4)$

Equations (3) & (4) can be solved by one of the following ways

- substitution method
- Inverse method
- Cramers rule
- Gaussian Elimination method

Let us now use the Cramers rule

$$X = \frac{|A_x|}{|A|} \quad \begin{bmatrix} 0.08 & 0.01 \\ 0.01 & 0.02 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 11 \\ 7 \end{bmatrix}$$

$|A| = (0.08)(0.02) - (0.01)(0.01)$

$= 0.0016 - 0.0001 = 0.0015$

$|A_x| = \begin{vmatrix} 11 & 0.01 \\ 7 & 0.02 \end{vmatrix} = 11(0.02) - 7(0.01) = 0.22 - 0.07 = 0.15$

Hence $x = \frac{0.15}{0.0015} = 100$

Second Order Sufficient Condition

$$\pi_{xx} = -0.08 \quad \pi_{xy} = -0.01 = \pi_{yx}, \pi_{yy} = -0.02$$

$$H = \begin{vmatrix} \pi_{xx} & \pi_{xy} \\ \pi_{yx} & \pi_{yy} \end{vmatrix}$$

$$= \begin{vmatrix} -0.08 & -0.01 \\ -0.01 & -0.02 \end{vmatrix} \quad \begin{aligned} |H_1| &= -0.08 < 0 \\ |H_2| &= |H| = 0.0015 > 0 \end{aligned}$$

$$Y = \frac{|A_y|}{|A|}$$

$$|A_y| = \begin{vmatrix} 0.08 & 11 \\ 0.01 & 7 \end{vmatrix} = 0.08(7) - 0.01(11)$$

$$= 0.56 - 0.11$$

$$= 0.45$$

$$\text{Therefore } Y = \frac{0.45}{0.0015} = \underline{\underline{300}}$$

The value (amounts) of x and y that maximize the profit of the firm are 100 and 300 respectively.

2. Given production firm $Q = f(L, K) = 6\sqrt{k^3} \sqrt{L}$

$$P_Q = 0.5$$

$$P_k = 0.1$$

$$P_L = 1$$

$$c = 0.1K + L$$

$$R = 0.5/6\sqrt{K^3} \sqrt{L}$$

$$R = 3k^{3/2} L^{1/2}$$

$$\Pi = 3\sqrt{k^3} \sqrt{L} - 0.1k - L$$

$$= 3k^{3/2} L^{1/2} - 0.1k - L$$

F.O.C.

$$\frac{\partial \Pi}{\partial L} = \frac{3}{2} k^{3/2} L^{-1/2} - 1 = 0 \dots \dots \dots (1)$$

$$\frac{\partial \Pi}{\partial k} = 3 \cdot \frac{3}{2} K^{1/2} L^{1/2} - 0.1 = 0 \dots \dots \dots (2)$$

$$\text{from (1), } \frac{\frac{3}{2} k^{3/2}}{L^{1/2}} = 1$$

$$\frac{2}{3} \times \frac{3}{2} k^{3/2} = \left(L^{1/2} \right) \frac{2}{3}$$

$$k^{3/2} = \frac{2}{3} L^{1/2}$$

$$k^{\frac{3}{2} \times \frac{2}{3}} = \left(\frac{2}{3} L^{\frac{1}{2}} \right)^{\frac{2}{3}}$$

$$k = \left(\frac{2}{3} \right)^{\frac{2}{3}} L^{\frac{1}{3}}$$

$$= \left(\frac{4}{9} \right)^{\frac{1}{3}} L^{\frac{1}{3}}$$

$$k = \left(\frac{4}{9} L \right)^{\frac{1}{3}} \dots\dots\dots (3)$$

Substitute the value of k from equation (3) into (2) to find the value of L.

$$3 \frac{3}{2} \left[\left(\frac{4}{9} L \right)^{\frac{1}{3}} \right]^{\frac{1}{2}} L^{\frac{1}{2}} = 0.1$$

$$\frac{9}{2} \left[\frac{4}{9} L \right]^{\frac{1}{6}} L^{\frac{1}{2}} = 0$$

$$\frac{9}{2} \left(\frac{4}{9} \right)^{\frac{1}{6}} L^{\frac{1}{6}} L^{\frac{1}{2}} = 0.1$$

$$\frac{9}{2} \left(\frac{4}{9} \right)^{\frac{1}{6}} L^{\frac{2}{3}} = 0.1$$

CHAPTER 3

3, Solutions to the self test exercises

Given: constant unit price=P

Inputs of x_1 and x_2 are purchased at 4 constant unit factor prices of w_1 and w_2 respectively.

The firm faces competitive input and output market and the production function is given by

Note that $\frac{\partial Q(x_1, x_2)}{\partial x_1} = MP_{x_1}$ when MP = marginal product

$$\frac{\partial Q(x_1, x_2)}{\partial x_2} = MP_{x_2}$$

Thus $\frac{P \partial Q(x_1, x_2)}{\partial x_1} = VMP_{x_1}$

$$\frac{P \partial Q(x_1, x_2)}{\partial x_2} = VMP_{x_2}$$

Dividing equation (1) by (2) yields

$$\frac{P \partial Q(x_1, x_2) / \partial x_1}{P \partial Q(x_1, x_2) / \partial x_2} = \frac{w_1}{w_2}$$

$$\frac{VMP_{x_1}}{VMP_{x_2}} = \frac{w_1}{w_2}$$

This means that optimal factor utilization is achieved when the ratios of the value marginal products of the respective factors are equal to the ratios of the input prices.

The second order condition requires that the slope of the VMP of the X's be lower than of the slope of the input (factor) prices i.e.

$$Q(x_1, x_2)$$

$$\Pi = PQ - C$$

$$Q = Q(x_1, x_2)$$

$$C = w_1 x_1 + w_2 x_2$$

$$\Pi = PQ(x_1, x_2) - w_1 x_1 - w_2 x_2$$

$$\Pi(x_1, x_2) = PQ(x_1, x_2) - w_1 x_1 - w_2 x_2$$

Taking the F.O.C

$$\frac{\partial \Pi}{\partial x_1} = 0 \quad \frac{P \partial Q(x_1, x_2)}{\partial x_1} - w_1 = 0 \dots \dots \dots (1)$$

$$\frac{\partial \Pi}{\partial x_2} = 0 \quad \frac{P \partial Q(x_1, x_2)}{\partial x_2} - w_2 = 0 \dots \dots \dots (2)$$

Given $\Pi = PQ - C$

$$\Pi(x_1, x_2) = PQ(x_1, x_2) - w_1 x_1 - w_2 x_2$$

F.O.C

$$\frac{\partial \Pi(x_1, x_2)}{\partial x_1} = 0, \quad \frac{P \partial Q(x_1, x_2)}{\partial x_1} - w_1 = 0 \dots \dots \dots (1)$$

$$\frac{\partial \Pi(x_1, x_2)}{\partial x_2} = 0, \quad \frac{P \partial Q(x_1, x_2)}{\partial x_2} - w_2 = 0 \dots \dots \dots (2)$$

S.O.C

$$\frac{\partial^2 \Pi(x_1, x_2)}{\partial x_1^2} < 0, \quad \frac{P \partial^2 (x_1, x_2)}{\partial x_1^2} < 0 \quad \text{b/c } \frac{\partial w_1}{\partial x_1} = 0$$

$$\frac{\partial^2 \Pi(x_1, x_2)}{\partial x_2^2} < 0, \quad \frac{P \partial^2 (x_1, x_2)}{\partial x_2^2} < 0 \quad \text{b/c } \frac{\partial w_2}{\partial x_2} = 0$$

4. Consider a monopolist firm

Given products Q_1, Q_2

The demands facing the monopolist firm are given as follows :

$$P_1 = 55 - Q_1 - Q_2$$

$$P_2 = 70 - Q_1 - 2Q_2$$

$$C(Q_1, Q_2) = Q_1^2 + Q_1 Q_2 + Q_2^2$$

$$\Pi(PQ) = TR_1 + TR_2 - TC$$

$$= P_1 Q_1 + P_2 Q_2 - TC$$

$$\begin{aligned} &= (55 - Q_1 - Q_2)Q_1 + (70 - Q_1 - 2Q_2)Q_2 - Q_1^2 - Q_1 Q_2 - Q_2^2 \\ &= 55Q_1 - Q_1^2 - Q_1 Q_2 + 70Q_2 - Q_1 Q_2 - 2Q_2^2 - Q_1^2 - Q_1 Q_2 - Q_2^2 \\ &= 55Q_1 + 70Q_2 - 3Q_1 Q_2 - 2Q_1^2 - 3Q_2^2 \end{aligned}$$

F.O.C

$$\frac{\partial \pi(Q_1, Q_2)}{\partial Q_1} = 55 - 3Q_2 - 4Q_1 = 0 \dots \dots \dots (1)$$

$$\frac{\partial \pi(Q_1, Q_2)}{\partial Q_2} = 0, \quad 70 - 3Q_1 - 6Q_2 = 0 \dots \dots \dots (2)$$

From equation (1), we have

$$4Q_1 + 3Q_2 = 55$$

$$3Q_1 + 6Q_2 = 70$$

$$A = \begin{bmatrix} 4 & 3 \\ 3 & 6 \end{bmatrix} \begin{bmatrix} Q_1 \\ Q_2 \end{bmatrix} = \begin{bmatrix} 55 \\ 70 \end{bmatrix} \text{ is matrix form}$$

Using Cramer's rule

$$Q_1 = \frac{|A_{Q_1}|}{|A|}$$

$$|A| = 6(4) - 3(3) = 15$$

$$\begin{aligned} |A_{Q_1}| &= \begin{vmatrix} 55 & 3 \\ 70 & 6 \end{vmatrix} = 55(6) - 70(3) \\ &= 330 - 320 \\ &= 10 \end{aligned}$$

$$Q_1 = \frac{|A_{Q_1}|}{|A|} = \frac{120}{15} = 8$$

$$|A_{Q_2}| = \begin{vmatrix} 4 & 55 \\ 3 & 70 \end{vmatrix} = 4(70) - 3(55) = 280 - 165 = 115$$

$$Q_2 = \frac{|A_{Q_2}|}{|A|} = \frac{115}{15} \approx \underline{\underline{7.67}}$$

S.O.C

$$|H| = \begin{vmatrix} \Pi_{11} & \Pi_{12} \\ \Pi_{21} & \Pi_{22} \end{vmatrix}$$

$$\Pi_{11} = -4 \qquad \Pi_{21} = -3$$

$$\Pi_{12} = -3 \qquad \Pi_{22} = -6$$

$$\text{Thus, } |H| = \begin{vmatrix} -4 & -3 \\ -3 & -6 \end{vmatrix}$$

$$\begin{aligned} |H_1| &= -4 < 0, |H_2| = |H| = -4(-6) - [-3(-3)] \\ &= 24 - 9 \\ &= 15 > 0 \end{aligned}$$

Since the values of the Hessian determinant alternate in signs, profit is maximized at the value of output where $Q_1 = 8$ & $Q_2 = \underline{\underline{7.67}}$

5. The demands functions facing the monopolist are given by

$$P_1 = 63 - 4Q_1$$

$$P_2 = 105 - 5Q_2$$

$$P_3 = 75 - 6Q_3$$

The total cost function is $C(Q) = 20 + 15Q$

Where $Q = Q_1 + Q_2 + Q_3$

$$\Pi = TR - TC$$

$$TR = TR_1 + TR_2 + TR_3$$

Profit (Π) is maximized when $MR_1 = MC_1$

$$MR_2 = MC_2$$

$$MR_3 = MC_3$$

$$\begin{aligned}
 \Pi &= TR_1 + TR_2 + TR_3 - TC \\
 &= P_1 Q_1 + P_2 Q_2 + P_3 Q_3 - TC \\
 &= (63 - 4Q_1)Q_1 + (105 - 5Q_2)Q_2 + (75 - 6Q_3)Q_3 - [20 - 15(Q_1 + Q_2 + Q_3)] \\
 &\quad \text{because } Q = Q_1 + Q_2 + Q_3 \\
 &= 63Q_1 - 4Q_1^2 + 105Q_2 - 5Q_2^2 + 75Q_3 - 6Q_3^2 - 20 - 15Q_1 - 15Q_2 - 15Q_3 \\
 &= 48Q_1 - 4Q_1^2 + 90Q_2 - 5Q_2^2 + 60Q_3 - 6Q_3^2 - 20
 \end{aligned}$$

F.O.C

$$\frac{\partial \Pi}{\partial Q_1} = 48 - 8Q_1 = 0 \dots \dots \dots (1) \quad Q_1 = 6$$

$$\frac{\partial \Pi}{\partial Q_2} = 90 - 10Q_2 = 0 \dots \dots \dots (2) \quad Q_2 = 9$$

$$\frac{\partial \Pi}{\partial Q_3} = 60 - 12Q_3 = 0 \dots \dots \dots (3) \quad Q_3 = 5$$

S.O.C

$$\begin{aligned}
 H &= \begin{vmatrix} \Pi_{11} & \Pi_{12} & \Pi_{13} \\ \Pi_{21} & \Pi_{22} & \Pi_{23} \\ \Pi_{31} & \Pi_{32} & \Pi_{33} \end{vmatrix} \\
 &= \begin{vmatrix} -8 & 0 & 0 \\ 0 & -10 & 0 \\ 0 & 0 & -12 \end{vmatrix}
 \end{aligned}$$

$$|H_1| = -8 < 0$$

$$|H_2| = 80 > 0$$

$$|H_3| = -960 < 0$$

Thus, the monopolist maximizes profit when it produce $Q_1=6$, $Q_2=9$ and $Q_3=5$ and total $Q = Q_1 + Q_2 + Q_3$

$$= 6 + 9 + 5$$

$$= \underline{\underline{20}}$$

$$\begin{aligned}
 P_1 &= 63 - 4Q_1 \\
 &= 63 - 4(6) \\
 &= 63 - 24 \\
 &= \underline{\underline{39}}
 \end{aligned}$$

$$\begin{aligned}
 P_2 &= 105 - 5(Q_2) \\
 &= 105 - 5(9) \\
 &= 105 - 45 \\
 &= \underline{\underline{60}}
 \end{aligned}$$

$$\begin{aligned}
 P_3 &= 75 - 6Q_3 \\
 &= 75 - 6(5) \\
 &= 75 - 30 \\
 &= \underline{\underline{45}}
 \end{aligned}$$

6. $P_1 = 100 - Q_1$

$$P_2 = 80 - Q_2$$

$$c = 6Q \quad Q = Q_1 + Q_2$$

$$\begin{aligned}
 \Pi &= TR_1 + TR_2 - TC \\
 &= P_1 Q_1 + P_2 Q_2 - c \\
 &= (100 - Q_1)Q_1 + (80 - Q_2)Q_2 - 6(Q_1 + Q_2) \text{ because } Q = Q_1 + Q_2 \\
 &= 100Q_1 - Q_1^2 + 80Q_2 - Q_2^2 - 6Q_1 - 6Q_2 \\
 \Pi &= 94Q_1 - Q_1^2 + 74Q_2 - Q_2^2
 \end{aligned}$$

F.O.C

$$\frac{\partial \Pi}{\partial Q_1} = 94 - 2Q_1 = 0 \quad Q_1 = 47$$

$$\frac{\partial \Pi}{\partial Q_2} = 74 - 2Q_2 = 0, \quad Q_2 = 37$$

i.

$$\begin{aligned}
 a) \quad Q &= Q_1 + Q_2 \\
 &= 47 + 37 \\
 &= \underline{\underline{84}}
 \end{aligned}$$

$Q_1 = 47$ and $Q_2 = 37$ would be sold in the two markets.

$$\begin{aligned}
 b) \quad P_1 &= 100 - Q_1 & P_2 &= 80 - Q_2 \\
 &= 100 - 47 & &= 80 - 37 \\
 &= \underline{\underline{53}} & &= \underline{\underline{43}}
 \end{aligned}$$

$$\begin{aligned}
 c) \quad \Pi &= 94Q_1 - Q_1^2 + 74Q_2 - Q_2^2 \\
 &= 94(47) - (47)^2 + 74(37) - (37)^2
 \end{aligned}$$

ii) If price discrimination is made illegal

a) In the absence of price discrimination $\underline{\underline{P_1 = P_2 = P}}$

$$\text{Thus, } P_1 = 100 - Q_1 \Rightarrow P = 100 - Q_1$$

$100 - P = Q_1$

$$P_2 = 80 - Q_2 \Rightarrow P = 80 - Q_2$$

$80 - P = Q_2$

$$\begin{aligned}
 \text{Beside } Q &= Q_1 + Q_2 \\
 &= 100 - P + 80 - P \\
 Q &= 180 = 2P \\
 P &= 90 - \frac{1}{2}Q
 \end{aligned}$$

Therefore,

$$\Pi = R - C \quad R = PQ$$

$$= \left(90 - \frac{1}{2}Q\right)Q - 6Q$$

$$= 90Q - \frac{1}{2}Q^2 - 6Q$$

$$\Pi = 84Q - \frac{1}{2}Q^2$$

F.O.C.

$$\frac{\partial \Pi}{\partial Q} = 84 - Q = 0$$

$$Q = \underline{\underline{84}}$$

b) Since $P = 90 - \frac{1}{2}Q$

$$= 90 - \frac{1}{2}(84)$$

$$= 90 - 42$$

$$= \underline{\underline{48}}$$

c) $\Pi = R - C$

$$= 84Q - \frac{1}{2}Q^2$$

$$= 84(84) - \frac{1}{2}(84)^2$$

$$= 7056 - \frac{1}{2}(7056)$$

$$= 7056 - 3528$$

$$= \underline{\underline{3528}}$$

This is the amount of profit that the monopolist gets when it does not practice price discrimination.

The amount of profit loss = profit obtainable with price discrimination - profit obtainable without price discrimination.

Exercise 4.1

1. $y = 3x_1^2 - 5x_1 - x_1x_2 + 6x_2^2 - 4x_2 + 2x_2x_3 + 4x_3^2 + 2x_3 - 3x_1x_3$,

First order condition

$$\frac{\partial y}{\partial x_1} = 6x_1 - 5 - x_2 - 3x_3 = 0 \Rightarrow 6x_1 - x_2 - 3x_3 = 5$$

$$\frac{\partial y}{\partial x_2} = -x_1 + 6x_2 + 2x_3 - 4 = 0 \Rightarrow -x_1 + 6x_2 + 2x_3 = 4$$

$$\frac{\partial y}{\partial x_3} = -3x_1 + 2x_2 + 8x_3 + 2 = 0 \Rightarrow -3x_1 + 2x_2 + 8x_3 = -2$$

$$\begin{vmatrix} 6 & -1 & -3 \\ -1 & 6 & 2 \\ -3 & 2 & 8 \end{vmatrix} \begin{vmatrix} x_1 \\ x_2 \\ x_3 \end{vmatrix} = \begin{vmatrix} 5 \\ 4 \\ -1 \end{vmatrix}$$

Using crammers' rule

$$x_1 = \frac{\begin{vmatrix} 5 & -1 & -3 \\ 4 & 6 & 2 \\ -1 & 2 & 8 \end{vmatrix}}{\begin{vmatrix} 6 & -1 & -3 \\ -1 & 6 & 2 \\ -3 & 2 & 8 \end{vmatrix}}, x_2 = \frac{\begin{vmatrix} 6 & 5 & -3 \\ -1 & 4 & 2 \\ -3 & -1 & 8 \end{vmatrix}}{\begin{vmatrix} 6 & -1 & -3 \\ -1 & 6 & 2 \\ -3 & 2 & 8 \end{vmatrix}}, x_3 = \frac{\begin{vmatrix} 6 & -1 & 5 \\ -1 & 6 & 4 \\ -3 & 2 & -1 \end{vmatrix}}{\begin{vmatrix} 6 & -1 & -3 \\ -1 & 6 & 2 \\ -3 & 2 & 8 \end{vmatrix}} =$$

Second order condition ;

$$f_{11} = 6, f_{12} = -1, f_{13} = -3, f_{21} = -1, f_{22} = 6, f_{23} = 2, f_{31} = -3, f_{32} = 2, f_{33} = 8$$

So, the third order hessian matrix will be $H = \begin{vmatrix} 6 & -1 & -3 \\ -1 & 6 & 2 \\ -3 & 2 & 8 \end{vmatrix}$.

Principal minors $|H_1| = 6 > 0, |H_2| = \begin{vmatrix} 6 & -1 \\ -1 & 6 \end{vmatrix} = 5 > 0, |H_3| = 214 > 0$ so, the point is maximum.

3. Maximize $c = 45x^2 + 90xy + 90y^2$, s.t $2x + 3y = 60$,

$$L = 45x^2 + 90xy + 90y^2 + \lambda(60 - 2x - 3y)$$

$$\frac{\partial L}{\partial x} = 90x + 90y - 2\lambda = 0$$

$$\frac{\partial L}{\partial y} = 90x + 180y - 3\lambda = 0$$

$$\frac{\partial L}{\partial \lambda} = 2x + 3y = 60$$

$$\begin{vmatrix} 45 & 45 & -1 \\ 30 & 60 & -1 \\ 2 & 3 & 0 \end{vmatrix} \begin{vmatrix} x \\ y \\ \lambda \end{vmatrix} = \begin{vmatrix} 0 \\ 0 \\ 60 \end{vmatrix}, x = \frac{\begin{vmatrix} 0 & 45 & -1 \\ 0 & 60 & -1 \\ 60 & 3 & 0 \end{vmatrix}}{\begin{vmatrix} 45 & 45 & -1 \\ 30 & 60 & -1 \\ 2 & 3 & 0 \end{vmatrix}} = 6.66, y = \frac{\begin{vmatrix} 45 & 0 & -1 \\ 30 & 0 & -1 \\ 2 & 60 & 0 \end{vmatrix}}{\begin{vmatrix} 45 & 45 & -1 \\ 30 & 60 & -1 \\ 2 & 3 & 0 \end{vmatrix}} =$$

$$\lambda = \frac{\begin{vmatrix} 45 & 45 & 0 \\ 30 & 60 & 0 \\ 2 & 3 & 60 \end{vmatrix}}{\begin{vmatrix} 45 & 45 & -1 \\ 30 & 60 & -1 \\ 2 & 3 & 0 \end{vmatrix}} =$$

$$\text{s.o.c } H|H| = \begin{vmatrix} 90 & 90 & 2 \\ 90 & 180 & 3 \\ 2 & 3 & 0 \end{vmatrix} \Rightarrow |H1| = 90 > 0, |H2| = \begin{vmatrix} 90 & 90 \\ 90 & 180 \end{vmatrix} > 0, |H3| > 0 \text{ so, the}$$

critical values are maximum points.

5. Maximize $u(x, y) = x^{0.25}y^{0.4}$, subject to, $2x + 8y = 104$

$$l = x^{0.25}y^{0.4} + \lambda(104 - 2x - 8y)$$

$$\frac{\partial l}{\partial x} = 0.25x^{-0.75}y^{0.4} - 2\lambda = 0$$

$$\frac{\partial l}{\partial y} = 0.4x^{0.25}y^{-0.6} - 8\lambda = 0$$

$$\frac{\partial l}{\partial \lambda} = 104 - 2x - 8y = 0$$

$$x \approx 4.5, y \approx 12$$

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